

Pega Platform 26.1.1 on Azure Kubernetes Service

Externalized Kafka on Confluent Cloud and search on managed OpenSearch

Design, Helm configuration, testing and operations for six environments built from a cloned and upgraded Pega Platform 8.8 database

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Document control

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Owner	Customer platform engineering lead
Review cycle	Before each rehearsal, before the production cutover, and after any change to a vendor page listed in Section 3

Revision history

Version	Date	Change	Author
1.0	1 October 2026	Kafka and search runbook for a three-release upgrade path.	Implementation team
2.0	1 October 2026	Rewritten for the clone-and-upgrade path, six environments, shared non-production services, managed OpenSearch and Okta.	Implementation team

Approvals

Approval confirms that the approver has reviewed the sections relevant to the role and accepts the design, the recommendations and the open decisions as stated.

Role	Name	Signature	Date
Customer business owner			
Customer enterprise architect			
Customer security architect			
Customer platform engineering lead			
Customer database administration lead			
Pega Lead System Architect			
Kafka and search engineering lead			

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1 Executive summary

1.1 Purpose

This runbook tells the customer's IT teams how to design, configure, test and run Kafka and search for Pega Platform 26.1.1 on Azure Kubernetes Service (AKS). Kafka is provided by Confluent Cloud. Search is provided by the Pega Search and Reporting Service (SRS) on top of a managed OpenSearch service. Both are configured through the Pega Helm charts in six environments: DEV, SIT, UAT, PERF, PREPROD and PROD.

1.2 The clone-and-upgrade approach

The customer is moving from Pega Platform 8.8 without upgrading the running 8.8 servers. A new 26.1.1 platform is built on AKS with Kafka and search outside Pega from the start. The 8.8 database is cloned, and the clone is upgraded to 26.1.1 by the Pega installer, run as the Helm chart installer job. Pega 26.1.1 then starts on the upgraded clone. Each lower environment is built the same way, which gives the programme repeated rehearsals. Production uses a final clone taken after 8.8 is stopped. The 8.8 system and its database stay untouched until the rollback window closes. Figure 1 shows the flow.

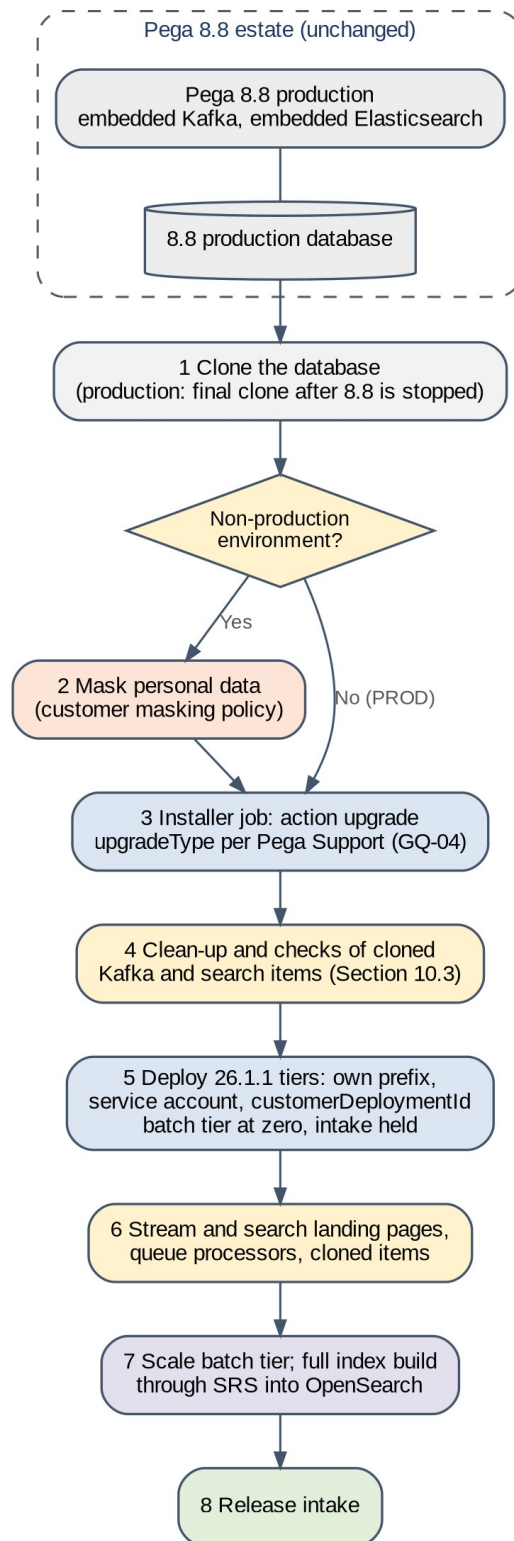


Figure 1. Clone-and-upgrade flow for each environment
 Prepared for this implementation. Not a Pega-published diagram.

1.3 Kafka and search design in brief

Table 1 summarises the design. Production has its own Kafka cluster and its own OpenSearch service. The five non-production environments share two sets of services, with isolation enforced per environment.

Table 1. Kafka and search design by environment group

Layer	PROD	PERF and PREPROD (group NP2)	DEV, SIT and UAT (group NP1)
Kafka	Confluent Cloud cluster cc-prd, Enterprise or Dedicated, Azure Private Link	Shared cluster cc-np2, same type and settings as PROD	Shared cluster cc-np1, Enterprise with Private Link
Isolation in Kafka	Dedicated cluster	Topic prefix, service account, prefixed ACLs and client quota per environment	Same as NP2
Search storage	OpenSearch service os-prd	Shared service os-np2, same provider and settings as PROD	Shared service os-np1
Isolation in search	Dedicated service	SRS per environment, customerDeploymentId per environment, OpenSearch user and index-scoped role per environment	Same as NP2
Pega-to-SRS tokens	Okta custom authorization server for PROD only	Shared non-production authorization server, one client per environment	Same as NP2
Secrets	Key Vault per environment, delivered by the External Secrets Operator	Same	Same

1.4 Sharing services to reduce cost

Six dedicated Kafka clusters and six OpenSearch services would cost the most and need the most operation. The recommended model uses three sets of services instead of six. DEV, SIT and UAT share one set. PERF and PREPROD share a second set that matches production, so load tests and production rehearsals run on production-like services without affecting functional testing. Section 5 compares this model with the alternatives and gives a cost model that the customer completes with its own prices.

1.5 Answer to the migration question

Decision. No Pega stream topic and no search index is copied from 8.8. The 8.8 system uses embedded Kafka and embedded Elasticsearch, whose data lives on the 8.8 nodes, not in the database. Pega states that existing stream data cannot be moved to a new Kafka [\[R13\]](#), and SRS indexes all searchable data after Pega connects to it [\[R15\]](#).

Before the production clone is taken, intake on 8.8 is held and queues are allowed to empty. Queue items that are held in the database, such as delayed and broken items, travel with the clone and are handled as listed in Section 10.3. Application Kafka data sets on the customer's own Kafka clusters are decided one by one (Section 11.3).

1.6 Questions for Pega Support

Pega documentation does not settle three points that the plan depends on. Section 10.1 lists seven questions in full. The three that gate the production plan are these:

- GQ-01: Pega Support must confirm that a direct upgrade of an 8.8 database to 26.1.1 is supported, and name any required 8.8 patch level.
- GQ-03: Pega states that Hazelcast must be removed before updating to '25 or later [\[R5\]](#). Pega Support must confirm how this applies to a cloned database that is upgraded offline and never started on 8.8 again.
- GQ-04: Pega Support must confirm the installer upgrade type for a cloned database ([in-place](#) or the out-of-place types the Helm chart documents [\[R23\]](#)).

1.7 Decisions needed

Table 2 lists the decisions that block the first build. Section 18.2 lists all open decisions with owners and milestones.

Table 2. Decisions needed before the first build

ID	Decision	Recommended answer	Needed by
OD-01	Pega Support answers to the gating questions	Raise the case now; plan with the conservative answer until the reply arrives	M2 DEV build
OD-02	Confluent cluster types	Enterprise with Private Link for NP1; PROD type for NP2 and PROD after partition measurement	M2 DEV build
OD-03	OpenSearch provider	Provider that passes the selection in Section 7.3	M2 DEV build
OD-05	Okta design	Dedicated custom authorization server for PROD; one shared non-production server with a per-client guid claim, if proven in DEV	M2 DEV build
OD-07	Masking approach	Mask before the upgrade, so the index build uses masked data only	M2 DEV build

1.8 Readiness statement

This document is ready for customer review as a design and runbook. Durations for draining, cloning, upgrading and index building depend on data volume, so they come from the rehearsals in Section 14.2, not from estimates. Points that no source settles are marked as tests, Pega Support questions or open decisions. Conflicts between sources are logged in Section 18.4.

2 Scope, dependencies and reading guide

2.1 Scope

Table 3. Scope of this document

In scope	Out of scope
Confluent Cloud and managed OpenSearch design for DEV, SIT, UAT, PERF, PREPROD and PROD. Configuration in the Pega Helm charts: the <code>stream</code> and <code>pegasearch</code> sections of the <code>pega</code> chart, and the SRS section of the <code>backingservices</code> chart. What the cloned 8.8 database brings that affects Kafka and search, and how it is cleaned up before and after the first 26.1.1 start. Whether Kafka topics and search indexes are migrated from 8.8, and how 8.8 in-flight work is handled at cutover. Shared non-production services with isolation between environments, and the cost model. Okta as the token issuer for Pega-to-SRS authorization. Testing, failure scenarios, known issues, troubleshooting and operations for Kafka and search.	Application functional regression content. Cassandra and Decision Data Store. Constellation. Pega Diagnostic Center beyond the network path it needs. General AKS build-out not related to Kafka or search. Commercial terms and prices from Confluent and the OpenSearch provider.

2.2 Dependencies covered only to the depth Kafka and search need

The database clone, the installer upgrade and Pega's '25 and '26 prerequisites belong to other workstreams. Table 4 names the owner of each, the gate it must pass before Kafka and search work can rely on it, and the source.

Table 4. Dependencies and their gates

Dependency	Why it matters for Kafka and search	Owner	Gate	Source
Database clone and masking	The clone carries 8.8 stream and search settings and queue items (Section 10.3). Masking decides what data the index build uses.	DBA team	Clone restored and masked; row counts recorded	Customer policy
Installer upgrade (<code>installer.upgrade</code> , <code>upgradeType</code>)	The upgrade must finish before Pega starts against Kafka and SRS.	Platform team with Pega LSA	Installer job completed; log reviewed	[R23]
Hazelcast removal	From '25, cluster messaging that Hazelcast carried uses Kafka, so a Kafka outage affects more than queue processing.	Pega LSA	GQ-03 answered	[R4, R5, R7]
Tomcat 10.1 and Jakarta EE	Custom JARs that fail to load can stop nodes before Kafka and search checks run.	Application team	Custom JARs rebuilt and tested	[R3, R9]
Primary keys on all tables	Required before an update to '25 or later.	DBA team	primaryKeyUtility dry run clean	[R3, R10]
Queue processor changes in '26	Partitioned, multi-threaded processing; DSS <code>delayeditems/dataflowbased/threadspersnode</code> must be removed (PEGA0179).	Pega LSA	DSS removed on the clone	[R3]

2.3 Confirmed customer inputs

Table 5. Confirmed customer inputs

Input	Value	Effect on this document
Environments	DEV, SIT, UAT, PERF, PREPROD, PROD	Six columns in every per-environment table; six prefixes, service accounts, deployment IDs, Okta clients and Key Vaults.
8.8 Kafka	Embedded Kafka (Pega stream service inside 8.8)	Nothing in it can be moved to Confluent Cloud. Queues are emptied on 8.8 before the final clone (Section 11).
8.8 search	Embedded Elasticsearch	Indexes are on the 8.8 nodes' disks and cannot be carried over. 26.1.1 builds all indexes through SRS (Section 7.8).
Token issuer for Pega to SRS	Okta	Okta design in Section 7.6.
OpenSearch provider	Not chosen yet	Selection method in Section 7.3; Helm values written so only the endpoint, port, credentials and CA depend on the provider.
Pega Support answers	Not received	Every gating question is open with an owner and a milestone (Section 10.1).

2.4 Where each question is answered

The customer asked thirteen questions. Table 6 shows where each one is answered, so a reviewer can check that none is missed.

Table 6. Traceability from customer questions to sections

#	Question	Answered in
1	Is the direct clone-and-upgrade path from 8.8 to 26.1.1 supported, and on what conditions?	Section 10.1
2	How is Confluent Cloud designed per environment?	Sections 5 and 6
3	How is the managed OpenSearch service designed per environment?	Sections 5 and 7
4	How is each one configured in the Helm charts for the upgrade run and the deploy run?	Section 9 and Appendix B
5	What in the cloned 8.8 database conflicts with the new configuration, and how is it cleaned up?	Section 10.3
6	How is a cloned non-production environment kept away from production Kafka, integrations and other environments?	Sections 5.3, 10.4 and 14.3
7	Must topics or indexes be migrated from 8.8, and how is in-flight work handled at cutover?	Sections 11 and 13
8	How do several non-production environments share one cluster and one OpenSearch service safely?	Sections 5.3 and 5.4
9	How much does sharing save, and what risk does it add?	Section 5.5
10	How is it tested?	Section 14
11	Which failures are tested, and how?	Section 14.6
12	What known issues should be avoided?	Section 15
13	How is it run day to day, including refresh and retirement?	Section 17

2.5 Reading guide by role

Table 7. Reading guide by role

Reader	Read first	Then
Business owner, programme manager	Sections 1, 5.5, 13 and 18	Section 14.2
Enterprise and security architects	Sections 4, 5, 7.6 and 8	Section 18
Pega Lead System Architect	Sections 10 and 11	Sections 13 to 15
Database administrators	Sections 10.1, 10.3 to 10.5 and 12.7	Section 13
Platform, Kafka and search engineers	Sections 6 to 9 and 12	Sections 16, 17, Appendices A to C
Test lead	Section 14	Section 15

2.6 Conventions and naming

- **Must** marks a vendor requirement or a fixed decision. **Should** marks a recommendation that the customer can change through the decision process in Section 18.2.
- Values in angle brackets, such as `<bootstrap-host>`, are environment-specific. Each one is listed in Appendix A.
- References in square brackets, such as [R11], point to the numbered sources in Appendix F.
- Figures marked "Prepared for this implementation" were drawn for this document. Figures from Pega, Pega Academy, Confluent and OpenSearch carry a source line, and Appendix G lists them.

Every environment uses the same naming pattern, shown in Table 8. Prefixes end with a hyphen, and no deployment ID is the start of another, so a prefixed ACL or an index pattern for one environment can never match another environment's topics or indexes.

Table 8. Environment naming

Environment	Code	Group	Topic prefix	customerDeploymentId	Service account	Okta client
DEV	dev	NP1	pega-dev-	pega26-dev	sa-pega-dev	pega-srs-dev
SIT	sit	NP1	pega-sit-	pega26-sit	sa-pega-sit	pega-srs-sit
UAT	uat	NP1	pega-uat-	pega26-uat	sa-pega-uat	pega-srs-uat
PERF	perf	NP2	pega-perf-	pega26-perf	sa-pega-perf	pega-srs-perf
PREPROD	ppd	NP2	pega-ppd-	pega26-ppd	sa-pega-ppd	pega-srs-ppd
PROD	prd	PRD	pega-prd-	pega26-prd	sa-pega-prd	pega-srs-prd

3 Evidence baseline

3.1 How the evidence was gathered

Every product statement in this document comes from one of these sources, in this order of authority:

1. Pega documentation for Pega Platform '26, and for '25 where the topic belongs to that release.
2. The Pega Helm charts repository on GitHub, chart version 4.13.0, for configuration keys and defaults.
3. Pega Academy for architecture illustrations.
4. Confluent Cloud documentation.
5. OpenSearch documentation, and the chosen provider's documentation once OD-03 is decided.
6. Okta developer documentation.
7. Microsoft Learn and the External Secrets Operator documentation.

Where two sources disagree, the higher one is followed and the conflict is logged in Section 18.4. Where no source settles a point, the document says so and turns it into a test, a Pega Support question or an open decision.

3.2 Facts this design depends on

Table 9 records the facts as published on the evidence cut-off date. Re-check them before each rehearsal, because vendors update these pages.

Table 9. Version and support facts used in this design

Topic	Fact	Source
Externalized services	From Pega Platform '24.2, embedded Kafka, Elasticsearch and Cassandra are not supported. Kafka and search must be external.	[R2]
Kubernetes	AKS is supported. x86-64 nodes only. Database in the same region as the worker nodes. Helm 3.0 or later.	[R20, R27]
Kafka role	"All Queue Processors and Job Schedulers in Pega Platform require Kafka." Without it, Pega Platform is not fully functional.	[R11]
Kafka client	Pega '26 ships the Kafka client library 4.0.0, backward compatible with 3.9.2.	[R3]
Kafka settings	message.max.bytes 5000000; replica.fetch.max.bytes 5000100; unclean.leader.election.enable false; auto.create.topics.enable false.	[R11]
Kafka security	Generic Kafka security only; vendor-specific features such as IAM roles are not supported.	[R11]
Kafka ACLs	TOPIC and GROUP on the stream name prefix (ALL, PREFIXED); TRANSACTIONAL_ID * (READ and WRITE, LITERAL); CLUSTER IDEMPOTENT_WRITE.	[R11]
Stream data	Existing stream data cannot be moved to a new Kafka provider.	[R13]
Helm stream keys	<code>stream.saslMechanism</code> accepts PLAIN, SCRAM-SHA-256 and SCRAM-SHA-512. External secret keys: STREAM_TRUSTSTORE_PASSWORD, STREAM_KEYSTORE_PASSWORD, STREAM_JAAS_CONFIG.	[R23]
SRS role	SRS replaces embedded search and the legacy plug-in, which are not available from '24.2.	[R14]
SRS matrix	SRS 1.44.3 or later. Image <code>search-n-reporting-service-os</code> supports OpenSearch 1.3, 2.15 and 2.19; best practice OpenSearch 2.15.	[R14]
SRS sharing	SRS can support more than one Pega deployment; it isolates data with a unique CUSTOMER_DEPLOYMENT_ID. Do not share the search service with non-Pega software.	[R14]
Deployment ID	The customerDeploymentId is the prefix of all indexes and must equal the <code>guid</code> claim when OAuth is used. It defaults to the namespace name.	[R23]
Index build	After Pega connects to SRS, SRS indexes all searchable data, which requires a downtime period.	[R15]

Topic	Fact	Source
Search settings	Index auto-creation must be off (SRS sets it if its user has the manage cluster privilege). destructive_requires_name false.	[R14]
SRS tokens	OAuth client credentials with private_key_jwt or client_secret_basic; scope pega.search:full; private key in SRS_OAUTH_PRIVATE_KEY.	[R23]
Hazelcast	Removed in '25. "Before updating to Pega Platform '25, remove Hazelcast from your deployment." Affects embedded Hazelcast and the Clustering Service.	[R5]
Queue processing in '26	Partitioned, multi-threaded queue processing. Remove DSS delayeditems/dataflowbased/threadspernode (PEGA0179). SystemPulse topic uses 6 partitions.	[R3]
Installer actions	install, deploy, install-deploy, upgrade, upgrade-deploy. Upgrade types include in-place, out-of-place-rules, out-of-place-data, zero-downtime, custom.	[R23]
Confluent quotas	Client quotas are supported on Enterprise, Freight and Dedicated clusters, applied per service account or identity pool, not per API key.	[R36]
Okta	Custom scopes and custom claims need a custom authorization server. The org authorization server cannot be customized.	[R42, R43, R44]

4 Architecture

4.1 Pega reference architecture for '26

Pega's Kubernetes architecture for '26 has three layers: the Pega application namespace (web and batch deployments and an installer job), backing services that Pega provides (SRS and Constellation), and services the client provides (search, Kafka, Cassandra for Customer Decision Hub only, and the database) [R21]. Hazelcast does not appear, because '25 removed it.

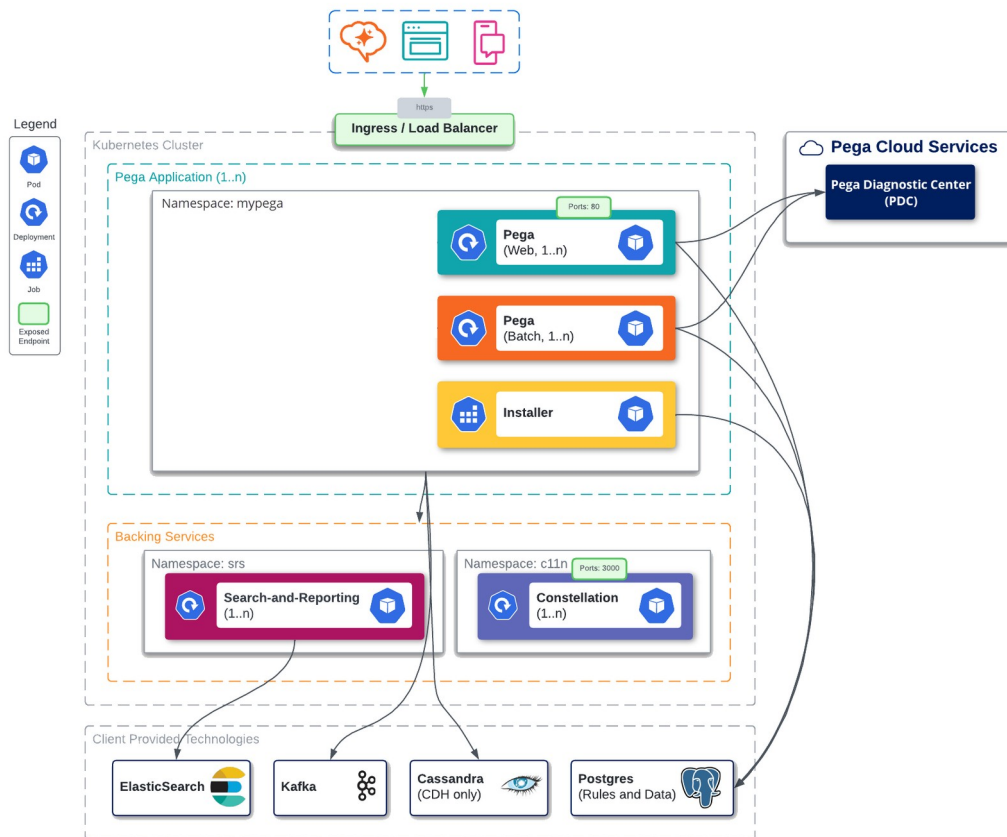


Figure 2. Pega Platform Kubernetes architecture for '26

Source: Pega Documentation, "Understanding the Pega Platform deployment architecture" [R21]. © Pegasystems Inc. Reproduced with attribution. The database shown is an example.

The 8.8 source system runs Kafka and Elasticsearch inside the Pega cluster. Pega Academy shows that older layout and the externalized layout side by side [R29, R30]. The comparison explains why nothing in the 8.8 Kafka or search storage can be moved: it lives on the 8.8 nodes.

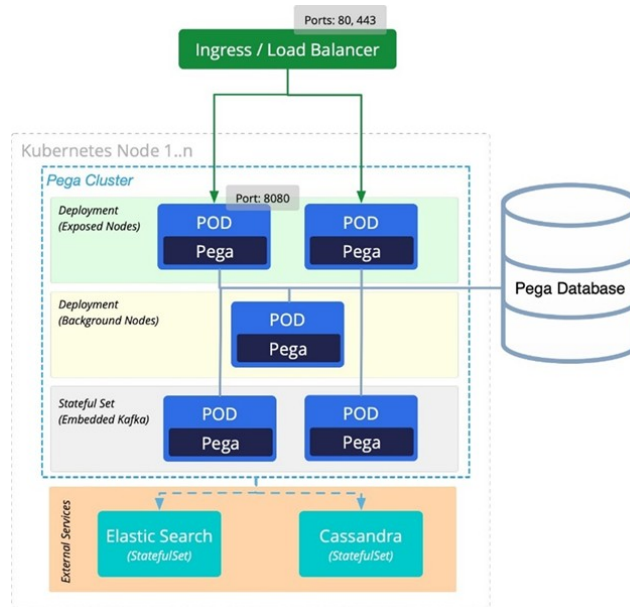


Figure 3. Earlier Pega cluster layout with embedded Kafka and in-cluster Elasticsearch

Source: Pega Academy, "Cloud deployment architecture" [R30]. © Pegasystems Inc. Reproduced with attribution. Shown to explain the 8.8 source; this layout is not supported from '24.2.

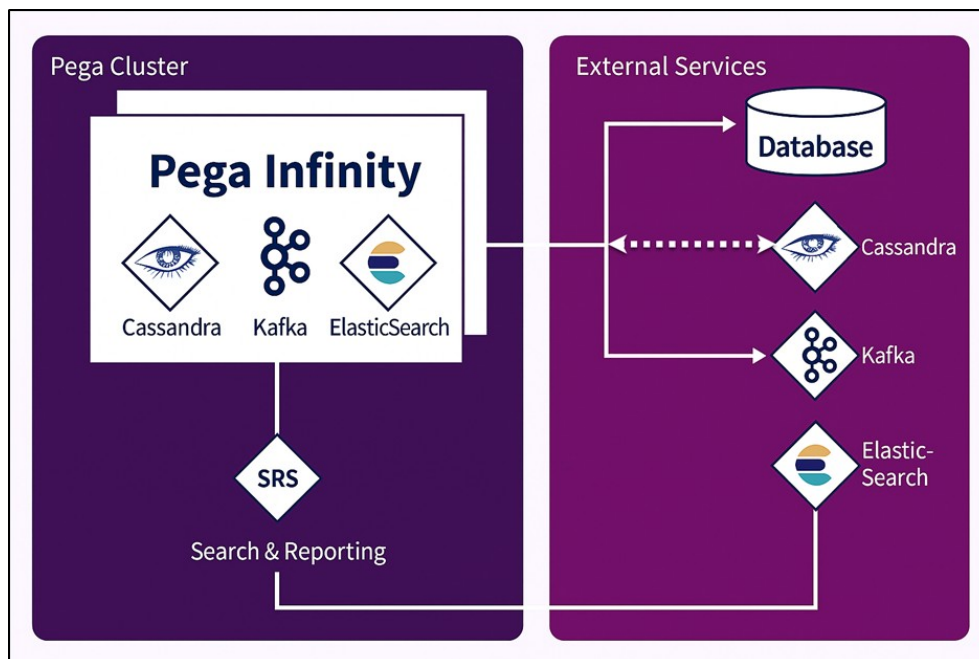


Figure 4. Externalized services in a Pega deployment

Source: Pega Academy, "Deployment architecture with external services" [R29]. © Pegasystems Inc. Reproduced with attribution.

4.2 Target architecture on Azure

Figure 5 shows one environment. The Pega web and batch tiers, the installer job and SRS run in a private AKS cluster. Confluent Cloud, the OpenSearch service, the upgraded database, Key Vault and the container registry are reached through private endpoints. Users reach Pega through Application Gateway with Web Application Firewall. Okta is a public SaaS service, reached through the hub firewall on an allow-list.

Table 10. Target architecture components

Component	Role	Design choice	Source
Web tier	User and API traffic (nodeType WebUser)	HPA on CPU; pod disruption budget	[R23]
Batch tier	Queue processors, job schedulers, search indexing, data flows	No ingress; held at zero replicas for the first start of a clone (Section 10.4)	[R21, R23]
Installer job	Upgrades the cloned database	Runs with <code>global.actions.execute: upgrade</code> (Section 9.4)	[R23]
SRS	Search and Reporting Service between Pega and OpenSearch	One deployment per environment; network policy; OAuth enabled	[R14, R24]
External Secrets Operator	Copies Key Vault secrets into Kubernetes secrets	Workload identity to Key Vault	[R23, R48]
Confluent Cloud	Kafka for the Pega stream service	Private Link; replication factor 3; prefixed ACLs	[R11, R31, R32]
Managed OpenSearch	Index storage for SRS	Version on the SRS matrix; private endpoint	[R14]
Okta	Issues tokens Pega presents to SRS	Custom authorization server; one client per environment	[R42, R43]
Pega database	Upgraded clone of the 8.8 database	Same region as AKS; private endpoint; TLS	[R20]

Figure 6 shows the five flows that matter for Kafka and search. Flow 1 is the user request. Flow 2 is queue processing: producers on any tier write to Kafka, and the batch tier consumes. Flow 3 is search: the batch tier

indexes changed records through SRS, and the web tier sends queries through SRS. Flow 4 is the token path to Okta. Flow 5 is telemetry.

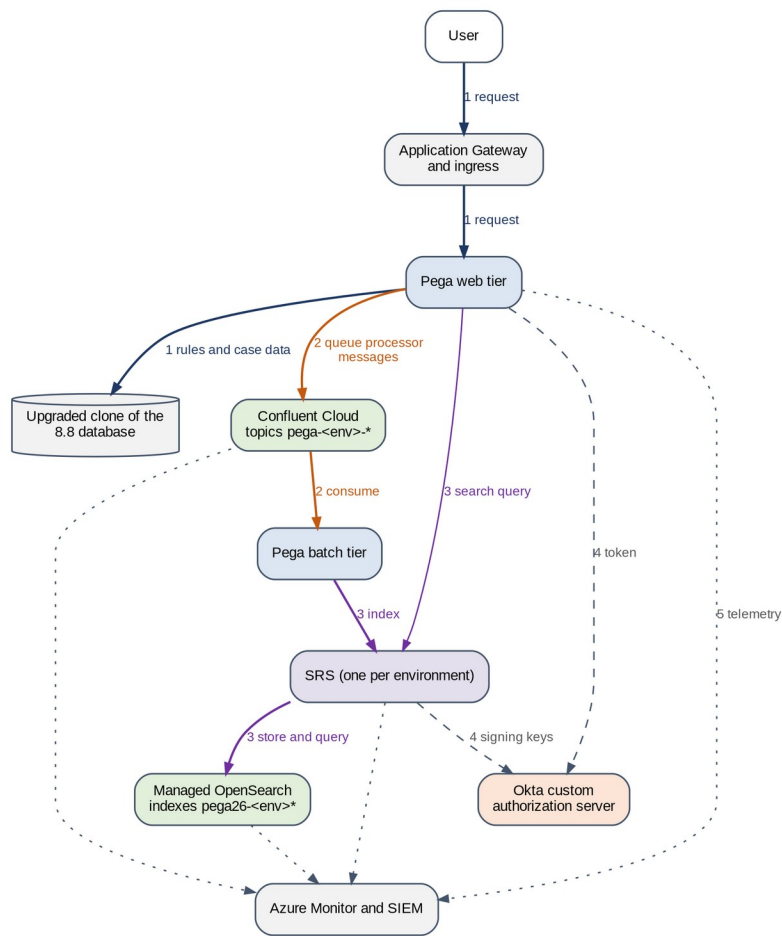


Figure 6. Logical flows between Pega, Kafka, SRS, OpenSearch and Okta

Prepared for this implementation. Not a Pega-published diagram.

4.4 Network and DNS

Every private endpoint needs a DNS record that resolves to its private IP from inside AKS [R47]. Confluent Private Link needs the wildcard and zonal records that Confluent lists for the cluster's network [R32]. A missing zonal record is the most common reason a Kafka client reaches the bootstrap server and then fails on broker connections. Figure 7 shows the name resolution and traffic paths, and Table 11 lists the flows the firewall and network policies must allow.

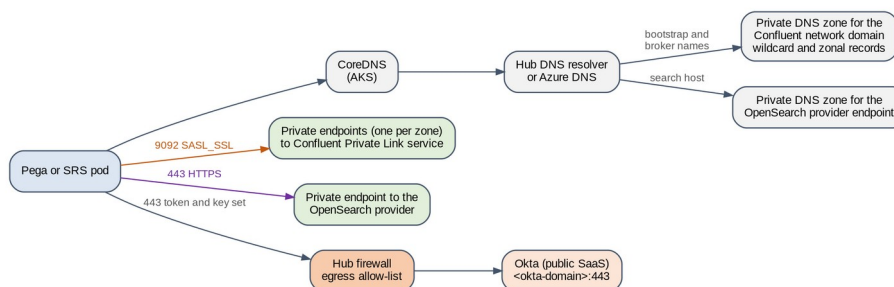


Figure 7. Name resolution and traffic paths for Kafka, OpenSearch and Okta

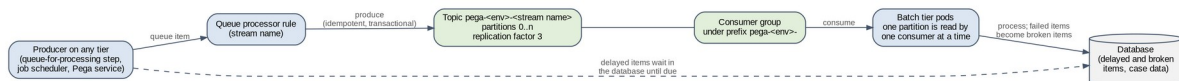
Prepared for this implementation. Not a Pega-published diagram.

Table 11. Network flows to allow and to deny

From	To	Port	Protocol	Purpose
Pega pods (web, batch)	Confluent private endpoints	9092	SASL_SSL	Stream service [R32]
Pega pods	SRS service in srs-<env>	SRS service port	HTTPS with bearer token	Indexing and search
SRS pods	OpenSearch private endpoint	443 or the provider port	HTTPS	Index storage and queries
Pega pods and installer job	Database private endpoint	Engine port	TLS	Rules and data
Pega pods	Okta <okta-domain>	443	HTTPS through the firewall	Token request
SRS pods	Okta <okta-domain>	443	HTTPS through the firewall	Key set download
External Secrets Operator	Key Vault private endpoint	443	HTTPS	Secret synchronization
Any pod in a non-production namespace	Production Kafka, production OpenSearch, production integrations	Any	Any	Denied by firewall and network policy (Section 5.3)

4.5 Message path inside Pega

Each queue processor writes to a Kafka topic named from the stream name pattern, for example `pega-sit-<stream name>`. The batch tier reads each topic through a consumer group under the same prefix, and one partition is read by one consumer at a time. Items that are not yet due wait in the database as delayed items. Items that fail after their retries become broken items, also held in the database. This split matters for the cloned database: the topics are empty in 26.1.1, but the database-held items are not (Section 10.3). Figure 8 shows the path.

**Figure 8. Message path from producer to batch tier**

Prepared for this implementation. Not a Pega-published diagram.

Note. Pega '26 moves queue processing to a partitioned, multi-threaded model [R3]. The number of partitions per topic therefore limits how many batch threads can work on one queue processor at once. Measure partitions after the first start (Section 6.7) rather than estimating them.

5 Shared and dedicated service model

Pega allows one Kafka cluster and one search service to serve more than one Pega deployment. Kafka isolation comes from the stream name prefix and prefixed ACLs [R11]. Search isolation comes from the customerDeploymentId, which SRS uses as the index prefix [R14, R23]. This section uses those features to cut the number of service sets from six to three, and states the controls that keep each environment's data apart.

5.1 Grouping options

Four groupings were assessed. Table 12 compares them. Option B is recommended.

Table 12. Non-production grouping options

Option	Service sets	Strengths	Weaknesses
A. Dedicated per environment	6 Kafka clusters, 6 OpenSearch services	No shared blast radius; simplest isolation	Highest cost; six sets to patch, monitor and rotate
B. Two shared groups plus PROD (recommended)	NP1: DEV, SIT, UAT. NP2: PERF, PREPROD. PROD dedicated. 3 sets in total	Load tests and rehearsals run on production-like services without touching functional testing; PREPROD matches PROD	PERF and PREPROD runs must be scheduled so they do not overlap
C. One shared non-production set plus PROD	2 sets	Lowest cost	A PERF load test or PREPROD rehearsal affects DEV, SIT and UAT; non-production services cannot match PROD type without paying PROD price for all
D. PREPROD alone, others shared	DEV, SIT, UAT, PERF shared; PREPROD; PROD. 3 sets	PREPROD fully separate	PERF load tests affect functional testing; same number of sets as B with less benefit

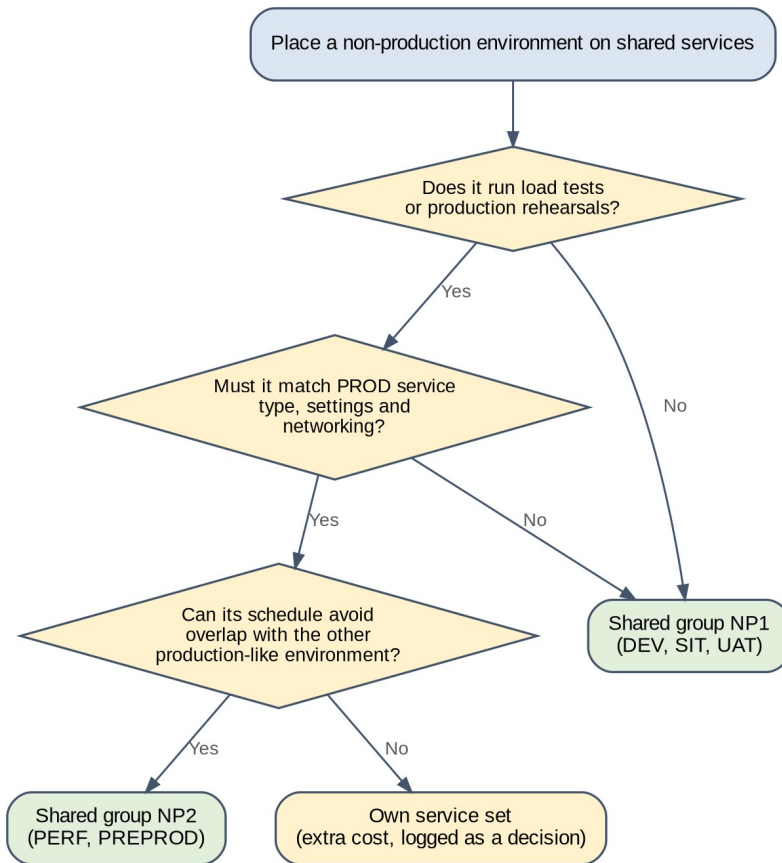


Figure 9. Decision: placing a non-production environment on shared services

Prepared for this implementation. Not a Pega-published diagram.

5.2 Recommended model

Figure 10 shows the two non-production groups and Figure 11 shows production. Each environment has its own namespace, topic prefix, service account, customerDeploymentId, SRS deployment, OpenSearch user, Okta client and Key Vault. Only the Kafka cluster and the OpenSearch service are shared within a group.

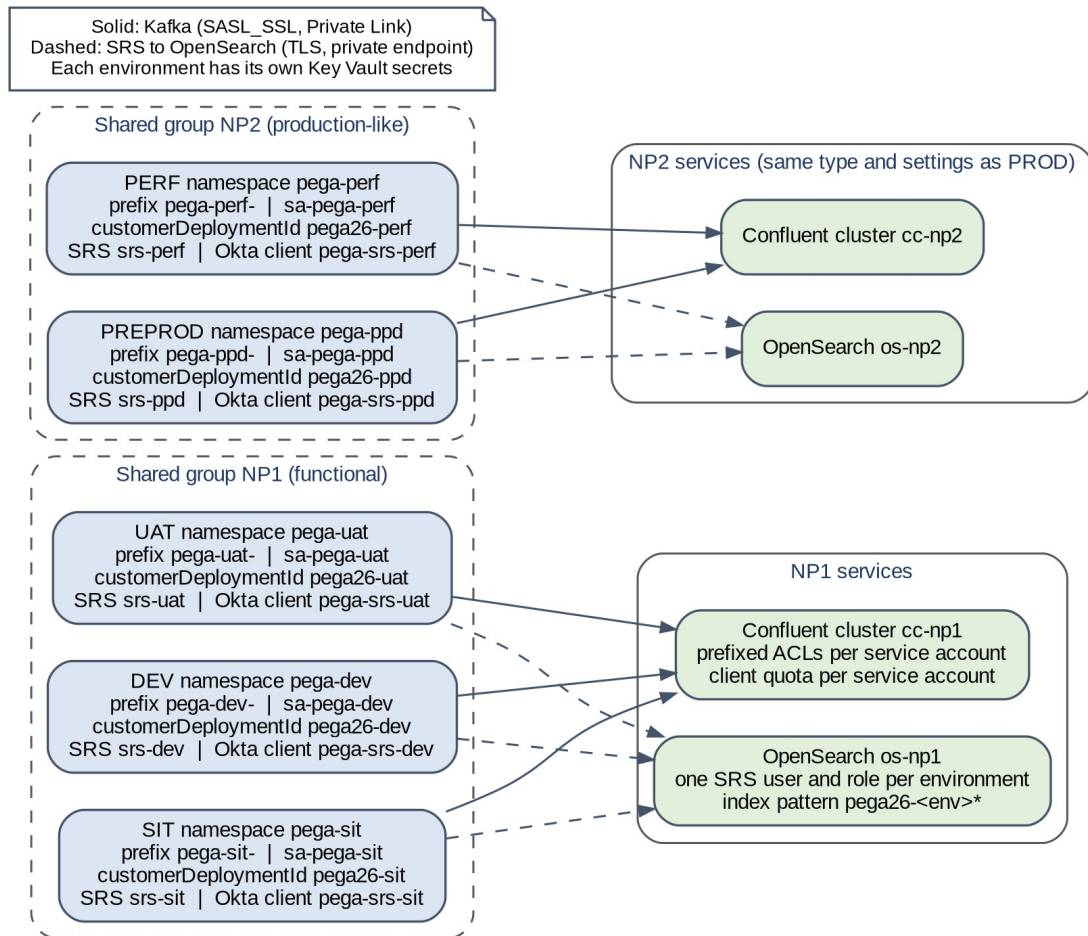


Figure 10. Shared non-production topology and per-environment boundaries

Prepared for this implementation. Not a Pega-published diagram.

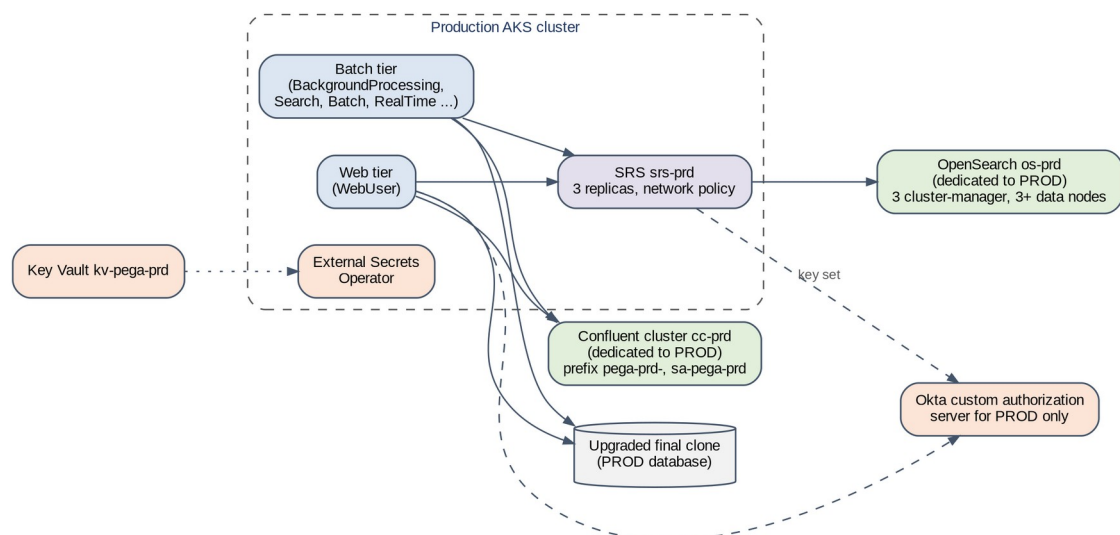


Figure 11. Production topology with dedicated services

Prepared for this implementation. Not a Pega-published diagram.

5.3 Isolation controls

Isolation is enforced at every layer, so a single mistake does not expose another environment. Table 13 lists the control at each layer and the test that proves it.

Table 13. Isolation controls by layer

Layer	Control	What it prevents	Proof (Section 14.3)
Kafka topics	Prefix <code>pega-</code> per environment through <code>stream.streamNamePattern</code> ; TOPIC ACL ALL PREFIXED on that prefix only [R11, R23]	Reading, writing or deleting another environment's topics	IT-01, IT-02
Kafka consumer groups	GROUP ACL ALL PREFIXED on the same prefix [R11]	Joining or resetting another environment's consumer groups	IT-03
Kafka credentials	One service account per environment; API keys in that environment's Key Vault only [R36]	Credential reuse across environments	IT-04
Kafka capacity	Client quota per service account on Enterprise or Dedicated clusters [R36]	One environment's load starving the others	PT-03
Search indexes	customerDeploymentId <code>pega26-</code> , used as the index prefix [R23]	Index name collisions	IT-05
Search credentials	One SRS deployment and one OpenSearch user per environment, with a role limited to index pattern <code>pega26-*</code> if SRS accepts it (Section 7.5) [R40, R41]	One environment's SRS reading or deleting another's indexes	IT-06
Search tokens	Okta client per environment; SRS checks that <code>guid</code> equals its own customerDeploymentId [R23]	A token for one environment being accepted by another environment's SRS	IT-07
Secrets	Key Vault per environment; External Secrets Operator identity per namespace [R48]	Pods reading another environment's secrets	IT-08
Network	Firewall allow-list per environment; Kubernetes network policies deny by default	A cloned environment reaching production Kafka, production OpenSearch or production integrations	IT-09, IT-10
Cloned data	Application Kafka data sets and other endpoints reported or disabled before the batch tier starts (Section 10.4)	Test environments consuming production topics or calling production systems	IT-10, FS-27

Caution. Wildcard ACLs are forbidden on a shared cluster. Pega's ACL list includes `TRANSACTIONAL_ID *` [R11], which every environment needs. That ACL does not give access to topics or groups, but it means transactional IDs are not separated by environment. Record this as an accepted residual risk (RK-07).

Never grant TOPIC or GROUP with a literal `*` or with the bare prefix `pega-`, which would match every environment.

5.4 Noisy neighbours and quotas

Confluent supports client quotas on Enterprise, Freight and Dedicated clusters. Quotas apply to a service account or an identity pool, not to an API key, and the Metrics API labels throughput by principal [R36]. One service account per environment therefore gives both a quota boundary and per-environment metrics.

- Set a produce and a consume quota per service account in NP1 and NP2, sized from the measured peak of that environment plus headroom (OD-09).
- Leave PROD without quotas unless another workload is added to its cluster.
- In NP2, PERF load tests and PREPROD rehearsals are booked in a shared calendar. A run that overlaps the other environment's run needs both test leads' approval.

- If a load test in one environment affects another, stop the test, check the Metrics API by principal, lower that service account's quota and rerun (FS-22).

OpenSearch has no equivalent quota per user. In the shared search services, limit the effect of one environment on the others by running full index builds outside the other environments' test hours, and by sizing the data nodes for the largest concurrent build (Section 7.4).

5.5 Cost comparison model

Prices change and depend on the customer's agreements, so this document gives no figures. Table 14 lists the inputs the customer supplies, and Table 15 shows how each option is costed from them.

Table 14. Cost model inputs supplied by the customer

Input	Where it comes from	Notes
Confluent cluster base price per type	Confluent price list or contract	Enterprise and Dedicated differ in pricing unit (eCKU or CKU) [R31]
Capacity units per cluster	Measured partitions and throughput (Section 6.7)	Shared clusters add the needs of each environment
Private Link and data transfer	Confluent and Azure price lists	One set of private endpoints per cluster
OpenSearch nodes per service	Sizing in Section 7.4	Cluster-manager and data nodes
OpenSearch storage and snapshots	Provider price list	Measured searchable data volume
Operations effort per service set	Customer operating model	Patching, monitoring, on-call, key rotation
Support plans	Confluent and provider contracts	Production support level for PROD and NP2

Table 15. Cost model by option (multiply each line by the customer's unit prices)

Cost line	Option A	Option B (recommended)	Option C
Kafka clusters	6 x (base + units)	NP1 + NP2 + PROD: 3 x (base + units)	2 x (base + units)
Private Link sets	6	3	2
OpenSearch services	6 x (nodes + storage)	3 x (nodes + storage)	2 x (nodes + storage)
SRS deployments	6 (inside AKS)	6 (inside AKS)	6 (inside AKS)
Okta authorization servers	Up to 6	2 (PROD and shared non-production)	2
Operations effort	6 service sets	3 service sets	2 service sets
Risk cost	Lowest	Shared blast radius within NP1 and NP2; scheduling in NP2	Load tests affect all non-production; no production-like test services

Option B removes three Kafka clusters and three OpenSearch services compared with Option A. The saving is the base price of those services plus their private endpoints and operations effort, less any extra capacity units the shared clusters need. Option C saves one more set but removes the production-like test services, which raises the chance of a production problem that rehearsals did not show.

5.6 Production

Production always has its own Confluent cluster, its own OpenSearch service, its own Okta authorization server and its own Key Vault. Production credentials, endpoints and DNS zones are not linked to any non-production network.

6 Confluent Cloud design and configuration

6.1 What Pega uses Kafka for

Pega uses Kafka in two ways. The stream service carries every queue processor and job scheduler, so Pega is not fully functional without it [R11]. This is what the Helm `stream` section configures. Separately, a Kafka data set is an application feature that reads from or writes to the customer's own Kafka topics. Pega requires that Kafka data sets use a different Kafka service from the stream service [R11]. Section 11.3 covers data sets.

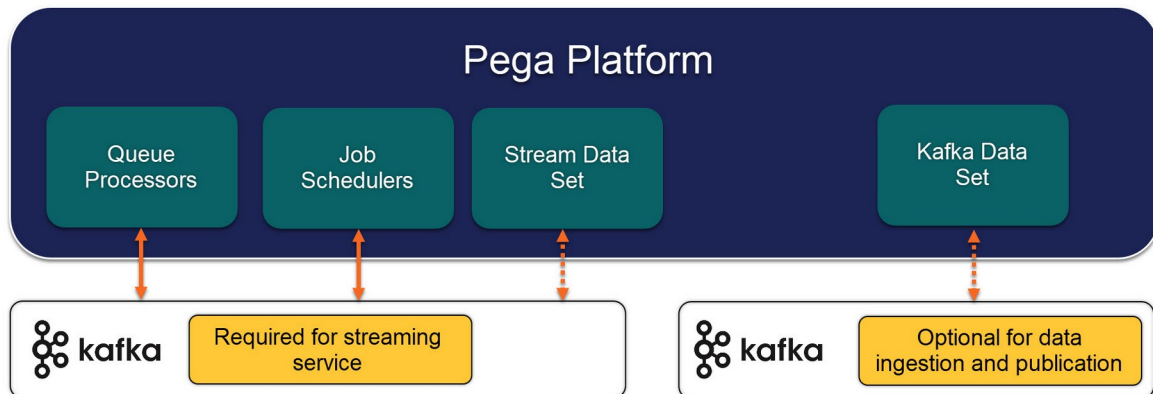


Figure 12. The two Kafka use cases in Pega Platform

Source: Pega Documentation, "External Kafka in your deployment" [R11]. © Pegasystems Inc. Reproduced with attribution.

Important. From '25, Pega uses its streaming function and the external Kafka service for the cluster messaging that Hazelcast used to carry, and Pega asks clients to review the Kafka configuration before the change [R7]. A Kafka outage on 26.1.1 therefore affects more than queue processing. Treat the Kafka service as a tier-1 dependency and test its loss explicitly (FS-01, FS-03).

6.2 Pega requirements mapped to Confluent Cloud

Confluent Cloud is a managed service, so some broker settings Pega lists cannot be changed by the customer. Table 16 shows each requirement, what Confluent Cloud allows, and what the runbook does.

Table 16. Pega Kafka requirements mapped to Confluent Cloud

Pega requirement	Confluent Cloud position	Action
Kafka client 4.0.0 in '26, compatible with 3.9.2 [R3]	Fully managed Kafka	Produce, consume and transactional checks from a Pega pod in every environment (Section 14.4)
Confluent 5.4.x or later [R11]	Current release line	None beyond the checks above
message.max.bytes 5000000 [R11]	Topic max.message.bytes default 2,097,164; editable up to 20,971,520 on Enterprise and Dedicated [R33]	After each start, set max.message.bytes to 5,000,000 on every Pega topic (command K-6)
replica.fetch.max.bytes and replica.fetch.response.max.bytes 5000100 [R11]	Not in the list of editable cluster settings [R34]	Ask Confluent support to confirm in writing (RC-03)
unclean.leader.election.enable false [R11]	Not customer-editable [R34]	Ask Confluent support to confirm in writing (RC-03)
auto.create.topics.enable false [R11]	Editable on Dedicated; default false [R34]	Keep false. Pega creates topics through its admin client, which needs CREATE in the topic ACL
Generic Kafka security only [R11]	SASL_SSL with API keys (PLAIN) is generic Kafka	Use SASL_SSL with PLAIN (Section 6.5)

Pega requirement	Confluent Cloud position	Action
ACLs for TOPIC, GROUP, TRANSACTIONAL_ID, CLUSTER IDEMPOTENT_WRITE [R11]	Supported on service accounts	Create before the first start (Section 6.5)
Production sizing supports about 1000 partitions [R11]	Partition limits per capacity unit are published [R31]	Measure after the first start (Section 6.7)

6.3 Cluster type per group

Confluent offers Basic, Standard, Enterprise, Dedicated and Freight clusters [R31]. Basic and Standard do not offer private networking, so they cannot hold cloned or masked production data. Freight does not support idempotent producers or transactions [R31], and Pega's ACL profile needs both [R11]. That leaves Enterprise and Dedicated.

Table 17. Confluent Cloud cluster limits relevant to Pega (source [R31])

Dimension (per unit)	Basic eCKU	Standard eCKU	Enterprise eCKU	Dedicated CKU
Ingress MBps	5	25	60	60
Egress MBps	15	75	180	180
Partitions (before replication)	30	250	3,000	4,500
Private networking	No	No	Yes (Private Link)	Yes (Private Link or VNet peering)
Client quotas [R36]	No	No	Yes	Yes

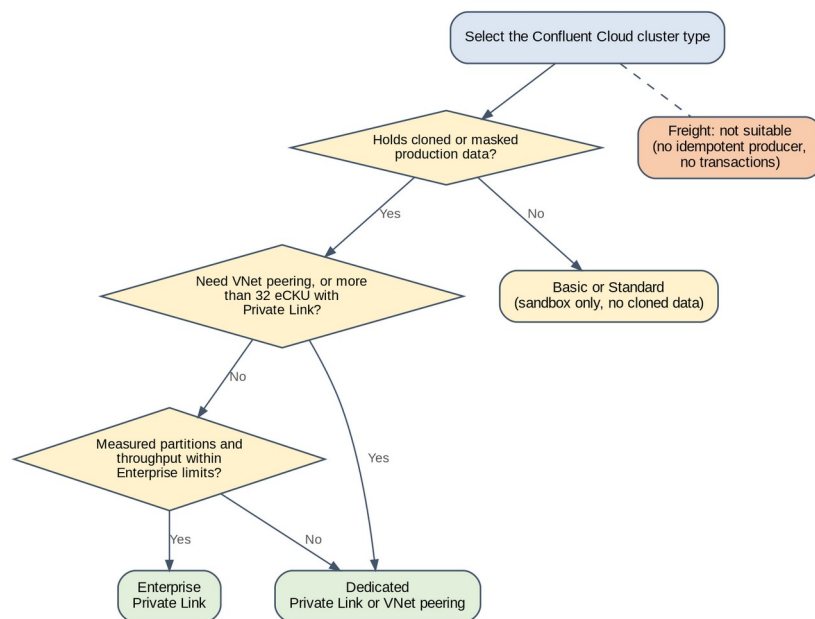


Figure 13. Decision: Confluent Cloud cluster type

Prepared for this implementation. Not a Pega-published diagram.

Table 18. Cluster type by group

Group	Recommended type	Reason
NP1 (DEV, SIT, UAT)	Enterprise with Private Link	Holds masked production data; quotas available; elastic capacity suits uneven test load

Group	Recommended type	Reason
NP2 (PERF, PREPROD)	Same type as PROD	Load tests and rehearsals must show production behaviour
PROD	Enterprise with Private Link, or Dedicated if measured partitions or throughput exceed Enterprise limits, or VNet peering is required	Decided after the first PERF measurement (OD-02)

Caution. Enterprise clusters that use Private Link on Azure are limited to 32 eCKU [R31]. Add up the measured partitions of every environment in the group before confirming Enterprise.

6.4 Private networking

Azure Private Link gives one-way private access from the customer VNet to Confluent Cloud. Confluent's Azure steps are: create the Confluent network, add a Private Link access for the customer subscription, create one private endpoint per zone, create the DNS records, and test [R32]. Each group's cluster needs its own set. The NP1 and NP2 private DNS zones are linked only to non-production networks, and the PROD zone only to the production network.

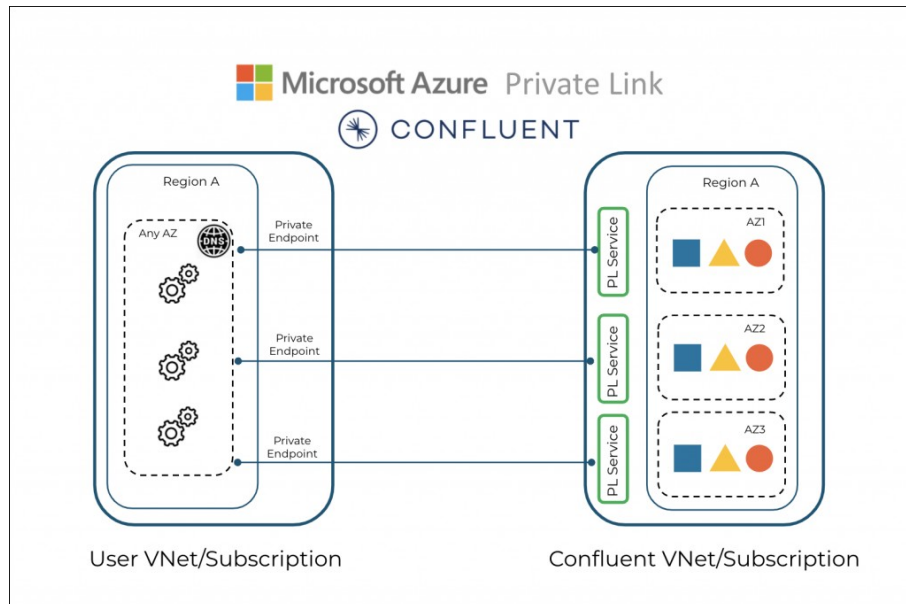


Figure 14. Azure Private Link between a customer VNet and Confluent Cloud

Source: Confluent Documentation, "Use Azure Private Link for Dedicated clusters on Confluent Cloud" [R32]. © Confluent, Inc. Reproduced with attribution.

Name	Type	TTL	Value
*	A	60	172.16.128.4 172.16.128.5 172.16.128.6
@	SOA	3600	Email: azureprivatedns-host.microsoft.com Host: azureprivatedns.net Refresh: 3600 Retry: 300 Expire: 2419200 Minimum TTL: 10 Serial number: 1
*.az1	A	3600	172.16.128.4
*.az2	A	3600	172.16.128.5
*.az3	A	60	172.16.128.6

Figure 15. DNS records that Confluent lists for Azure Private Link

Source: Confluent Documentation [R32]. © Confluent, Inc. Reproduced with attribution.

Table 19. Private Link set-up steps per cluster

Step	Action	Check
N-1	Create the Confluent network in the AKS region and register the customer subscription for Private Link.	Network ID and subscription recorded in Appendix A
N-2	Create one private endpoint per zone in the private endpoint subnet, using the service aliases Confluent shows.	Connection state Approved for each
N-3	Create the private DNS zone for the Confluent network domain with the wildcard and zonal records Confluent shows; link it to the right VNet or resolver.	Zone export kept as evidence
N-4	From a pod in <code>pega-<code><code></code></code> , resolve the bootstrap name and a broker name.	Both return private IPs (K-1)
N-5	From the same pod, open TLS to the bootstrap name on port 9092.	Certificate chain shown (K-2)
N-6	Run a metadata request.	Every broker name resolves privately (K-3)

Note. Confluent states that the Confluent CLI might need internet access to the Confluent control plane even when the cluster uses Private Link [R32]. Allow that only from the administration jump host, never from Pega pods.

6.5 Service accounts, authentication and ACLs

Pega lists several SASL options [R11], but the Helm `stream.saslMechanism` key accepts PLAIN, SCRAM-SHA-256 and SCRAM-SHA-512 [R23]. Confluent Cloud API keys use SASL/PLAIN. Figure 16 shows the choice.

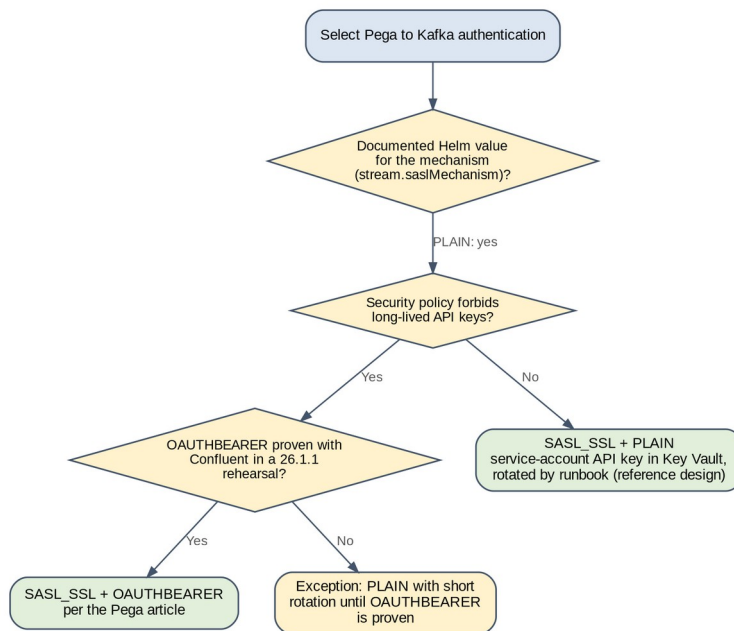


Figure 16. Decision: Pega to Kafka authentication

Prepared for this implementation. Not a Pega-published diagram.

Create one service account per environment and grant it only the ACLs Pega lists, scoped to that environment's prefix. Table 20 shows the ACLs for one environment; repeat them for each environment on its own cluster.

Table 20. Kafka ACLs for one environment's service account (source [R11])

Resource type	Resource name	Operation	Pattern	Why
TOPIC	pega -<code>-	ALL	PREFIXED	Create, write, read and delete Pega topics
GROUP	pega -<code>-	ALL	PREFIXED	Consumer groups for queue processors and data flows
TRANSACTIONAL_ID	*	READ, WRITE	LITERAL	Transactional producers
CLUSTER	the cluster	IDEMPOTENT_WRITE	LITERAL	Idempotent producers

Creating and listing one environment's service account and ACLs

```
# Confluent CLI. Check flags with: confluent kafka acl create --help
confluent iam service-account create sa-pega-<code> --description "Pega 26.1.1 <ENV>"
confluent kafka acl create --allow --service-account <sa-id> --operations all \
  --topic "pega-<code>-" --prefix --cluster <lkc-id>
confluent kafka acl create --allow --service-account <sa-id> --operations all \
  --consumer-group "pega-<code>-" --prefix --cluster <lkc-id>
confluent kafka acl create --allow --service-account <sa-id> --operations read,write \
  --transactional-id "*" --cluster <lkc-id>
confluent kafka acl create --allow --service-account <sa-id> \
  --operations idempotent-write --cluster-scope --cluster <lkc-id>
confluent api-key create --resource <lkc-id> --service-account <sa-id>
confluent kafka acl list --service-account <sa-id> --cluster <lkc-id>
```

To remove an environment's access, delete its ACLs with `confluent kafka acl delete` using the same flags, delete its API keys, then delete the service account (Section 17.5).

6.6 Topic design

Table 21. Topic design settings

Setting	Value	Reason
streamNamePattern	pega- <code><stream.name></code>	Separates environments on a shared cluster. Chart default is <code>pega-<code><stream.name></code></code> [R23].
replicationFactor	3	Pega's recommended value; cannot exceed the number of brokers [R23].
Topic creation	By Pega through its admin client (CREATE in the topic ACL)	Pega calls broker auto-creation a legacy method [R11]. If policy forbids CREATE, pre-create topics from the rehearsal list (OD-11).
max.message.bytes	5,000,000 on every Pega topic	Matches Pega's maximum message size [R11]; Confluent default is lower [R33].
SystemPulse_SystemPulseTopicName	6 partitions	'26 prerequisite; automatic if dynamic topic creation is allowed [R3].
Cluster messaging topics after Hazelcast removal	Covered by the prefixed ACL if Pega applies the pattern to them	Pega names five topics in the Hazelcast removal prerequisites [R6] but does not say whether the pattern applies. List the topics after the first DEV start and add literal ACLs for any outside the prefix (FS-04).

When an environment is refreshed from a new clone, keep its prefix and delete its old topics and consumer groups before the first start (Section 17.4). Keeping the prefix keeps ACLs, quotas and dashboards unchanged. A new prefix is only needed if old topics cannot be deleted, for example during an investigation; in that case create a new prefix such as `pega-sit2-` with its own ACLs.

6.7 Partition budget

Partitions drive both Confluent cost and limits. Measure them instead of estimating:

1. After the first full start of each environment, count partitions under its prefix (command K-5).
2. Record the count in Table 22 and add the headroom agreed for new queue processors (30 % is a sound starting point).
3. Add up the environments in each group and compare with the cluster limit for the chosen type (Table 17).
4. Repeat after each application release that adds queue processors.

Table 22. Partition budget worksheet

Environment	Group	Measured partitions	With 30 % headroom	Group total	Cluster limit	Within limit
DEV	NP1	<measured>	<x 1.3>	<sum for group>	<from Table>	<yes or no>
SIT	NP1	<measured>	<x 1.3>	<sum for group>	<from Table>	<yes or no>
UAT	NP1	<measured>	<x 1.3>	<sum for group>	<from Table>	<yes or no>
PERF	NP2	<measured>	<x 1.3>	<sum for group>	<from Table>	<yes or no>
PREPROD	NP2	<measured>	<x 1.3>	<sum for group>	<from Table>	<yes or no>
PROD	PRD	<measured>	<x 1.3>	<sum for group>	<from Table>	<yes or no>

6.8 Helm stream configuration

The `stream` section connects every tier to Kafka [R12]. Passwords and the JAAS string are left empty in the values file and supplied by an external secret with the keys `STREAM_TRUSTSTORE_PASSWORD`, `STREAM_KEYSTORE_PASSWORD` and `STREAM_JAAS_CONFIG` [R23].

values-`<code>`.yaml: stream section

```
stream:
  enabled: true
  bootstrapServer: "<bootstrap-host>:9092"
  securityProtocol: SASL_SSL
  saslMechanism: PLAIN
  trustStore: ""
  keyStore: ""
  jaasConfig: ""
  streamNamePattern: "pega-<code>-{stream.name}"
  replicationFactor: "3"
  external_secret_name: "pega-stream-secret"
```

Value of `STREAM_JAAS_CONFIG` in Key Vault (stored as one line)

```
org.apache.kafka.common.security.plain.PlainLoginModule required
  username="<api-key>" password="<api-secret>";
```

6.9 prpcUtils on 26.1.1

From '25, prpcUtils needs its own connection to Kafka for clustering. Pega documents a `-DNodeSettings` argument in `custom.jvm.args` of prpcUtils.properties for '25 and '26 [R8]. Pipelines that run prpcUtils against an upgraded environment must pass that environment's bootstrap server, prefix and JAAS value, generated at run time from Key Vault and deleted after the run.

7 Managed OpenSearch and SRS design and configuration

7.1 How search works in 26.1.1

SRS is a Pega backing service between Pega Platform and the search cluster. Pega sends indexing and query requests to SRS, and SRS manages the indexes in OpenSearch [\[R14\]](#). SRS isolates each Pega environment's data with a customerDeploymentId, which becomes the index prefix [\[R14, R23\]](#). The search service can serve several Pega deployments, but Pega advises against sharing it with non-Pega software [\[R14\]](#).

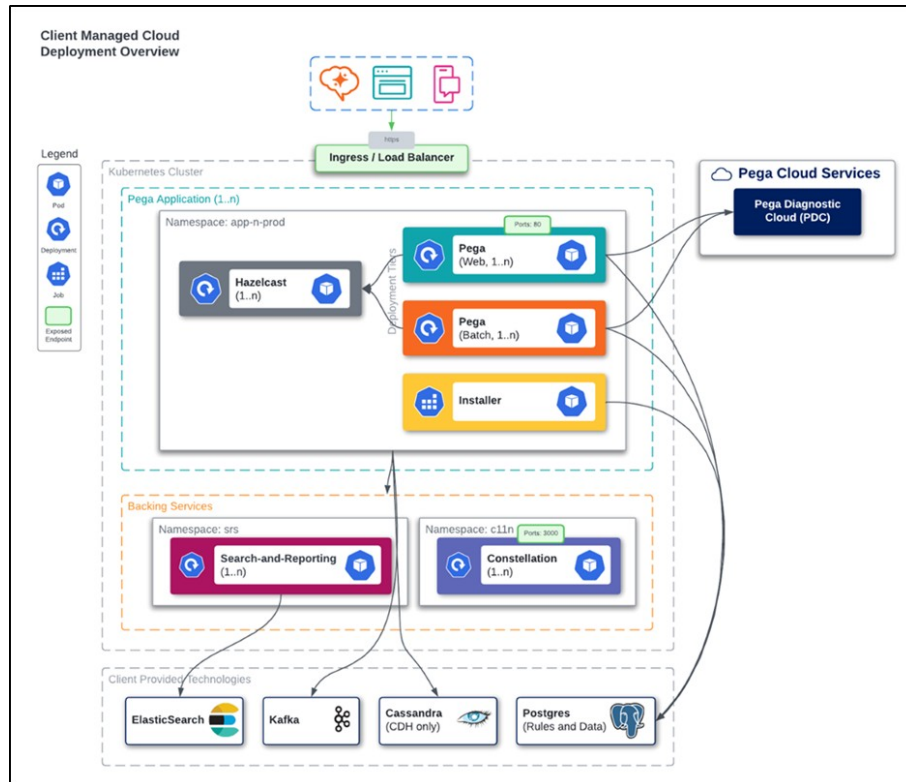


Figure 17. Client-managed cloud deployment with SRS as a backing service

Source: Pega Academy, "Search and Reporting Service" [\[R28\]](#). © Pegasystems Inc. Reproduced with attribution. The Hazelcast pod in this illustration does not apply to '25 and later.

7.2 SRS and OpenSearch versions

Table 23. SRS compatibility with OpenSearch for Pega 8.6 and later, SRS 1.44.3 or later (source [\[R14\]](#))

SRS image	Authentication	Certified OpenSearch versions	Pega best practice
search-n-reporting-service-os	Enabled	AWS OpenSearch service with Elasticsearch 7.10; OpenSearch 1.3, 2.15, 2.19	OpenSearch 2.15

- Choose OpenSearch 2.15 or 2.19. Version 1.3 is on the matrix but is an older line, and the provider may not offer it.
- Use the newest `search-n-reporting-service-os` tag that the SRS chart README lists for chart 4.13.0 [\[R24\]](#). The README lists a newer SRS than the Pega page (RC-01).
- Use only official OpenSearch images if the customer runs OpenSearch itself; custom images are not supported [\[R14\]](#).
- The search cluster that the backingservices chart can provision is for development and test only, and cannot be OpenSearch [\[R14, R24\]](#).

7.3 Provider selection

Azure has no first-party managed OpenSearch service, so the choice is between third-party providers that run OpenSearch in Azure regions and a self-managed cluster on AKS. Pega accepts a cloud subscription, a licensed product or open source [R14]. This document names no provider: a provider becomes a candidate only when its own documentation shows an Azure region and OpenSearch 2.15 or 2.19, and that check is part of the selection.

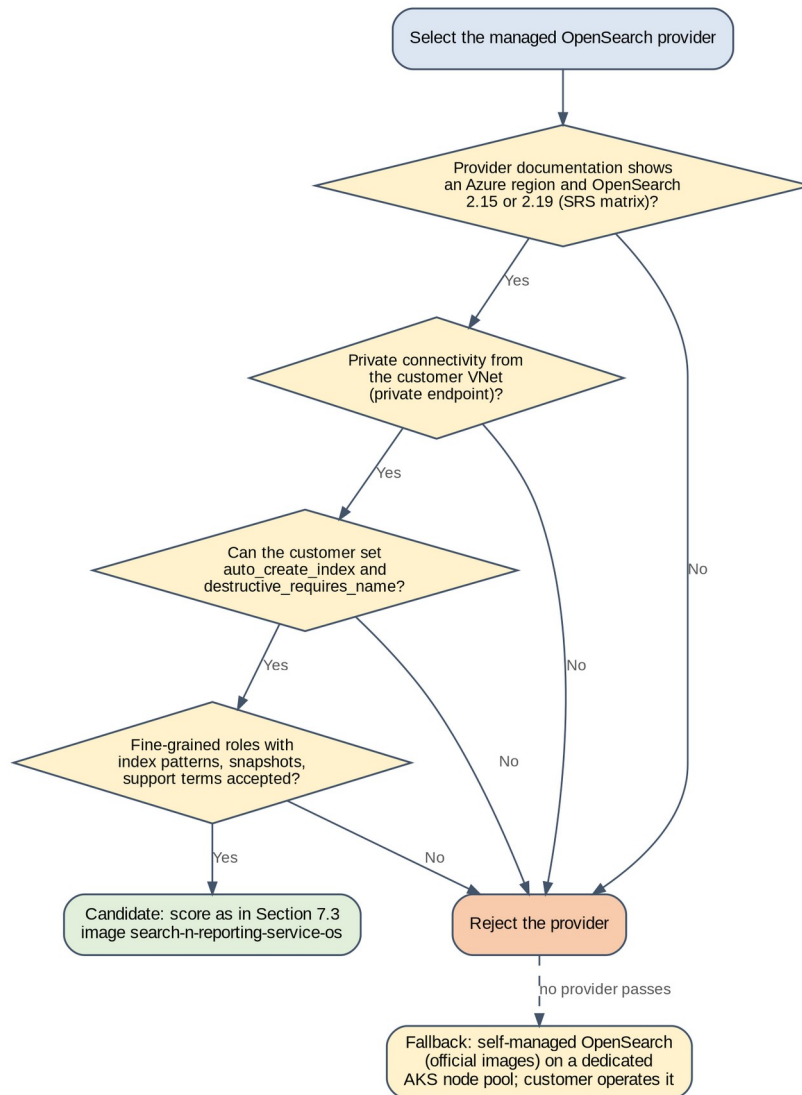


Figure 18. Decision: managed OpenSearch provider

Prepared for this implementation. Not a Pega-published diagram.

Score each candidate in Table 24. A provider that fails a mandatory criterion is rejected whatever its score.

Table 24. OpenSearch provider selection worksheet

Criterion	Type	Weight	Provider 1	Provider 2	Self-managed on AKS
Azure region same as AKS, shown in provider documentation	Mandatory				Yes
OpenSearch 2.15 or 2.19 available and pinnable	Mandatory				Yes
Private endpoint into the customer VNet	Mandatory				Yes

Criterion	Type	Weight	Provider 1	Provider 2	Self-managed on AKS
Customer can set <code>action.auto_create_index</code> and <code>action.destructive_requires_name</code> [R14]	Mandatory				Yes
Fine-grained access control with index patterns [R40, R41]	Mandatory				Yes
Snapshots with customer-chosen retention	Scored	15			
Support terms and response times for PROD	Scored	20			
Metrics export to Azure Monitor	Scored	15			
Version upgrade process and notice	Scored	15			
Operations effort for the customer	Scored	20			
Data residency and certifications	Scored	15			

Only four values depend on the provider: the endpoint host, the port, the SRS user credentials and the CA certificate. The Helm values in Appendix B keep those in angle brackets and in Key Vault, so the rest of the design does not change when OD-03 is decided.

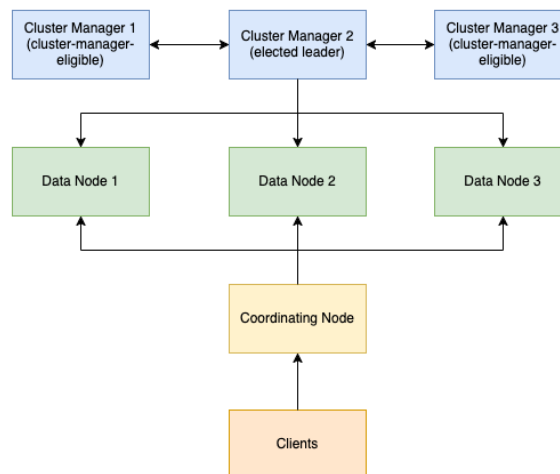


Figure 19. OpenSearch cluster roles: cluster-manager and data nodes

Source: OpenSearch Documentation, "Creating a cluster" [R37]. © OpenSearch contributors. Reproduced with attribution.

7.4 Cluster settings and sizing

SRS has its own index naming, so index auto-creation must be off. If the SRS user has the manage cluster privilege, SRS sets this itself; otherwise apply it manually [R14]. SRS also deletes indexes by pattern, which needs `destructive_requires_name` set to false [R14]. With an index-scoped SRS user (Section 7.5), apply both settings as an administrator before SRS starts.

Cluster settings required by SRS (run once per OpenSearch service)

```

PUT _cluster/settings
{
  "persistent": {
    "action": {
      "auto_create_index": "false",
      "destructive_requires_name": "false"
    }
  }
}

```

Caution. `destructive_requires_name: false` lets any user with delete rights remove indexes by wildcard. On a shared service, give delete rights only through the per-environment index-scoped roles, and keep administrator access to a break-glass group.

Pega publishes default sizing for search and SRS. Start from it and adjust after measuring, as Pega advises [R14].

Table 25. Pega default sizing for search and SRS (source [R14])

Service	Landscape	Instances	CPU each	RAM GB each	Storage GB each
Cluster-manager nodes	Production, stage	3	2	8	N/A
Data nodes	Production, stage	3	4	16	100
Data nodes	Testing, development	1	2	8	100
SRS	Production, stage	3 (autoscaled)	2	2	N/A
SRS	Testing, development	1 (autoscaled)	2	2	N/A

Table 26. Starting sizes per OpenSearch service

Service	Starting size	Basis	Adjust when
os-np1 (DEV, SIT, UAT)	3 cluster-manager, 3 data nodes at the Pega production size	Three environments' indexes plus one full build at a time; three nodes give zone spread	Disk above 60 % after all three builds
os-np2 (PERF, PREPROD)	Same as os-prd	Production-like behaviour for load tests and rehearsals	Never smaller than os-prd
os-prd	3 cluster-manager, 3 or more data nodes; storage from measured volume	Pega production sizing, scaled by the searchable data measured in the upgraded clone	Index build or query latency misses target in PERF

- **Searchable data volume.** Measure it from the first upgraded clone: run the full index build in DEV and record index sizes with `_cat/indices` (command S-4). Scale PROD storage from that, plus growth and one rebuild's working space.
- **Shards and replicas.** SRS creates and names the indexes itself [R14]. After the first build, read the shard and replica count of each index with `_cat/indices` and record it in Appendix A. Do not change index settings directly unless Pega Support advises it.
- **Shard budget.** Add up the shards of every environment on the service and compare with the provider's per-node shard limit.
- **Disk watermarks.** Record the provider's low, high and flood-stage watermark values in Appendix A, and alert before the low watermark is reached (Section 17.1). At the flood stage, indexes become read-only and indexing stops (FS-17).

7.5 Isolation in the shared OpenSearch services

Each environment has its own `customerDeploymentId` (Section 2.6). Set it explicitly in every values file, because the chart defaults it to the namespace name [R23], and treat it as immutable, as Pega asks when several environments share an SRS [R15]. A cloned environment must never reuse another environment's ID: its index build would overwrite or mix with the other environment's indexes, and its tokens would carry the other environment's `guid`.

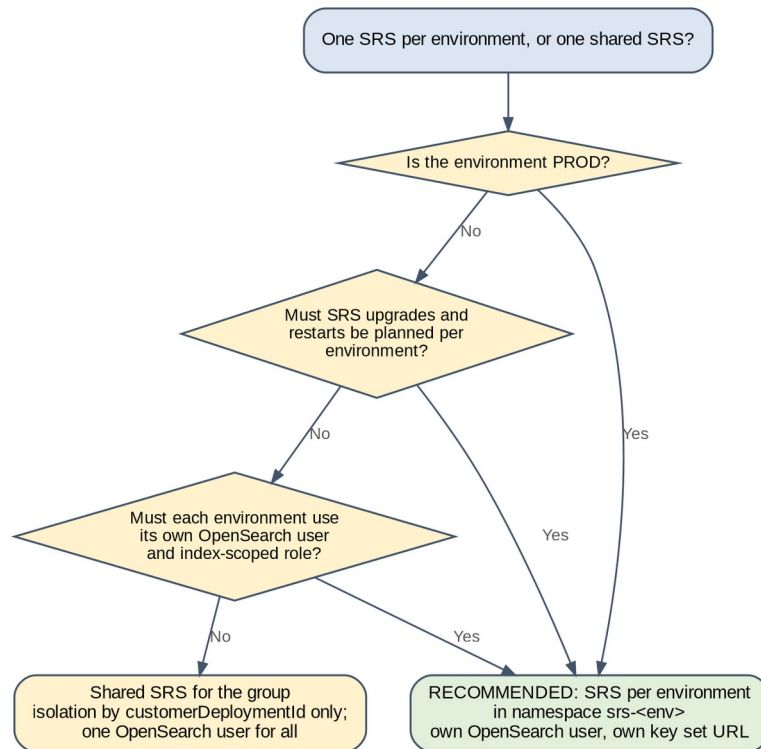


Figure 20. Decision: one SRS per environment or one shared SRS

Prepared for this implementation. Not a Pega-published diagram.

Table 27. SRS per environment compared with a shared SRS

Aspect	SRS per environment (recommended)	One SRS per group
OpenSearch credentials	One user per environment, index-scoped role	One user for the whole group
Blast radius	An SRS fault or bad upgrade affects one environment	Affects every environment in the group
Upgrades	Upgrade SRS per environment, in step with that environment's tests	One upgrade for all; all environments retest
Token key set	SRS points at that environment's Okta key set	One key set; every environment's tokens come from the same server
Cost	Small: 1 pod in DEV, SIT, UAT; 3 in PERF, PREPROD, PROD [R14]	Slightly smaller

Give each SRS its own OpenSearch user, mapped to a role limited to its index pattern [\[R40, R41\]](#). Pega says the manage cluster privilege is optional when the cluster settings are applied manually [\[R14\]](#). Pega does not say whether SRS needs any other cluster-level permission, so test the role in DEV (IT-06) and widen it only by the permissions the SRS log shows as missing.

Index-scoped role for one environment's SRS user (OpenSearch roles.yml format [\[R40, R41\]](#))

```

pega26->-srs:
  cluster_permissions:
    - cluster_composite_ops
  index_permissions:
    - index_patterns:
      - "pega26->*"
      allowed_actions:
        - "<index action group confirmed in DEV, for example indices_all>"
  
```

When an environment is refreshed or retired, delete its indexes with `DELETE /pega26->* as the environment's own SRS user, which cannot touch any other environment's indexes (Sections 17.4 and 17.5).`

7.6 Pega-to-SRS tokens with Okta

Pega obtains a token with the OAuth client credentials grant, authenticating with `private_key_jwt` or `client_secret_basic`, and asks for the scope `pega.search:full` [R16, R23]. SRS checks the token signature with the key set at `OAuthPublicKeyURL`, and checks that the `guid` claim equals the `customerDeploymentId` [R23]. Figure 21 shows the flow.

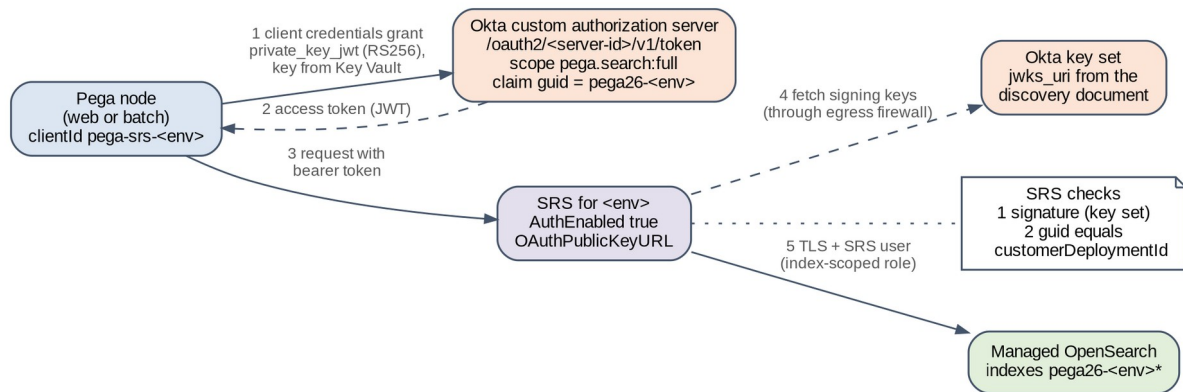


Figure 21. Token flow between Pega, Okta and SRS

Prepared for this implementation. Not a Pega-published diagram.

7.6.1 Why a custom authorization server is required

Okta's org authorization server cannot be customized: its audience, claims, policies and scopes are fixed, and its tokens are meant for Okta's own APIs [R42]. The client credentials flow has no user, so it cannot use OpenID scopes and needs a custom scope [R43]. Custom claims can only be added to a custom authorization server [R44]. Pega's scope and the `guid` claim are both custom, so Okta tokens for SRS must come from a custom authorization server. Custom authorization servers are part of Okta API Access Management [R42]. Confirm that the customer's Okta licence includes it, and how many servers it allows, before M2 (OD-05).

7.6.2 Issuing a different guid per environment

Table 28. Options for the guid claim

Option	How it works	Strengths	Weaknesses
1. One authorization server per environment	Each server has a claim <code>guid</code> with a fixed value, for example <code>"pega26-sit"</code> , included in access tokens	Simplest claim; full separation of keys and policies	Six servers; licence may limit the number
2. One shared server, claim based on the client	One claim <code>guid</code> whose value is an Okta expression on the requesting app, for example a conditional on <code>app.clientId</code> [R46]	One server for all non-production; fewer objects	Every non-production client shares a signing key set; must prove that the expression is evaluated for client credentials tokens

Okta's expression language exposes `app.id`, `app.clientId` and `app.profile` for custom claims [R46], and supports conditional expressions of the form `[Condition] ? [Value if TRUE] : [Value if FALSE]` [R46]. An example claim value for option 2 is shown below. Okta's pages do not state whether `app` attributes are available when the token is issued to a client with no user, so option 2 is accepted only after the DEV test.

Claim value expression for option 2 (written as one line in Okta)

```

app.clientId == "<dev-client-id>" ? "pega26-dev" :
app.clientId == "<sit-client-id>" ? "pega26-sit" :
app.clientId == "<uat-client-id>" ? "pega26-uat" :
app.clientId == "<perf-client-id>" ? "pega26-perf" :
app.clientId == "<ppd-client-id>" ? "pega26-ppd" : "none"

```

Decision. Recommendation (OD-05): PROD uses its own custom authorization server with a fixed `guid` claim of `pega26-prd`. The five non-production environments use one shared custom authorization server with the client-based claim (option 2), if the DEV test proves it. If the test fails, fall back to option 1 for non-production. Proof in DEV: request a token for the DEV client, decode it (command O-2), and confirm `scp` contains `pega.search:full` and `guid` equals `pega26-dev`. Repeat with the SIT client and confirm `pega26-sit`.

7.6.3 Okta set-up per environment

1. In the authorization server, create the scope `pega.search:full` and an access policy with a rule that allows the client credentials grant for the environment's client.
2. Create the claim `guid` for access tokens (fixed value for option 1, expression for option 2) [R44].
3. Create one API services application per environment, named `pega-srs-<code>`, with client authentication by public key / private key (`private_key_jwt`) [R45].
4. Generate an RSA key pair in a controlled workstation or pipeline. Register the public key in the application's key set (JWKS) [R45]. Store the private key, in PKCS8 form encoded with base64 as the chart requires, in Key Vault as `SRS_OAUTH_PRIVATE_KEY` [R23]. Delete local copies.
5. Record the client ID, the token endpoint and the key set URL in Appendix A.

Table 29. Okta values used in the Helm charts

Setting	Value	Source
Issuer	<code>https://<okta-domain>/oauth2/<auth-server-id></code>	[R42]
Token endpoint (<code>pegasearch.srsAuth.url</code>)	<code>https://<okta-domain>/oauth2/<auth-server-id>/v1/token</code>	[R43]
Key set URL (<code>srsRuntime.env.OAuthPublicKeyURL</code>)	The <code>jwtks_uri</code> value from <code>https://<okta-domain>/oauth2/<auth-server-id>/well-known/openid-configuration</code>	[R42]
Client ID (<code>pegasearch.srsAuth.clientId</code>)	Client ID of <code>pega-srs-<code></code>	[R23]
Authentication (<code>pegasearch.srsAuth.authType</code>)	<code>private_key_jwt</code>	[R23, R45]
Key algorithm (<code>pegasearch.srsAuth.privateKeyAlgorithm</code>)	RS256 (chart default)	[R23]
Scope (<code>pegasearch.srsAuth.scopes</code>)	<code>pega.search:full</code>	[R23]

7.6.4 Network path, token lifetime and rotation

- Pega pods call the Okta token endpoint, and SRS pods fetch the Okta key set. Both go to `<okta-domain>` on port 443 through the hub firewall. Add an FQDN allow rule for `<okta-domain>` from the Pega and SRS subnets only.
- Token lifetime is set in the access policy rule of the authorization server [R42]. Use the same lifetime in all environments and record it in Appendix A. Pega requests a new token when needed; measure how often in DEV by counting token requests in the Okta system log.
- Pega does not state whether SRS checks the issuer or audience. Test it in DEV with a token from another authorization server (FS-12) and record the result.
- Rotate the client key by adding a second public key to the application, updating `SRS_OAUTH_PRIVATE_KEY` in Key Vault, restarting the Pega tiers, checking search, and then removing

the old public key. Okta signing key rotation is handled by Okta; SRS reads the current key set from the key set URL.

7.7 SRS deployment (backingservices chart)

Each environment has its own SRS release in namespace `srs-<code>`. The keys come from the SRS chart README [R24].

backingservices values for one environment (complete files in Appendix B)

```
global:
  k8sProvider: "aks"
  imageCredentials:
    registry: "<acr-name>.azurecr.io"
srs:
  enabled: true
  deploymentName: "srs-<code>"
  srsRuntime:
    replicaCount: <1 in DEV, SIT, UAT; 3 in PERF, PREPROD, PROD>
    srsImage: "<acr-name>.azurecr.io/platform-services/search-n-reporting-service-os:<tag>"
    env:
      AuthEnabled: true
      OAuthPublicKeyURL: "<okta-jwks-url>"
    ssl:
      enabled: true
      clientAuthentication: "want"
      keystore:
        file: "srs-keystore.p12"
        type: "PKCS12"
      truststore:
        file: "srs-truststore.jks"
        type: "JKS"
      certsSecret: "srs-runtime-certs"
  srsStorage:
    provisionInternalESCluster: false
    domain: "<search-host>"
    port: <search-port>
    protocol: https
    tls:
      enabled: true
    basicAuthentication:
      enabled: true
    authSecret: "srs-search-credentials"
    requireInternetAccess: false
    networkPolicy:
      enabled: true
```

7.8 Index build planning

After Pega connects to SRS, SRS indexes all searchable data, and Pega states that this needs a downtime period. The length depends on the data model, the resources given to Pega and SRS, the number of queue processors and the amount of searchable data [R15]. For a cloned environment, the build runs during the first start with intake held (Section 10.4), so its length sets part of the production outage.

1. In every rehearsal, record the start and end of the build, the number of batch pods, SRS replicas and OpenSearch data nodes, and the row counts of the main indexed classes.
2. Plot build time against searchable data volume across DEV, PERF and PREPROD. PREPROD, built from a recent full-size clone on production-like services, is the best predictor.
3. Estimate production time as the PREPROD time scaled by the ratio of production to PREPROD data volume, plus 25 % contingency. Use the larger of this and the measured PREPROD time if the volumes are close.
4. If the estimate does not fit the outage window, add batch pods and OpenSearch data nodes for the cutover and measure again.

The build is complete when the search landing page shows every class indexed with no errors [R19], the document count of each main index matches the row count of its class in the database within the tolerance agreed with the application team, and no class shows CONFLICTS FOUND [R19]. Pega documents the rebuild steps and screens [R17].

7.9 Backups and restore

Every index can be rebuilt from the Pega database, so OpenSearch snapshots are optional. They shorten recovery but add storage cost.

Table 30. Recovery choices for search

Choice	Recovery after loss of the OpenSearch service	Recovery time
No snapshots	Recreate the service, apply the cluster settings, restart SRS, run a full index build	Measured full build time plus provisioning time
Provider snapshots	Restore the latest snapshot, then reindex classes changed since the snapshot	Restore time plus a partial build; measure in PERF

Elasticsearch 8.x snapshots cannot be restored into OpenSearch [R38, R39], and the 8.8 embedded indexes are not in a snapshot-ready form, so snapshots play no part in moving from 8.8.

8 Secrets, certificates and identity

No password, key or JAAS string is written in a Helm values file. The Pega Helm charts support external secrets for the database, stream, SRS OAuth and certificates [R23]. Key Vault is the system of record, one vault per environment, and the External Secrets Operator copies secrets into Kubernetes [R48].

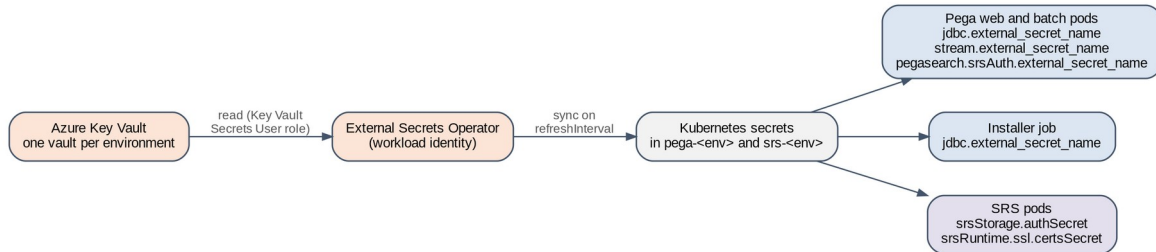


Figure 22. Secret delivery from Key Vault to Pega, installer and SRS pods

Prepared for this implementation. Not a Pega-published diagram.

8.1 Secret inventory

Table 31 lists every secret. Key names marked as fixed are required by the Pega charts; the secret names are this design's convention.

Table 31. Secret inventory (one set per environment, in kv-pegas-<code>)

Kubernetes secret	Keys	Used by	Key Vault secrets	Key names
pega-db-secret	DB_USERNAME, DB_PASSWORD	Pega tiers and installer job (global.jdbc.external_secret_name)	pega-db-username, pega-db-password	Fixed [R23]
pega-stream-secret	STREAM_TRUSTSTORE_PASSWORD, STREAM_KEYSTORE_PASSWORD, STREAM_JAAS_CONFIG	Pega tiers (stream.external_secret_name)	stream-truststore-password, stream-keystore-password, confluent-jaas	Fixed [R23]
pega-srs-oauth	SRS_OAUTH_PRIVATE_KEY	Pega tiers (pegasearch.srsAuth.external_secret_name)	okta-srs-private-key	Fixed [R23]
srs-search-credentials	username, password	SRS (srs.srsStorage.authSecret)	opensearch-srs-username, opensearch-srs-password	Per SRS README [R24]
srs-runtime-certs	keystore and truststore files, keystorePassword, truststorePassword	SRS (srs.srsRuntime.ssl.certsSecret)	srs-tls-keystore, srs-tls-truststore, passwords	Per SRS README [R24]
pega-tier-tls	TOMCAT_KEYSTORE_CONTENT, TOMCAT_KEYSTORE_PASSWORD, ca.crt	Web tier TLS (tier.service.tls.external_secret_names)	pega-tls-keystore, pega-tls-password, pega-tls-ca	Fixed [R23]

8.2 External Secrets Operator manifests

SecretStore and ExternalSecrets for the Pega namespace (Azure Key Vault provider [\[R48\]](#))

```
apiVersion: external-secrets.io/v1beta1
kind: SecretStore
metadata:
  name: keyvault
  namespace: pega-<code>
spec:
  provider:
    azurekv:
      authType: WorkloadIdentity
      vaultUrl: "https://kv-pega-<code>.vault.azure.net"
      serviceAccountRef:
        name: eso-keyvault-reader
---
apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: pega-stream-secret
  namespace: pega-<code>
spec:
  refreshInterval: 1h
  secretStoreRef:
    name: keyvault
    kind: SecretStore
  target:
    name: pega-stream-secret
  data:
    - secretKey: STREAM_TRUSTSTORE_PASSWORD
      remoteRef: { key: stream-truststore-password }
    - secretKey: STREAM_KEYSTORE_PASSWORD
      remoteRef: { key: stream-keystore-password }
    - secretKey: STREAM_JAAS_CONFIG
      remoteRef: { key: confluent-jaas }
---
apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: pega-srs-oauth
  namespace: pega-<code>
spec:
  refreshInterval: 1h
  secretStoreRef:
    name: keyvault
    kind: SecretStore
  target:
    name: pega-srs-oauth
  data:
    - secretKey: SRS_OAUTH_PRIVATE_KEY
      remoteRef: { key: okta-srs-private-key }
```

Create the same pattern for `pega-db-secret` and `pega-tier-tls` in `pega-<code>`, and for `srs-search-credentials` and `srs-runtime-certs` in `srs-<code>`. Each namespace has its own SecretStore and workload identity, and that identity can read only its own environment's vault.

Note. Check which External Secrets Operator API version is installed (`v1beta1` or `v1`) and use it. The stream truststore and keystore password keys can hold empty values with SASL/PLAIN and a public CA, but keep the keys in the order the chart expects [\[R23\]](#).

8.3 Rotation

Table 32. Secret rotation

Secret	Steps	Restart needed
Confluent API key	Create a second key for the same service account; write the new JAAS value to Key Vault; wait for or force the External Secrets refresh; rolling restart of the Pega tiers; check the Stream landing page; delete the old key	Yes, Pega tiers

Secret	Steps	Restart needed
OpenSearch SRS user password	Change the password on the provider; update Key Vault; refresh; restart SRS; check search (S-5)	Yes, SRS
Okta client key	Add the new public key to the Okta app; update <code>okta-srs-private-key</code> ; refresh; restart Pega tiers; check search; remove the old public key	Yes, Pega tiers
Database password	Per the DBA process; update Key Vault; refresh; restart Pega tiers	Yes

Caution. Pods read these values at start. A new Key Vault version reaches the Kubernetes secret on the next refresh, but running pods keep the old value until they restart. A rotation without a restart works until the old credential is deleted, then fails (FS-20).

8.4 Certificates

Table 33. Certificates

Certificate	Issued by	Used for	Renewal owner
Public TLS certificate for the Pega host name	Customer public CA	Application Gateway listener	Platform team
Web tier backend certificate	Customer private CA	Application Gateway to web pods	Platform team
SRS server certificate	Customer private CA	Pega to SRS TLS	Platform team
Confluent broker certificates	Confluent (public CA)	SASL_SSL from Pega	Confluent
OpenSearch endpoint certificate	Provider or customer CA	SRS to OpenSearch TLS	OpenSearch provider
Okta endpoint certificate	Okta (public CA)	Token and key set calls	Okta

Caution. Do not route Pega-to-Confluent, SRS-to-OpenSearch or Okta traffic through a TLS-inspecting proxy. Re-signed certificates break the trust chain that the clients check.

9 Helm configuration per environment

9.1 Chart version and the three runs

This design uses Pega Helm charts 4.13.0: the `pega` chart for Pega Platform and the installer job, and the `backingservices` chart for SRS [R23, R24]. Every key in this section and in Appendix B was checked against the README and default values file of that chart version. Each environment built from a clone goes through three Helm runs of the `pega` chart, shown in Table 34.

Table 34. Helm runs of the `pega` chart for an environment built from a clone

Run	<code>global.actions.execute</code>	What runs	Why
U: upgrade	<code>upgrade</code>	Installer job only; no Pega pods	Upgrades the cloned database. Pega must not start until the clean-up in Section 10.3 is done.
F: first start	<code>deploy</code>	Web tier only; batch tier <code>replicas: 0</code> with HPA off	No queue processor, job scheduler or real-time data flow runs while cloned items are checked (Section 10.4).
D: steady state	<code>deploy</code>	Web and batch tiers at normal size	Index build, then normal operation.

The chart's actions are `install`, `deploy`, `install-deploy`, `upgrade` and `upgrade-deploy` [R23]. `upgrade-deploy` is not used, because it would start Pega straight after the upgrade, before the cloned items are checked.

9.1.1 Upgrade type

The chart supports the upgrade types `in-place` (rules and data in one run, with downtime), `out-of-place-rules` and `out-of-place-data` (a two-step out-of-place upgrade), `zero-downtime` and `custom` [R23]. The chart default is `in-place`. The zero-downtime type needs REST credentials for the running pre-upgrade system to perform pre- and post-actions [R23], and a cloned database has no running 8.8 system attached. The design therefore uses `in-place` on the clone, which is disposable until the cutover, and asks Pega Support to confirm it (GQ-04).

9.2 Pega chart keys per environment

Table 35 lists every `pega` chart key this design sets, with its value per environment. Keys not listed keep the chart default. Values in angle brackets are recorded in Appendix A.

Table 35. Pega chart keys per environment (chart 4.13.0)

Key	Meaning	DEV	SIT	UAT	PERF	PREPROD	PROD	Source
<code>global.provider</code>	Kubernetes provider	aks	aks	aks	aks	aks	aks	[R23]
<code>global.actions.execute</code>	Run type (Table 34)	U/F/D	U/F/D	U/F/D	U/F/D	U/F/D	U/F/D	[R23]
<code>global.customerDeploymentId</code>	Index prefix and guid value	pega26-dev	pega26-sit	pega26-uat	pega26-perf	pega26-ppd	pega26-prd	[R23]
<code>global.jdbc.url</code>	Upgraded clone for this environment	<jdbc-url>	<jdbc-url>	<jdbc-url>	<jdbc-url>	<jdbc-url>	<jdbc-url>	[R23]
<code>global.jdbc.external_secret_name</code>	DB_USERNAME, DB_PASSWORD	pega-db-secret	pega-db-secret	pega-db-secret	pega-db-secret	pega-db-secret	pega-db-secret	[R23]
<code>global.jdbc.rulesSchema, dataSchema</code>	Schemas of the clone	<rules-schema>	<rules-schema>	<rules-schema>	<rules-schema>	<rules-schema>	<rules-schema>	[R23]
<code>global.docker.pega.image</code>	Pega image	platform / pega:26.1.1	platform / pega:26.1.1	platform / pega:26.1.1	platform / pega:26.1.1	platform / pega:26.1.1	platform / pega:26.1.1	[R23]

Key	Meaning	DEV	SIT	UAT	PERF	PREPROD	PROD	Source
tier[web].replicas and HPA min / max	Web pods	1 / 2	1 / 3	2 / 3	3 / 6	3 / 6	3 / 6	[R23]
tier[batch].replicas and HPA min / max	Batch pods (run F: 0, HPA off)	1 / 2	1 / 3	1 / 3	3 / 6	3 / 6	3 / 6	[R23]
tier[*].pdb.enabled	Pod disruption budget	true	true	true	true	true	true	[R23]
tier[web].service.tls.external_secret_names	Backend TLS keystore	pega-tier-tls	pega-tier-tls	pega-tier-tls	pega-tier-tls	pega-tier-tls	pega-tier-tls	[R23]
tier[web].ingress.domain	Host name	<pega-host>	<pega-host>	<pega-host>	<pega-host>	<pega-host>	<pega-host>	[R23]
cassandra.enabled	No Decision Data Store	false	false	false	false	false	false	[R23]
hazelcast.enabled, clusteringServiceEnabled	Hazelcast removed in '25	false	false	false	false	false	false	[R5, R23]
stream.enabled	Stream service on	true	true	true	true	true	true	[R23]
stream.bootstrapServer	Confluent bootstrap	cc-np1	cc-np1	cc-np1	cc-np2	cc-np2	cc-prd	[R23]
stream.securityProtocol	TLS and SASL	SASL_SSL	SASL_SSL	SASL_SSL	SASL_SSL	SASL_SSL	SASL_SSL	[R23]
stream.saslMechanism	API key	PLAIN	PLAIN	PLAIN	PLAIN	PLAIN	PLAIN	[R23]
stream.streamNamePattern	Topic prefix	pega-dev-{stream.name}	pega-sit-{stream.name}	pega-uat-{stream.name}	pega-perf-{stream.name}	pega-ppd-{stream.name}	pega-prd-{stream.name}	[R23]
stream.replicationFactor	Replicas per partition	3	3	3	3	3	3	[R23]
stream.external_secret_name	STREAM_* keys	pega-stream-secret	pega-stream-secret	pega-stream-secret	pega-stream-secret	pega-stream-secret	pega-stream-secret	[R23]
pegasearch.externalSearchService	Use SRS	true	true	true	true	true	true	[R23]
pegasearch.externalURL	SRS service	sr-s-dev	sr-s-sit	sr-s-uat	sr-s-perf	sr-s-ppd	sr-s-prd	[R23]
pegasearch.srsAuth.enabled	OAuth to SRS	true	true	true	true	true	true	[R23]
pegasearch.srsAuth.url	Okta token endpoint	non-prod server	non-prod server	non-prod server	non-prod server	non-prod server	PROD server	[R23, R43]
pegasearch.srsAuth.clientId	Okta client	pega-srs-dev	pega-srs-sit	pega-srs-uat	pega-srs-perf	pega-srs-ppd	pega-srs-prd	[R23]
pegasearch.srsAuth.scopes	Scope	pega.search:full	pega.search:full	pega.search:full	pega.search:full	pega.search:full	pega.search:full	[R23]
pegasearch.srsAuth.authType	Client authentication	private_key_jwt	private_key_jwt	private_key_jwt	private_key_jwt	private_key_jwt	private_key_jwt	[R23]
pegasearch.srsAuth.privateKeyAlgorithm	Key algorithm	RS256	RS256	RS256	RS256	RS256	RS256	[R23]
pegasearch.srsAuth.external_secret_name	SRS_OAUTH_PRIVATE_KEY	pega-srs-oauth	pega-srs-oauth	pega-srs-oauth	pega-srs-oauth	pega-srs-oauth	pega-srs-oauth	[R23]
pegasearch.srsMTLS.enabled	Mutual TLS to SRS	false	false	false	false	false	false	[R23]
installer.image	Installer image	installer:26.1.1	installer:26.1.1	installer:26.1.1	installer:26.1.1	installer:26.1.1	installer:26.1.1	[R23]
installer.upgrade.upgradeType	Upgrade type (GQ-04)	in-place	in-place	in-place	in-place	in-place	in-place	[R23]

Note. Starting replica counts are sized for function, not load. Set the PERF, PREPROD and PROD values from the PERF load test (Section 14.5), and keep PREPROD equal to PROD.

9.3 Backingservices chart keys per environment

Table 36. Backingservices chart keys per environment

Key	Meaning	DEV	SIT	UAT	PERF	PREPROD	PROD	Source
global.k8sProvider	Provider	aks	aks	aks	aks	aks	aks	[R24]
srs.deploymentName	SRS release	srs-dev	srs-sit	srs-uat	srs-perf	srs-ppd	srs-prd	[R24]
srs.srsRuntime.replicaCount	SRS pods	1	1	1	3	3	3	[R14, R24]
srs.srsRuntime.srsImage	SRS image	- os:<tag>	- os:<tag>	- os:<tag>	- os:<tag>	- os:<tag>	- os:<tag>	[R14, R24]
srs.srsRuntime.env.AuthEnabled	Token checks	true	true	true	true	true	true	[R24]
srs.srsRuntime.env.OAuthPublicKeyURL	Okta key set	non-prod server	non-prod server	non-prod server	non-prod server	non-prod server	PROD server	[R24, R42]
srs.srsRuntime.ssl.*	TLS on SRS	enabled	enabled	enabled	enabled	enabled	enabled	[R24]
srs.srsStorage.provisionInternalEScluster	External search	false	false	false	false	false	false	[R24]
srs.srsStorage.domain	OpenSearch host	os-np1	os-np1	os-np1	os-np2	os-np2	os-prd	[R24]
srs.srsStorage.port, protocol	Endpoint	443, https	443, https	443, https	443, https	443, https	443, https	[R24]
srs.srsStorage.tls.enabled	TLS to OpenSearch	true	true	true	true	true	true	[R24]
srs.srsStorage.basicAuthentication.enabled	SRS user	true	true	true	true	true	true	[R24]
srs.srsStorage.authSecret	username, password	srs-search-credentials	srs-search-credentials	srs-search-credentials	srs-search-credentials	srs-search-credentials	srs-search-credentials	[R24]
srs.srsStorage.requireInternetAccess	No internet egress	false	false	false	false	false	false	[R24]
srs.srsStorage.networkPolicy.enabled	Network policy	true	true	true	true	true	true	[R24]

9.4 Differences between the runs

Only two places change between runs: the action, and the batch tier size. Table 37 shows them for SIT; PROD is the same with its own sizes. Appendix B has the complete files for SIT and PROD.

Table 37. Values that change between runs

Key	Run U (upgrade)	Run F (first start)	Run D (steady state)
global.actions.execute	upgrade	deploy	deploy
tier[batch].replicas	not used	0	1 in SIT, 3 in PROD
tier[batch].hpa.enabled	not used	false	true
installer.*	used	ignored	ignored

9.5 Pipeline rules

- Keep one values file per environment and run in Git. Each change goes through review by the platform team and the Pega LSA.
- Pin the chart version (`helm upgrade --install ... --version 4.13.0`) and record image digests in Appendix A.
- Run `helm template` and `helm lint` with the values file in the pipeline before every apply.

- Fail the pipeline if a values file contains a literal password, API secret, JAAS string or private key. Secrets come only through `external_secret_name` keys.
- Fail the pipeline if `customerDeploymentId` or `streamNamePattern` does not match the environment's row in Table 8.

10 The cloned database

A database cloned from 8.8 production carries 8.8 settings, records and work. Some of it refers to the embedded Kafka and embedded Elasticsearch that no longer exist, and some of it points at production systems. This section lists what must be confirmed with Pega Support, what must be cleaned up, and the controlled first start that every cloned environment follows.

10.1 Gating questions for Pega Support

Pega's pages say that Hazelcast must be removed before updating to '25 or later, and that this applies to embedded Hazelcast and the Clustering Service [R5]. They do not say how this applies to a database that is cloned, upgraded offline and only ever started on 26.1.1. Table 38 lists the questions to raise in one Pega Support case before M2. Any condition Pega sets becomes a mandatory step in Section 12.7.

Table 38. Gating questions for Pega Support

ID	Question	Why it matters	Evidence found	Answer	Date
GQ-01	Is a direct upgrade of an 8.8 database to 26.1.1 with the 26.1.1 installer supported?	The whole path depends on it	Not settled by the pages read	Open	
GQ-02	Which 8.8 patch level is required before the clone?	May need an 8.8 patch in production before the final clone	Not settled	Open	
GQ-03	Does the Hazelcast removal requirement apply to a clone upgraded offline? If so, which settings must be in the database before the upgrade?	Pega states removal is required before updating to '25 [R5]; the removal steps assume a running system [R6]	Not settled for this case	Open	
GQ-04	Which installer upgrade type suits a cloned database: <i>in-place</i> , or <i>out-of-place-rules</i> then <i>out-of-place-data</i> ?	Sets run U and the outage length	Chart types listed in [R23]	Open	
GQ-05	How does 26.1.1 treat 8.8 stream and search DSS and node records in an upgraded database, and which takes precedence: DSS in the database or Helm settings?	A stale DSS could override the Helm configuration	Not settled; tested as FS-24	Open	
GQ-06	How are 8.8 delayed and broken queue items processed after the upgrade, given the '26 queue processing changes?	Items may fail or run unexpectedly	'26 changes queue processing [R3]	Open	
GQ-07	Is any SRS or customerDeploymentId state stored in the database that a clone would carry?	Could link a clone to another environment's indexes	Not settled; tested as IT-05	Open	

Important. Until the answers arrive, the plan assumes the conservative answer for each question: rehearse the full path twice, apply the clean-up in Section 10.3, and treat any difference from expected behaviour as a stop condition at the first-start Go/No-Go.

10.2 How Helm settings and database settings interact

Pega 26.1.1 gets its stream and search configuration from the Helm chart. A cloned 8.8 database may also hold Dynamic System Settings (DSS) for the stream service or for search. Pega documents the stream setting names, for example `services/stream/provider`, `services/stream/broker/url` and `services/stream/name/pattern` [R8]. The pages read do not settle whether a DSS with such a name in the database would override the value the Helm chart supplies. The design therefore treats it as a risk: search the cloned DSS for any setting that contains `services/stream` or refers to search hosts, record them, and

prove in the first DEV start that the landing pages show the Helm values (FS-24). GQ-05 asks Pega Support for the rule.

10.3 Inventory and clean-up

Table 39 lists the items that affect Kafka and search. For each one it gives where it lives, what 26.1.1 does with it, the risk, the action, when the action happens and the source. Time points are: B (before the upgrade, on the clone), A (after the upgrade, before the first start), F (during the first start, with the batch tier at zero), and S (after the first start). No table names are given: Pega has not published them for these items, and each change uses the Pega screen named.

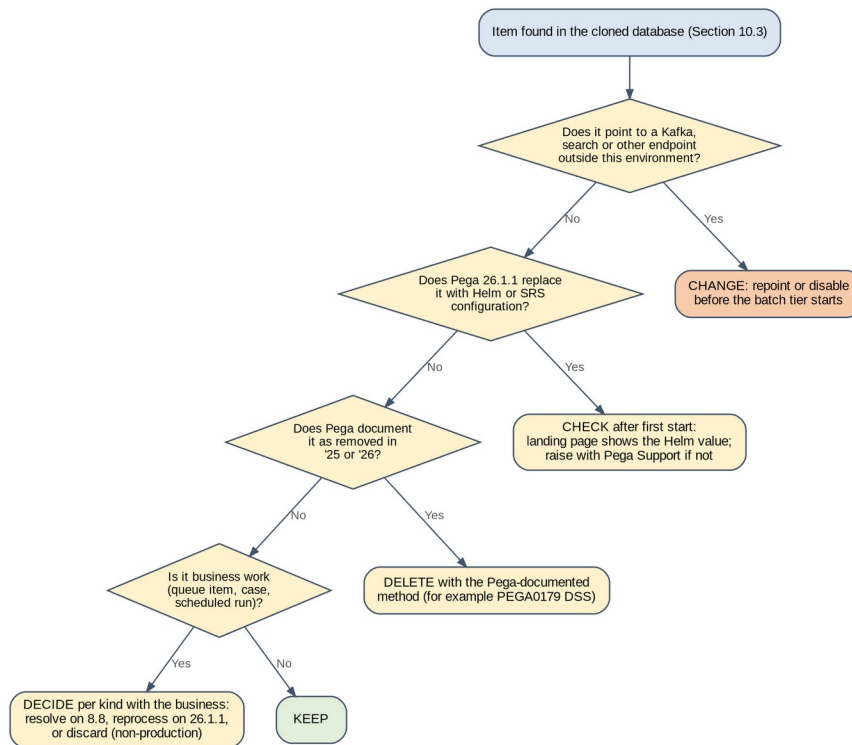


Figure 23. Decision: treatment of an item found in the cloned database

Prepared for this implementation. Not a Pega-published diagram.

Table 39. Kafka and search items in the cloned database

ID	Item	Where it lives	26.1.1 behaviour	Risk	Action	When	Source
CD-01	Stream service DSS from 8.8 (provider, broker URL, name pattern, replication)	DSS (Records > SysAdmin > Dynamic System Settings)	Helm supplies the stream settings; precedence not documented	Stale value overrides Helm	Record; check landing page shows Helm values; remove only as Pega Support advises (GQ-05)	F	[R8, R12]
CD-02	Embedded stream node records from 8.8	Stream service landing page (Configure > Decisioning > Infrastructure > Services > Stream)	External Kafka has no stream nodes	Landing page shows unavailable 8.8 nodes; confusion during checks	Check; remove stale entries from the landing page if shown	F	[R13]
CD-03	Search settings and index host node records from 8.8 embedded Elasticsearch	DSS and search landing page	SRS configured by Helm replaces embedded search	Stale setting points search at 8.8 nodes	Check search landing page shows SRS; record and remove stale settings as Pega Support advises	F	[R14, R16]
CD-04	Search index status from 8.8	Search landing page	Indexes are rebuilt through SRS	Status shows old indexes as present	Full index build; check status [R19]	S	[R15, R19]

ID	Item	Where it lives	26.1.1 behaviour	Risk	Action	When	Source
CD-05	Delayed queue items	Database (queue processor delayed items)	Become due and are queued to Kafka when their time arrives	Items run on 26.1.1 against changed rules, or call production systems from a clone	Count on 8.8 before the clone; in non-production, decide per queue processor to discard or keep; in PROD, keep and test in PREPROD (GQ-06)	B, S	[R3]
CD-06	Broken queue items	Database (queue processor broken items, shown in Admin Studio)	Stay broken until requeued or deleted	Requeue on 26.1.1 runs old work with new rules	Resolve on 8.8 before the final clone where possible; record counts; decide per queue processor	B, S	[R3]
CD-07	Job scheduler state	Job scheduler rules and their run state	Job schedulers run on batch nodes after start	Clone runs production jobs (email, extracts) on first start	Batch tier at zero in run F; disable jobs that call external systems in non-production before scaling the batch tier	F	[R11]
CD-08	Data flow runs, including real-time runs on Kafka data sets	Data flow landing pages	Runs restart against 26.1.1 topics or the customer's Kafka	Runs resume against old offsets or production topics	Stop or delete runs that refer to 8.8 stream partitions; restart from the agreed point	F, S	[R11]
CD-09	Kafka configuration instances and Kafka data sets for application integrations	Records > SysAdmin > Kafka; Records > Data Model > Data Set	Connect to the customer's own Kafka clusters	A non-production clone reads production topics and moves production consumer offsets	Repoint to test clusters or disable in every non-production environment; decide per data set in PROD (Section 11.3)	F	[R11]
CD-10	Other production endpoints: connectors, listeners, email accounts, file listeners	Integration rules and data instances	Run on start if enabled	A clone calls or polls production systems	Short checklist owned by the application team; firewall denies production endpoints from non-production	F	Customer
CD-11	DSS delayed items/data flow based/thread spawner	DSS	No longer used in '26	PEGA0179 alert	Delete	A	[R3]
CD-12	Hazelcast-related settings from 8.8	DSS and prconfig values	Hazelcast removed in '25	Unknown until GQ-03 is answered	Apply Pega Support's instructions	B or A	[R5]
CD-13	Tables without primary keys	Database	Required for '25 and later	Upgrade blocked	Run primaryKeyUtility as the DBA team plans	B	[R10]
CD-14	Custom search properties and reports that rely on search	Rules	Indexed by SRS after the build	Feature behaves differently on SRS	List them; include in the functional checks (Section 14.4)	S	[R14]

Caution. Do not change these items with direct SQL against Pega tables unless Pega Support provides the statement for this release. Use the Pega screens named in the table during run F, when no background processing is running.

10.4 First-start control

Every environment built from a clone follows the same scripted first start. Table 40 gives each step an owner, a check and the evidence to keep, and Figure 24 shows the Go/No-Go logic.

Table 40. First-start steps for an environment built from a clone

Step	Action	Owner	Check	Evidence
ST-1	Run U: upgrade the clone with the installer job	Platform team, DBA	Job completed; no errors in the log (P-5)	Installer log
ST-2	Apply time-point A actions from Table 39	Pega LSA, DBA	Each item signed off	Clean-up record (Appendix D)
ST-3	Run F: deploy the web tier with this environment's prefix, service account and customerDeploymentId; batch tier at zero	Platform team	Web pods Ready (P-1)	kubectl output
ST-4	Hold intake: no public DNS record, inbound listeners off, batch tier at zero	Platform team, application team	No user or inbound traffic in logs	Ingress and listener status

Step	Action	Owner	Check	Evidence
ST-5	Check the Stream landing page and the Search landing page; apply time-point F actions	Pega LSA	Provider ExternalKafka, status NORMAL, prefix <code>pega-</code> ; search uses SRS with <code>pega26-</code>	Screenshots
ST-6	Check the firewall log for denied connections to production or other environments	Platform team	No attempts, or each attempt traced to an item and fixed	Firewall log extract
ST-7	Run D: scale the batch tier; start the full index build	Platform team, Pega LSA	Index build running; batch pods Ready	Start time
ST-8	Check index completeness and counts (Section 7.8)	Pega LSA	All classes indexed; counts match	Count sheet
ST-9	Release intake: public DNS, listeners and job schedulers as agreed	Application team	Smoke tests pass	Test report

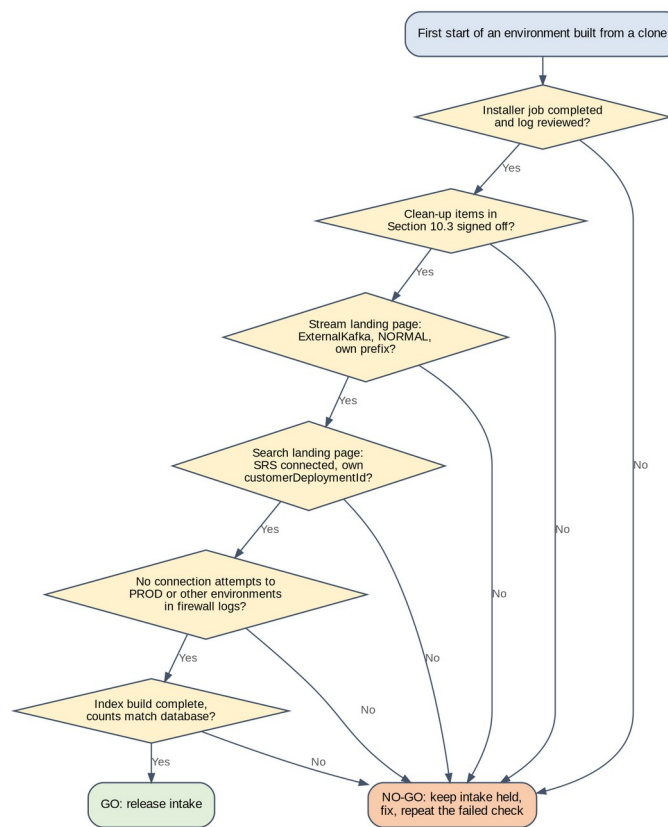


Figure 24. Decision: Go/No-Go at the first start
 Prepared for this implementation. Not a Pega-published diagram.

10.5 Data protection and masking

Non-production clones of production data must follow the customer's masking policy. Mask the clone before the upgrade (time point B). The index build then reads only masked data, and no unmasked value reaches OpenSearch, Kafka topics or logs. If masking runs after the upgrade, it must still finish before run D, because the index build copies searchable data into OpenSearch. Masking after the index build leaves unmasked data in the indexes until a full rebuild. The masking approach is OD-07.

Note. Masking rules must keep values that integrations and searches depend on in a usable form, for example case IDs and keys used in search. Agree the rules with the application team, and include search checks on masked data in the functional tests.

11 The 8.8 content question

11.1 Answer

Nothing in the 8.8 Kafka or search storage is migrated. Figure 25 shows the decision for each kind of content, and Table 41 gives the treatment and the evidence. Each statement was checked against the current Pega pages on the evidence cut-off date.

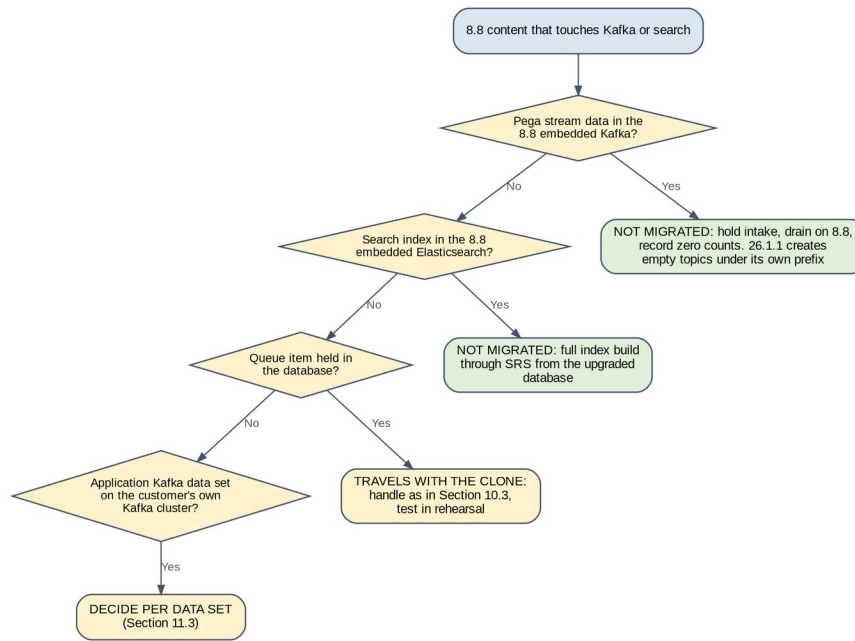


Figure 25. Decision: does 8.8 content need migrating?

Prepared for this implementation. Not a Pega-published diagram.

11.2 Treatment per content type

Table 41. Treatment of 8.8 content

Content	Answer	How it is handled	Evidence
Pega stream data in Kafka (queue processor messages, data flow partitions)	Not migrated	Pega states existing stream data cannot be moved to a new Kafka. 8.8 uses embedded Kafka, so there is no external cluster to drain to. Before the production clone, hold intake on 8.8, wait until queue processors show nothing ready to process, record the counts and any broken items, then stop 8.8. 26.1.1 starts with empty topics that it creates under <code>pega-prd-</code> .	[R13, R25]
Queue items held in the database (delayed, broken)	Travel with the clone	Database content, not Kafka content. Handled as items CD-05 and CD-06 (Section 10.3) and tested in rehearsal.	[R3]
Topic names and configuration	Not migrated	26.1.1 creates its own topics under its own prefix. Only the topic settings policy, such as <code>max.message.bytes</code> , carries over as configuration.	[R11, R23]
Search indexes (embedded Elasticsearch on 8.8)	Not migrated	The indexes live on the 8.8 nodes, not in the database. They are built from the upgraded database after Pega connects to SRS, which needs downtime; the time is measured in rehearsal (Section 7.8). Elasticsearch 8.x snapshots cannot be restored into OpenSearch.	[R15, R17, R38, R39]
Application Kafka data sets on the customer's own topics	Decided per data set	Keep the existing cluster and topic, point at a new cluster, or mirror with Confluent Cluster Linking, which keeps offsets. Agree the start offset so messages are neither skipped nor processed twice.	[R11, R35]

Content	Answer	How it is handled	Evidence
Custom search data (custom search properties, reports that depend on search)	Rebuilt and verified	Rebuilt by the full index build; verified with count comparisons and functional checks.	[R18, R19]

11.3 Application Kafka data sets

Application Kafka data sets read from or write to the customer's own Kafka clusters, which are separate from the Pega stream service [R11]. Each one is decided on its own, with the application owner (Figure 26). In every non-production environment, every data set is repointed to a test topic or disabled before the batch tier starts, so a clone never consumes from a production topic or moves a production consumer group's offsets.

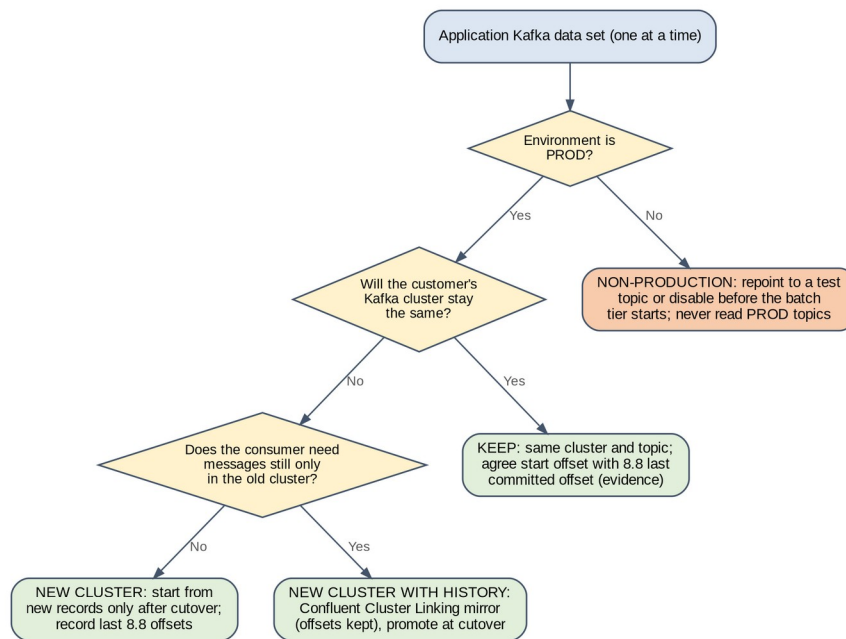


Figure 26. Decision: treatment of each application Kafka data set
Prepared for this implementation. Not a Pega-published diagram.

Table 42. Application Kafka data set register (one row per data set, completed with the application team)

Data set	Direction	Cluster and topic	PROD treatment	Start offset at cutover	Non-production treatment	Owner
<data set name>	<consume or produce>	<cluster / topic>	<keep, new cluster, Cluster Linking>	<offset rule and evidence>	<test topic or disabled>	<application owner>

11.4 Why the Stream Migration activity is not used

Pega provides a Stream Migration activity and a Helm chart guide for switching a running deployment from embedded Stream to an external Kafka, or between Kafka providers [R13, R25]. It drains the queues of a running system before the switch. This programme never switches the running 8.8 system to an external Kafka: 8.8 keeps its embedded Kafka until it stops, and 26.1.1 starts on a different system with its own Confluent cluster. Holding intake and letting the 8.8 queues empty gives the same result, an empty queue at the moment of the final clone, without changing 8.8.

11.5 Statement for approval

Decision. No Pega stream topic and no search index is copied from 8.8. Database-held queue items travel with the clone and are handled as listed in Table 39. Application Kafka topics are handled one by one as listed in Table 42.

12 Deployment runbook per environment

Every environment is built in the same order, with a check after each step. Shared services are created once per group (NP1, NP2, PROD); per-environment objects are created for each environment. Values in angle brackets come from Appendix A, and commands from Appendix C.

12.1 Build order

Table 43. Build order

Phase	Component	Scope	Depends on	Owner
1	Network and DNS: private endpoint subnet, firewall rules, private DNS zones	Per group and per environment	Landing zone	Cloud architect
2	Key Vault, External Secrets Operator identity, Okta authorization server and client	Per environment	Phase 1	Platform team, identity team
3	Confluent cluster and Private Link; service account, ACLs, quota, API key	Cluster per group; rest per environment	Phases 1, 2	Kafka engineer
4	OpenSearch service and private endpoint; cluster settings; SRS user and role	Service per group; user per environment	Phases 1, 2	Search engineer
5	SRS deployment (backingservices chart)	Per environment	Phases 2, 4	Search engineer
6	Database clone, masking, upgrade (run U)	Per environment	GQ answers or conservative plan	DBA team, platform team
7	Clean-up and first start (run F), index build (run D), intake release	Per environment	Phases 3, 5, 6	Pega LSA, platform team

12.2 Network and DNS

Table 44. Network and DNS steps

ID	Action	Owner	Check	Evidence
NW-1	Create the private endpoint subnet with network policies for private endpoints enabled.	Cloud architect	Subnet exists; NSG applies	Azure export
NW-2	For a new group, complete N-1 to N-6 (Section 6.4) for its Confluent cluster.	Kafka engineer	K-1 to K-3 pass from a pod	Command output
NW-3	For a new group, create the OpenSearch private endpoint and DNS record as the provider documents.	Search engineer	S-1 resolves to a private IP and returns the version	Command output
NW-4	Add firewall rules for this environment: its own Confluent and OpenSearch endpoints, <okta-domain>:443, registry and Pega Diagnostic Center if used. Deny production endpoints from non-production.	Cloud architect	Allowed tests pass; denied tests fail (IT-09)	Rule export, test output
NW-5	Apply Kubernetes network policies: deny by default; allow pega to srs, srs to the OpenSearch endpoint, pods to DNS.	Platform team	Test pod cannot reach a denied target	Test output

12.3 Secrets and identity

Table 45. Secrets and identity steps

ID	Action	Owner	Check	Evidence
SI-1	Create kv- <code>pega-<code><code></code></code> with RBAC, private endpoint and public access disabled.	Platform team	Reachable from the spoke only	Test output
SI-2	Create the workload identity for the External Secrets Operator in <code>pega-<code><code></code></code> and <code>srs-<code><code></code></code> ; grant Key Vault Secrets User on this vault only.	Platform team	SecretStore Ready	kubectl output (P-4)

ID	Action	Owner	Check	Evidence
SI-3	Create the Okta client <code>pega-srs-<code><code></code></code> and register its public key (Section 7.6); store the private key in Key Vault.	Identity team	Token request succeeds (O-1)	Decoded claims (O-2)
SI-4	Create the ExternalSecrets in Section 8.2.	Platform team	All secrets SecretSynced	kubectl output

12.4 Confluent Cloud

Table 46. Confluent Cloud steps

ID	Action	Owner	Check	Evidence
CC-1	For a new group, create the cluster (type per OD-02) in the AKS region with Private Link.	Kafka engineer	Cluster status Up	Cluster ID
CC-2	Create service account <code>sa-pega-<code><code></code></code> and the four ACLs (Section 6.5).	Kafka engineer	ACL list matches Table 20 (K-7)	ACL export
CC-3	Create an API key for the service account; write the JAAS value to <code>kv-pega-<code><code></code></code> as <code>confluent-jaas</code> .	Kafka engineer	Secret version exists; key not in any file or ticket	Secret metadata
CC-4	In NP1 and NP2, create the client quota for the service account (OD-09).	Kafka engineer	Quota listed	Quota export
CC-5	Run IT-01 to IT-04 against the other environments in the group.	Kafka engineer	All denied	Test output

12.5 OpenSearch

Table 47. OpenSearch steps

ID	Action	Owner	Check	Evidence
OS-1	For a new group, provision the OpenSearch service (OD-03) at the size in Section 7.4, version 2.15 or 2.19.	Search engineer	S-1 shows the version; S-2 green	Output
OS-2	For a new group, apply the cluster settings in Section 7.4 as administrator.	Search engineer	S-3 shows both settings	Output
OS-3	Create the SRS user for this environment and map it to the index-scoped role <code>pega26-<code><code></code>-srs</code> (Section 7.5).	Search engineer	User can create and delete <code>pega26-<code><code></code>-test</code> ; cannot touch another prefix	Test output
OS-4	Store the SRS user credentials in <code>kv-pega-<code><code></code></code> .	Search engineer	Secret synced to <code>srs-<code><code></code></code>	kubectl output

12.6 SRS

Table 48. SRS steps

ID	Action	Owner	Check	Evidence
SR-1	Create the SRS TLS keystore and truststore and store them in Key Vault.	Platform team	Secret <code>srs-runtime-certs</code> synced	kubectl output
SR-2	Deploy SRS: <code>helm upgrade --install srs-<code><code></code> pega/backingservices -n srs-<code><code></code> -f backingservices-<code><code></code>.yaml</code> .	Search engineer	SRS pods Ready (P-1)	Output
SR-3	From a pod in <code>pega-<code><code></code></code> , call the SRS service with and without a token.	Search engineer	Without token refused; with token accepted (S-5)	Output

12.7 Database clone and upgrade

Table 49. Database clone and upgrade steps

ID	Action	Owner	Check	Evidence
DB-1	For non-production, take the clone from the agreed production copy; for PROD, take the final clone at cutover step C-3.	DBA team	Clone restored; row counts recorded	Clone record
DB-2	Mask the clone (non-production) per OD-07.	DBA team	Masking report	Report
DB-3	Apply any pre-upgrade condition from Pega Support (GQ-02, GQ-03) and the time-point B items in Table 39.	DBA team, Pega LSA	Each item signed off	Clean-up record
DB-4	Run U: <code>helm upgrade --install pega pega/pega -n pega-<code><code></code> -f pega-<code><code></code>-upgrade.yaml --version 4.13.0.</code>	Platform team	Installer job completed (P-5)	Installer log, duration
DB-5	If the installer job fails, read the log, fix the cause, and follow Pega's guidance for rerunning the upgrade on this database. If in doubt, restore the clone and start again from DB-1 (FS-26).	Platform team, DBA	Rerun completes	Incident record

12.8 First start, index build and intake release

Follow the first-start control in Section 10.4 (steps ST-1 to ST-9). The Helm commands are:

Helm commands for runs F and D

```
# Run F: web tier only, batch tier held at zero
helm upgrade --install pega pega/pega -n pega-<code> -f pega-<code>-first.yaml \
--version 4.13.0
# Run D: steady state, batch tier at normal size (starts queue processing)
helm upgrade --install pega pega/pega -n pega-<code> -f pega-<code>-deploy.yaml \
--version 4.13.0
```

12.9 Per-environment notes

Table 50. Per-environment notes

Environment	Built from	Masked	Shared group	Notes
DEV	Production copy	Yes	NP1	First build. Proves the Okta claim design, the index-scoped role, DSS precedence and the full path; finds the cluster messaging topics.
SIT	Production copy (can be the DEV clone)	Yes	NP1	First environment with integrations to test systems; data set register applied.
UAT	Recent production copy	Yes	NP1	Business tests on search over masked data.
PERF	Full-size production copy	Yes	NP2	Load tests, partition and index build measurements; sets PROD sizes.
PREPROD	Most recent full-size production copy	Per policy	NP2	Rehearsal 1 and Rehearsal 2 of the production cutover, run by the cutover team.
PROD	Final clone at cutover	No	PRD	Section 13.

13 Production cutover and rollback

13.1 Principles

- The 8.8 system and its database are not changed by the cutover, apart from stopping 8.8. They stay available for rollback until the rollback window closes.
- Every step was run at least twice in PREPROD with production-sized data, by the people who will run it in production.
- The outage window comes from the rehearsal measurements plus 25 % contingency, not from estimates.
- Rollback is free of data loss until intake is released on 26.1.1. After that, rollback loses work done on 26.1.1, so the business decides.

13.2 Milestones

Table 51. Programme milestones used in this document

Milestone	Content	Exit criteria
M1 Design approval	This document approved; open decisions owned	Signatures on the approvals page
M2 DEV build	First environment from a clone	First-start Go/No-Go passed; Okta, role and DSS tests recorded
M3 PERF measurements	Load tests, partitions, index build time	PROD sizes and cluster type agreed (OD-02)
M4 Rehearsal 1 (PREPROD)	Full cutover sequence with timings	All steps completed; defects logged
M5 Rehearsal 2 (PREPROD)	Full cutover sequence by the cutover team	Timing fits the window; no open severity 1 or 2 defects
M6 Go/No-Go 1 and production cutover	Section 13.3	Go/No-Go 3 passed

13.3 Cutover sequence

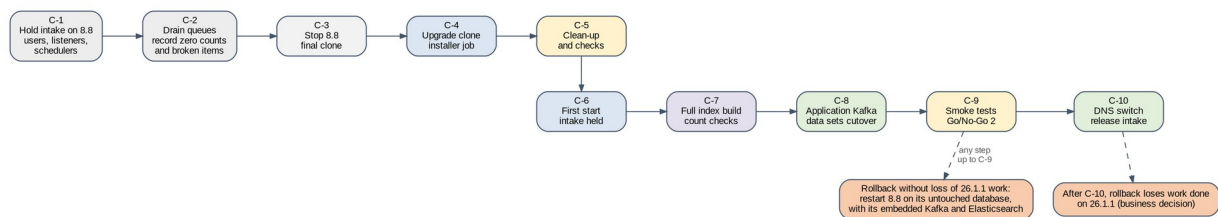


Figure 27. Production cutover sequence from the Kafka and search point of view

Prepared for this implementation. Not a Pega-published diagram.

Table 52. Production cutover steps

Step	Action	Owner	Duration from	Evidence
C-1	Hold intake on 8.8: block user access, stop inbound listeners and services, disable job schedulers that create work.	Application team	Rehearsal	Change record
C-2	Let queue processors finish. Record the count of items ready to process (must be zero) and the count of broken items per queue processor from Admin Studio. Record the last committed offset of each application Kafka data set consumer.	Pega LSA	Rehearsal	Screenshots, offset list
C-3	Stop 8.8. Take the final clone (DB-1).	DBA team	Rehearsal	Clone record

Step	Action	Owner	Duration from	Evidence
C-4	Run U on the clone (DB-3, DB-4).	Platform team	Rehearsal	Installer log
C-5	Apply time-point A clean-up; run F; time-point F checks (ST-2 to ST-6).	Pega LSA	Rehearsal	Clean-up record, screenshots
C-6	Go/No-Go at first start (Figure 24).	Cutover manager	Fixed	Gate record
C-7	Run D; full index build; completeness and count checks (ST-7, ST-8).	Pega LSA	Rehearsal	Count sheet
C-8	Application Kafka data sets cutover (Section 13.5).	Application team	Rehearsal	Offset evidence
C-9	Smoke tests; Go/No-Go 2.	Test lead	Rehearsal	Test report, gate record
C-10	Switch public DNS to Application Gateway; release intake (ST-9); Go/No-Go 3 after 30 minutes.	Platform team	DNS TTL	DNS change record

Important. Lower the TTL of the public DNS record to 300 seconds at least 48 hours before cutover, so a DNS switch or revert takes effect quickly. Restore the normal TTL after hypercare.

13.4 Go/No-Go gates

Table 53. Go/No-Go gates

Gate	When	Criteria	Decision owner
Go/No-Go 1	Seven days before	Two PREPROD rehearsals passed; window fits with 25 % contingency; Pega Support answers received or conservative plan accepted; rollback rehearsed	Business owner with platform lead
First-start gate	After C-5	All checks in Figure 24 up to the index build	Cutover manager
Go/No-Go 2	After C-9	Index build complete and counts match; application data sets started at agreed offsets; smoke tests pass	Cutover manager
Go/No-Go 3	30 minutes after C-10	No severity 1 or 2 incidents; queue backlog falling; search response within target	Cutover manager with business owner

13.5 Application Kafka data set cutover

1. At C-2, record the last committed offset of each 8.8 consumer group that reads the customer's Kafka topics (K-8 on the customer's cluster).
2. For data sets that keep the same cluster and topic, configure the 26.1.1 data flow to start from the recorded offset, or reuse the same consumer group if the application team confirms it, so no message is skipped or processed twice.
3. For data sets that move to a new cluster without history, start from new records only, and keep the 8.8 offset list as evidence of what was processed.
4. For data sets mirrored with Cluster Linking, promote the mirror topics at cutover; Cluster Linking keeps offsets [R35].
5. Record the first offset processed by 26.1.1 for each data set and compare with step 1.

13.6 Rollback

Table 54. Rollback steps

Step	Action
RB-1	Declare rollback; record the reason and the time.
RB-2	Hold intake on 26.1.1. If intake was released, export the list of cases created or changed on 26.1.1 for reconciliation.
RB-3	Revert public DNS to 8.8.
RB-4	Start 8.8 on its untouched database. Its embedded Kafka and embedded Elasticsearch were stopped, not changed.
RB-5	Check the 8.8 stream and search landing pages and queue processors; release 8.8 intake.
RB-6	Reset application Kafka data set consumers on 8.8 to the offsets agreed for the rollback.
RB-7	Scale 26.1.1 to zero; keep it and its Kafka topics and indexes for analysis.

The point after which rollback loses work is C-10, when intake is released on 26.1.1. Before C-10, rollback loses nothing. After C-10, cases created or changed on 26.1.1 are not in the 8.8 database, and messages produced to application Kafka topics by 26.1.1 have been consumed downstream. The business owner decides at that point between rollback with reconciliation and fixing forward, using the reconciliation list from RB-2.

14 Test strategy, rehearsals and failure scenarios

14.1 Test stages

Table 55. Test stages

Stage	Entry criteria	Environment	Owner	Evidence
Connectivity	Network, DNS and secrets steps done	Every environment	Platform team	K-1 to K-3, S-1 to S-3, O-1 outputs
Configuration	Run F complete	Every environment	Pega LSA	Landing page screenshots; inventory sign-off
Functional	Run D and index build complete	DEV, SIT, UAT	Test lead	Checks in Section 14.4
Isolation	Two environments in a group running	NP1, NP2	Security architect	IT-01 to IT-10 records
Performance	PERF built at full size	PERF	Performance lead	PT-01 to PT-04 reports
Resilience and failure	Performance baseline recorded	PERF, PREPROD	Platform team	Failure catalogue records (Section 14.6)
Security	Isolation tests passed	SIT, PREPROD	Security architect	No public endpoints; secret scan of values; token claim checks; TLS versions
Operational acceptance	Monitoring live	PREPROD	Operations lead	Alerts reach on-call; runbooks used by operations staff
Full rehearsal	All above passed	PREPROD (twice)	Cutover manager	Timed cutover record (Appendix D)

14.2 Rehearsals and measurements

Rehearse the full clone-and-upgrade path at least twice in PREPROD on production-sized data before production. Record each measurement against data volume, because they set the production window.

Table 56. Rehearsal measurements

Measurement	Steps	Used for	Rehearsal 1	Rehearsal 2
Drain time on 8.8	C-1 to C-2	Window		
Clone time	C-3	Window		
Upgrade time	C-4	Window		
Clean-up time	C-5	Window		
First start time	ST-3 to ST-6	Window		
Full index build time	ST-7 to ST-8	Window; OpenSearch sizing		
Time to release intake	C-9 to C-10	Window		
Partitions created	K-5	Confluent capacity		
Searchable data volume	S-4	OpenSearch sizing		

14.3 Isolation tests

Each test uses environment A's credentials against environment B in the same group, and must be refused. Run them in NP1 and NP2 after every new environment joins a group.

Table 57. Isolation tests

ID	Test	Method	Pass criteria
IT-01	Read and write B's topics with A's key	kcat produce and consume to <code>pega--<topic></code> with A's credentials	Authorization error
IT-02	Create and delete topics under B's prefix	<code>confluent kafka topic create</code> and <code>delete</code> as A	Refused
IT-03	Join or reset B's consumer group	kcat consumer with group <code>pega-- . . .</code>	Authorization error
IT-04	Use A's key from B's namespace	Check B's Key Vault and secrets contain no A key	Not present
IT-05	Index names after the build	S-4 for <code>pega26-<A>*</code> ; list all indexes	A's indexes all carry A's prefix; none outside a known prefix
IT-06	A's SRS user on B's indexes; SRS works with the scoped role	Search, write and delete on <code>pega26-*</code> as A's user; full build as A	Refused on B; A's build succeeds
IT-07	A's token at B's SRS	Send a request to B's SRS with A's token	Refused (guid mismatch)
IT-08	A's External Secrets identity on B's vault	Create a test ExternalSecret in A pointing at B's vault	Access denied
IT-09	Non-production pod to production endpoints	From a pod in NP1 and NP2: DNS and TCP to cc-prd, os-prd and production integration hosts	No private resolution; connection refused by firewall
IT-10	Cloned environment to production integrations	Firewall log review during run F and the first day of run D	No connection attempts to production systems

14.4 Functional and integration checks

- Queue a test item to a test queue processor and confirm it is processed; repeat after every deployment.
- Run a job scheduler with a test activity and confirm it runs on the batch tier.
- Create a case, wait for indexing, and find it with case search; repeat for each class with custom search properties (CD-14).
- Run each report that depends on search and compare with the 8.8 baseline on the same clone data.
- For each application Kafka data set in SIT, run its data flow against the test topic and compare counts.
- Check the produce path with a message near 5,000,000 bytes, if the application produces large messages (FS-06).

14.5 Performance tests

Table 58. Performance tests

ID	Test	Target	Measured
PT-01	Peak user load plus peak queue processor load in PERF	Agreed response times; consumer lag returns to baseline after peak	
PT-02	Full index build throughput	Fits the production window after scaling (Section 7.8)	
PT-03	Quota enforcement: PERF load while PREPROD runs a light test	PREPROD response unchanged; PERF throttled at its quota	
PT-04	Consumer lag under peak per queue processor	Lag within the alert threshold	

14.6 Failure scenario catalogue

Table 59 lists the failures to test. Run Kafka, search and platform scenarios in PERF or PREPROD, and clone-specific scenarios in DEV during the first build and again in PREPROD. Where Pega does not document the exact behaviour, record what happens in the first run; that record becomes the expected result for later runs.

Table 59. Failure scenario catalogue

ID	Scenario	How to cause it safely	Expected behaviour	Detection	Recovery	Pass criteria
FS-01	Kafka bootstrap unreachable	Block 9092 to the Confluent endpoints with a network policy	Queue processing and cluster messaging stop; Stream landing page not NORMAL	Stream landing page; Pega log Kafka connection errors; alert	Remove the block; pods reconnect or restart	Processing resumes; no item lost
FS-02	Zonal private DNS record missing	Remove one zonal record in DEV	Some broker connections fail after bootstrap	K-3 shows a broker resolving publicly or not at all	Restore the record	All broker names resolve privately
FS-03	API key revoked	Delete the environment's API key in DEV	Authentication failures; stream down	Pega log SASL errors; Confluent audit log	Create a key; update Key Vault; restart tiers	Stream NORMAL within the rotation time
FS-04	ACL missing for a new topic	Remove the TOPIC prefixed ACL briefly in DEV	Topic creation or access refused	Authorization errors in Pega log	Restore the ACL; add literal ACLs for unprefix topics	All topics accessible
FS-05	Partition limit reached	Lower the cluster headroom in a sandbox or simulate with a test topic set	New topic creation fails	Confluent error; Pega log	Delete unused topics or add capacity	New queue processor starts
FS-06	Message larger than max.message.bytes	Produce a 6 MB test message through a test queue processor	Producer error for that item	RecordTooLargeException in Pega log	Set topic to 5,000,000; keep messages within it [R11]	Item within limit processed
FS-07	Broker rolling restart during Confluent maintenance	Observe a scheduled maintenance in PERF	Short retries; no loss	Confluent notice; brief lag rise	None	No broken items caused
FS-08	Consumer lag builds under load	PT-04 at 150 % of peak	Lag grows then drains	Lag alert	Add batch pods within partition count	Lag back to baseline
FS-09	One SRS pod lost	Delete one SRS pod in PERF	Searches continue on other pods	Pod restart event	Kubernetes recreates it	No search errors seen by users
FS-10	Whole SRS deployment lost	Scale SRS to zero in PERF	Search and indexing fail; other functions continue	Search errors; SRS alert	Scale back; check indexing backlog clears	Backlog cleared; counts match
FS-11	Okta token cannot be obtained	Block <code><okta-domain></code> at the firewall for the environment	Search fails once the current token expires	Pega log token errors	Restore the rule	Search resumes without restart, or with restart recorded
FS-12	Token from the wrong issuer	Configure a test client on another authorization server	SRS refuses if it checks issuer; record result	SRS log	Use the right server	Behaviour recorded in Section 18.4
FS-13	<code>guid</code> does not match customerDeploymentId	Change the claim value for the DEV client	SRS refuses requests	401 or 403 in SRS log	Restore the claim	Search resumes
FS-14	Client key rotated in Okta but not in Key Vault	Remove the current public key from the DEV client	Token requests fail	Okta system log; Pega log	Register the key or update Key Vault and restart	Search resumes
FS-15	OpenSearch status yellow	Stop one data node (provider tooling) in PERF	Searches continue	S-2 yellow; alert	Provider restores the node	Green; no Pega errors
FS-16	OpenSearch status red	Simulate in a sandbox service by closing an index copy	Searches on the affected index fail	S-2 red; alert	Restore or rebuild the index	Green; counts match
FS-17	Flood-stage disk watermark	Fill test data on a sandbox service	Indexes read-only; indexing fails	Disk alert; SRS errors	Add storage; clear the read-only block as the provider documents	Indexing resumes

ID	Scenario	How to cause it safely	Expected behaviour	Detection	Recovery	Pass criteria
FS-18	OpenSearch credentials rotated without SRS update	Change the SRS user password only	SRS calls refused	SRS log 401	Update Key Vault; restart SRS	Search resumes
FS-19	Certificate expires	Deploy an expired SRS server certificate in DEV	Pega to SRS TLS fails	TLS errors in Pega log; expiry alert should have fired earlier	Renew; restart	Expiry alert fires 30 days before
FS-20	Key Vault secret rotated without pod restart	Rotate the API key, delete the old key, skip the restart	Stream fails when the old key is deleted	SASL errors	Rolling restart	Runbook includes the restart
FS-21	AKS node drained or zone lost	Cordon and drain all nodes in one zone	Pods reschedule within PDB limits	Pod events	Kubernetes reschedules	Processing and search continue
FS-22	Non-production load test saturates the shared cluster	PERF load above its quota while PREPROD tests	PERF throttled; PREPROD unaffected	Metrics API by principal	Lower quota; reschedule	PREPROD response unchanged
FS-23	Index rebuild during business hours on a shared service	Full build in SIT while UAT tests search	UAT search slower	Latency metrics	Schedule builds out of hours	UAT latency within target
FS-24	Stale 8.8 stream or search DSS overrides Helm	In DEV, record cloned DSS; start with them in place	Landing pages show Helm values (expected) or DSS values (defect)	Stream and search landing pages	Follow Pega Support's answer to GQ-05	Helm values in effect
FS-25	Wrong customerDeployementId deployed	Deploy SIT with DEV's ID in a sandbox	SRS refuses tokens; if accepted, indexes mix	Pipeline check should block it; SRS log	Redeploy with the right ID; delete wrong indexes	Pipeline check blocks the deploy
FS-26	Installer upgrade fails halfway	Stop the installer job during a DEV rehearsal	Database left partly upgraded	Job status; installer log	Follow Pega's rerun guidance, or restore the clone and restart	Second run completes
FS-27	Clone starts with application Kafka data sets pointing to production	Leave one data set unchanged in DEV, with the firewall rule in place	Connection denied by the firewall	Firewall log; data set errors	Repoint or disable the data set	No production connection; IT-10 passes
FS-28	Delayed or broken items from 8.8 processed unexpectedly	Keep a sample of items in a DEV clone	Items run on 26.1.1 when due or queued	Queue processor statistics; application log	Apply the decision per queue processor (CD-05, CD-06)	Behaviour matches the decision
FS-29	Index build interrupted	Restart the batch tier during the build in DEV	Build stops or continues	Search landing page	Resume or restart the build as Pega documents [R17]	Complete build; counts match
FS-30	All batch pods lost	Scale the batch tier to zero in PERF under load	Queue processing stops; items accumulate in topics	Queue backlog alert	Scale back	Backlog drains; no loss

14.7 Exact checks

Table 60. Exact checks

Check	Command or screen	Expected result
Stream service	Dev Studio: Configure > Decisioning > Infrastructure > Services > Stream	Provider ExternalKafka; status NORMAL; bootstrap and prefix from Helm [R13]
Queue processors	Admin Studio > Queue processors	Ready to process falls to zero; broken items known
Search	Search landing page	SRS connected; all classes indexed; no CONFLICTS FOUND [R19]
Kafka DNS, TLS and metadata	K-1, K-2, K-3	Private IPs; trusted chain; all brokers listed
Topics, partitions, configuration, ACLs	K-4 to K-7	Topics under <code>pega-</code> only; max.message.bytes 5,000,000; ACLs as Table 20
OpenSearch health, settings, indexes	S-2, S-3, S-4	Green; both settings; indexes under <code>pega26-</code> only

Check	Command or screen	Expected result
SRS	S-5	Token required; healthy response
Okta token	O-1, O-2	<code>scp</code> includes <code>pega.search:full;guid</code> equals <code>pega26-<code><code></code></code>
Installer job	P-5	Completed without errors
Pods and secrets	P-1, P-4	All Ready; all ExternalSecrets SecretSynced

14.8 Acceptance criteria

Table 61. Acceptance criteria

ID	Criterion
AC-1	All exact checks pass in every environment, with evidence filed.
AC-2	All isolation tests pass in NP1 and NP2.
AC-3	All failure scenarios have been run at least once with results recorded; any defect is fixed or accepted by the business owner.
AC-4	Performance results meet the targets agreed by the business owner.
AC-5	Two PREPROD rehearsals completed, with the measured window within the agreed outage.
AC-6	Monitoring and alerts in Section 17.1 are live and tested.

15 Known issues and challenges

These are the problems that come up in this kind of implementation. Each one names the symptom, the cause, how to prevent it and how to fix it.

15.1 Cloned database

Table 62. Known issues: cloned database

Symptom	Cause	Prevention	Fix
Landing pages show 8.8 values instead of the Helm values	Stream or search DSS from 8.8 in the cloned database (CD-01, CD-03)	Record cloned DSS before the first start; FS-24 in DEV; GQ-05	Apply Pega Support's answer; restart the tiers
A production Kafka consumer group's offsets move while 8.8 is still live	A non-production clone started with its application Kafka data sets still pointing at production (CD-09)	Batch tier at zero in run F; data sets repointed before run D; firewall denies production clusters	Reset the production consumer group with the Kafka owner; repoint the data set
Stream landing page lists unavailable nodes	8.8 embedded stream node records in the clone (CD-02)	Check at ST-5	Remove stale entries on the landing page
Queue items from 8.8 fail on 26.1.1	Rules referenced by the item changed in the upgrade (CD-05, CD-06)	Drain fully on 8.8; resolve broken items before the clone; decide per queue processor	Fix and requeue, or discard with business approval
Index build longer than planned	Production-sized data; too few batch pods or data nodes	Measure in PERF and PREPROD; scale for the build (Section 7.8)	Add batch pods and data nodes; extend the window through the Go/No-Go
Emails or extracts sent from a test environment	Job schedulers and connectors in the clone (CD-07, CD-10)	Batch tier at zero in run F; checklist; firewall	Disable the job; inform recipients

15.2 Kafka and Confluent

Table 63. Known issues: Kafka and Confluent

Symptom	Cause	Prevention	Fix
Client reaches bootstrap, then broker timeouts	Broker names resolve to public IPs or not at all; zonal Private Link records missing [R32]	N-4 to N-6 for every new cluster; synthetic DNS check	Add the zonal records
Too many partitions on a shared cluster	Many queue processors per environment, multiplied by environments	Measured budget (Section 6.7); alert at 70 %	Add capacity or move an environment
RecordTooLargeException	Confluent topic default max.message.bytes below Pega's 5,000,000 [R33]	K-6 after every deployment	Raise the topic setting
SASL authentication fails after a secret change	JAAS value stored with wrong quoting or line breaks	Store as one line; test with K-3 before restarting Pega	Correct the Key Vault value; restart
Old credential still in use after rotation	Pods not restarted after the secret changed (FS-20)	Restart is a rotation step	Rolling restart
Topic creation denied	Security policy removed CREATE from the ACL	Decide OD-11 early; pre-create topics if needed	Restore CREATE or pre-create the topic
Kafka outage stops more than queue processing	From '25, Pega uses Kafka for cluster messaging that Hazelcast carried [R7]	Treat Kafka as tier-1; FS-01	Restore Kafka; check nodes reconnect

15.3 OpenSearch and SRS

Table 64. Known issues: OpenSearch and SRS

Symptom	Cause	Prevention	Fix
Indexes created with wrong names or settings, or index creation blocked	Auto-creation not disabled, or SRS user lacks rights [R14]	Apply cluster settings before SRS starts; IT-06	Apply settings; widen the role by the missing permission only; rebuild
SRS returns 401 or 403; Pega search fails	Okta token missing <code>pega.search:full</code> , missing or wrong <code>guid</code> , or issued by the org authorization server [R42, R44]	Custom authorization server; O-2 decode in every environment	Fix the scope, claim or server; restart Pega if the client changed
SRS will not start against OpenSearch	SRS image not on the matrix for the OpenSearch version [R14]	Pin versions per Section 7.2	Use a matrix pair
Old indexes remain after a refresh	Refresh skipped the index clean-up	Refresh checklist step (Section 17.4)	Delete <code>pega26-<code><code></code>*</code> as the environment's SRS user, then rebuild
Index deletion fails	<code>destructive_requires_name true</code> [R14]	OS-2	Set to false

15.4 Process

Every refresh of a non-production environment from a new production clone repeats the risky parts of the first build: masking, topic and index clean-up, repointed integrations and a full reindex. Treat each refresh as a change with the checklist in Section 17.4, not as a database task alone.

16 Troubleshooting

16.1 Method

Most failures between Pega and Kafka or search are connectivity failures. Work through the chain in Figure 28 in order: a failure at one step makes every later step fail, so the first failing step is the one to fix. For cloned environments, first rule out a stale 8.8 setting (CD-01, CD-03).

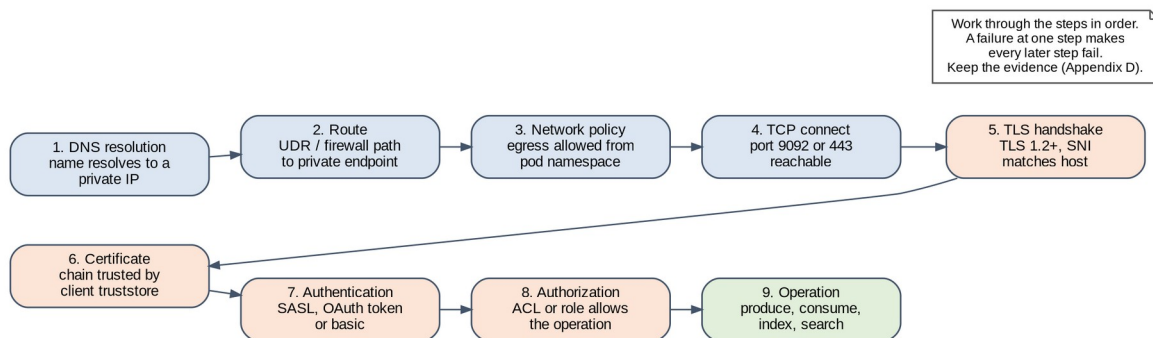


Figure 28. Connectivity chain for Kafka and search
Prepared for this implementation. Not a Pega-published diagram.

16.2 Kafka

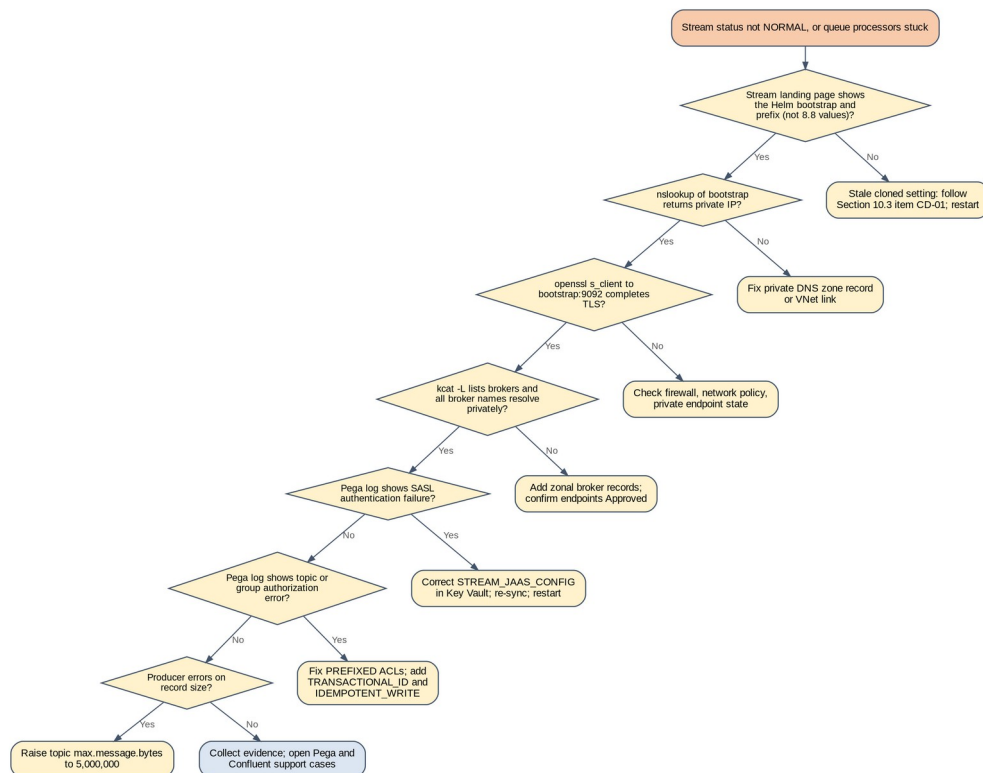
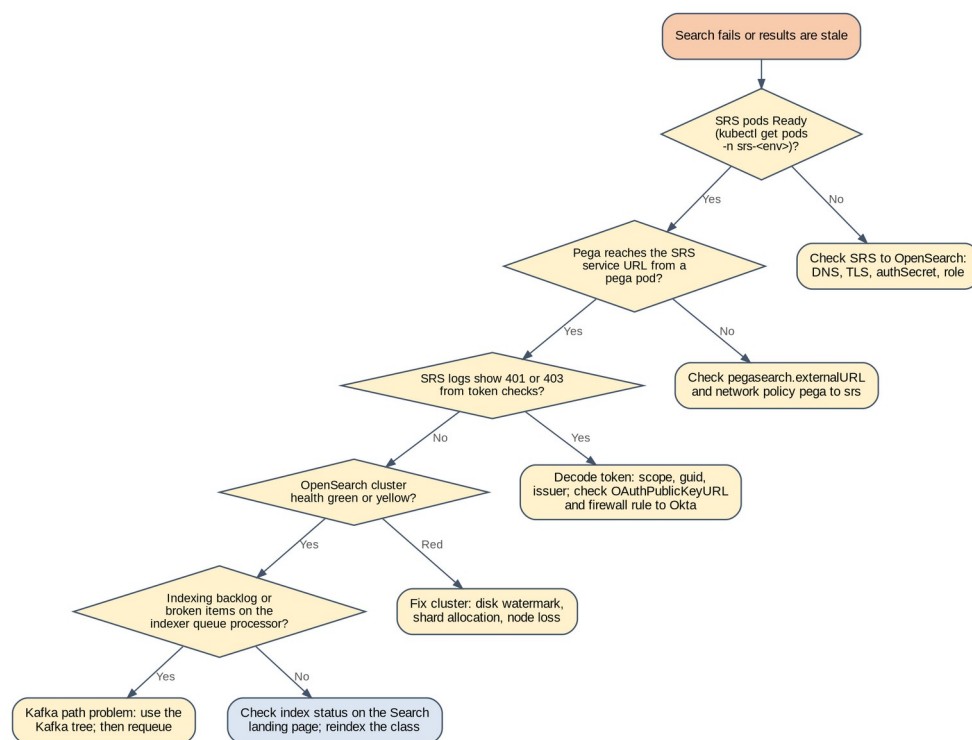


Figure 29. Troubleshooting tree: stream service
Prepared for this implementation. Not a Pega-published diagram.

Table 65. Kafka symptoms

Symptom	Likely cause	Check	Fix
Stream status not NORMAL after start	Stale cloned setting, DNS, network or authentication	Landing page values; K-1 to K-3; Pega log	Follow the tree
Bootstrap works, brokers time out	Zonal DNS records	K-3	Add records [R32]
Topic or group authorization errors	ACL prefix does not match <code>streamNamePattern</code>	K-4, K-7	Correct the prefix or ACL
Transactional or idempotent producer errors	Missing TRANSACTIONAL_ID or IDEMPOTENT_WRITE ACL	K-7	Add the ACL [R11]
RecordTooLargeException	Topic limit	K-6	Raise to 5,000,000
Ready to process keeps growing	Batch tier down or a failing processor	Admin Studio; batch pod logs (P-2)	Restore batch pods; fix the processor
Throughput drops in one shared environment	Quota reached, or another environment's load	Metrics API by principal	Adjust quota or schedule

16.3 Search

**Figure 30. Troubleshooting tree: search**

Prepared for this implementation. Not a Pega-published diagram.

Table 66. Search symptoms

Symptom	Likely cause	Check	Fix
SRS pods not Ready	OpenSearch unreachable or wrong credentials	SRS logs; S-1 from an SRS pod	Fix network or <code>authSecret</code> ; restart SRS
401 or 403 in SRS log	Token scope, <code>guid</code> or issuer	O-2	Fix Okta configuration
Token request fails	Firewall to Okta, client key mismatch	O-1; Okta system log	Fix rule or key

Symptom	Likely cause	Check	Fix
Results missing for recent changes	Indexing backlog	Queue processors; Kafka tree	Clear backlog
Class shows CONFLICTS FOUND	Mapping conflict in indexed properties [R19]	Search landing page	Fix the properties; reindex the class
Cluster red or read-only	Node loss or disk watermark	S-2; provider console	Restore capacity

16.4 Cloned database and upgrade

Table 67. Cloned database and upgrade symptoms

Symptom	Likely cause	Fix
Installer job fails	Database connectivity, permissions, schema names, or a missing prerequisite	Read the log (P-5); fix; rerun or restore the clone (FS-26)
primaryKeyUtility reports insufficient space	The utility copies each table missing a primary key	Add database storage; rerun [R10]
ClassNotFoundException for javax classes	Custom JAR not rebuilt for Jakarta EE	Import the rebuilt JAR into a higher CodeSet version [R9]
PEGA0179 alerts	Old queue processor DSS still present (CD-11)	Delete the DSS [R3]

16.5 Support cases

Pega Global Client Support handles externalized services as it handles databases: it determines the cause, then hands over to the third-party provider if the issue is not in Pega software [\[R22\]](#). Open cases with Pega logs from all tiers, Stream and search landing page screenshots, queue processor status, SRS logs, the Appendix C outputs, the decoded token claims, and the values file with secrets removed.

17 Operations, onboarding, refresh and retirement

17.1 Monitoring

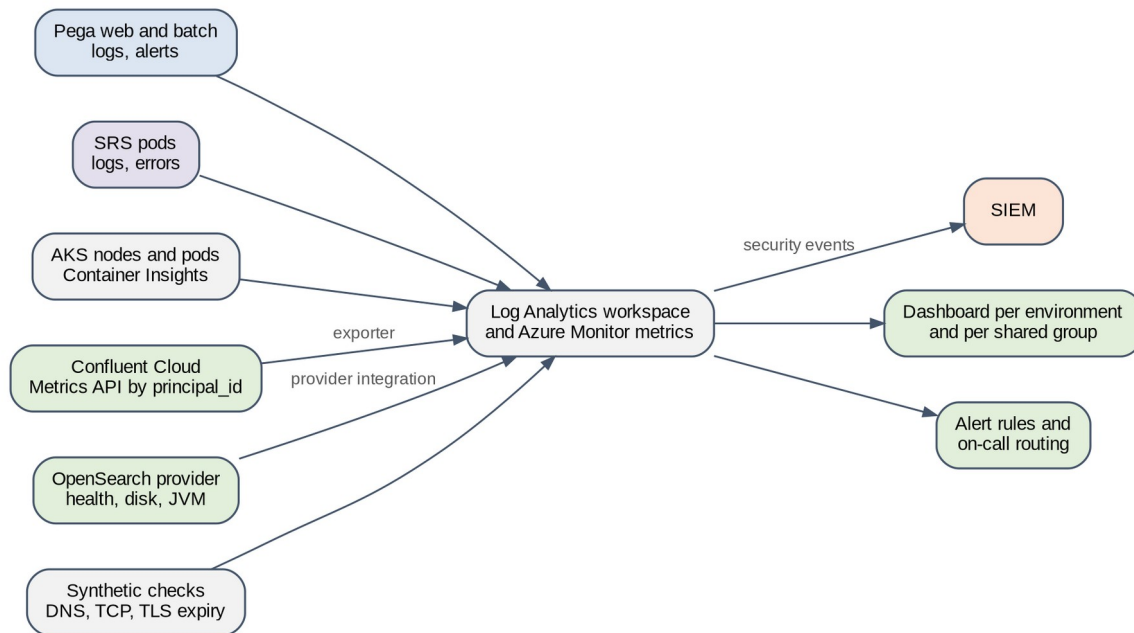


Figure 31. Telemetry flows

Prepared for this implementation. Not a Pega-published diagram.

Table 68. Monitoring signals, thresholds and responses

Signal	Source	Threshold	Response
Stream service status	Pega (Stream landing page, alerts)	Not NORMAL	Section 16.2
Queue backlog	Pega Admin Studio	Ready to process grows for longer than the agreed period	Check batch tier and consumer errors
Broken queue items	Pega Admin Studio	Above zero for business-critical processors	Investigate and requeue
Consumer lag and request errors per principal	Confluent Metrics API	Lag above baseline; authentication or authorization errors	Section 16.2
Partition count per cluster	Confluent Metrics API	Above 70 % of the cluster limit	Capacity review (Section 6.7)
Throughput per service account	Confluent Metrics API [R36]	At quota for longer than 15 minutes	Review quota or schedule
SRS errors and latency	SRS logs	Above baseline	Section 16.3
OpenSearch health	Provider metrics	Yellow for longer than 30 minutes; red at once	Section 16.3
OpenSearch disk	Provider metrics	Above the low watermark less 10 points	Add storage
Okta token failures	Pega log; Okta system log	Any repeated failure	Section 16.3
Certificate expiry	Synthetic checks	Less than 30 days	Renew (Section 8.4)
Private DNS resolution	Synthetic checks	A public IP or no answer	Section 4.4

17.2 Routine tasks

Table 69. Routine operations

Task	Frequency	Owner	Procedure
Confluent API key rotation	Per security policy	Kafka engineer	Section 8.3
OpenSearch SRS user password rotation	Quarterly	Search engineer	Section 8.3
Okta client key rotation	Yearly or per policy	Identity team	Section 8.3
Certificate renewal	Before expiry	Platform team	Section 8.4
Partition and shard capacity review	Monthly and after each release	Kafka and search engineers	K-5, S-4
Check new topics for max.message.bytes	After each deployment	Kafka engineer	K-4, K-6
SRS image update	With Pega updates or security fixes	Search engineer	Update <code>srsImage</code> per environment, NP1 first
OpenSearch version update	Per provider notice, within the SRS matrix	Search engineer	NP1, then NP2, then PROD; rebuild if advised
Cost review	Quarterly	Platform lead	Cost model in Section 5.5 with actual usage

17.3 Onboarding a new non-production environment

1. Approve the environment and its group with the decision in Section 5.1. Assign a code that is not the start of another code.
2. Add a row to Table 8 and to Appendix A.
3. Create the namespace, Key Vault, External Secrets identity and network policies (SI-1, SI-2, NW-5).
4. Create the Okta client and, for option 2, add the client to the `guid` expression (Section 7.6).
5. Create the service account, ACLs, quota and API key on the group's cluster (CC-2 to CC-4).
6. Create the OpenSearch user and index-scoped role (OS-3, OS-4).
7. Deploy SRS (SR-1 to SR-3).
8. Clone, mask and upgrade the database (DB-1 to DB-5), then follow the first-start control (Section 10.4).
9. Run the isolation tests against every other environment in the group.
10. Add the environment to monitoring and the cost review.

17.4 Refreshing an environment from a new clone

1. Agree the refresh window with the environment's users and the other environments in the group.
2. Scale the environment's Pega tiers to zero.
3. Delete its topics and consumer groups under `pega-<code>` (K-9), and its indexes `pega26-<code>*` (S-6). Keep the prefix, customerDeploymentId, ACLs, quota, Okta client and SRS user.
4. Take, mask and upgrade the new clone (DB-1 to DB-5).
5. Follow the first-start control (Section 10.4), including repointing application Kafka data sets and other endpoints.
6. Run the full index build and the functional checks.
7. Record the refresh in the environment log with the evidence.

17.5 Retiring an environment

1. Scale the Pega tiers and SRS to zero; uninstall the Helm releases.
2. Delete the topics and consumer groups under the prefix (K-9).
3. Delete the ACLs, API keys and service account (Section 6.5).
4. Delete the indexes `pega26-<code>* (S-6), then the OpenSearch user and role.`
5. Delete the Okta client and remove it from the `guid` expression.
6. Delete the Key Vault secrets, External Secrets objects and the namespace; purge the vault per retention policy.
7. Remove firewall rules and DNS records specific to the environment.
8. Record evidence of each deletion in the retirement record (Appendix D).

17.6 Roles and responsibilities

R = responsible, A = accountable, C = consulted, I = informed.

Table 70. RACI (Okta objects are owned by the customer identity team: A/R)

Activity	Platform team	DBA team	Kafka team	Search team	Pega team	Pega Support	Confluent	OpenSearch provider
Network, DNS, firewall	A/R	I	C	C	I	I	C	C
Confluent clusters, ACLs, quotas	C	I	A/R	I	C	I	C	I
OpenSearch services, roles	C	I	I	A/R	C	I	I	R
SRS deployment	R	I	I	A/R	C	C	I	I
Okta authorization servers and clients	C	I	I	C	C	I	I	I
Clone, masking, upgrade	R	A/R	I	I	C	C	I	I
Cloned item clean-up	C	R	C	C	A/R	C	I	I
Testing and rehearsals	R	R	R	R	A/R	I	C	C
Cutover and rollback	R	R	R	R	A/R	C	I	I
Operations after handover	A/R	R	R	R	R	C	C	C

18 Risks, open decisions, assumptions and source reconciliation

18.1 Risk register

Table 71. Risk register (L likelihood, I impact: H high, M medium, L low)

ID	Risk	L	I	Mitigation	Owner
RK-01	Pega Support does not confirm the direct 8.8 to 26.1.1 clone-and-upgrade path, or sets conditions late	M	H	Raise the case before M2 (OD-01); conservative plan; two rehearsals	Pega LSA
RK-02	A stale 8.8 DSS overrides the Helm stream or search configuration	M	H	FS-24 in DEV; GQ-05	Pega LSA
RK-03	A non-production clone touches production Kafka topics or systems	M	H	Batch tier at zero in run F; repointing; firewall deny; IT-09, IT-10	Platform team
RK-04	Index build longer than the outage window	M	H	Measure in PERF and PREPROD; scale for the build	Search team
RK-05	Okta licence lacks custom authorization servers, or the client-based claim does not work for client credentials	M	H	Check licence before M2; DEV test; fall back to option 1	Identity team
RK-06	SRS needs cluster permissions beyond the index-scoped role	M	M	IT-06 in DEV; widen only by named permissions	Search team
RK-07	TRANSACTIONAL_ID * ACL is shared by all environments on a cluster	L	M	Accepted residual risk; separate clusters for PROD	Kafka team
RK-08	Noisy neighbour in a shared group	M	M	Quotas; schedule; per-principal metrics	Kafka team
RK-09	No OpenSearch provider passes the selection	L	H	Self-managed fallback on AKS	Enterprise architect
RK-10	Delayed or broken items fail after the upgrade	M	M	Drain fully; resolve broken items; FS-28; GQ-06	Pega LSA
RK-11	Masking breaks search or integration keys	M	M	Masking rules agreed with the application team; search tests on masked data	DBA team
RK-12	Vendor documentation changes between rehearsal and cutover	M	M	Recheck Section 3 before each rehearsal	Enterprise architect

18.2 Open decisions

Each open decision has a recommended answer, an owner and the milestone by which it is needed (Section 13.2). If a decision is not taken by its milestone, the recommended answer applies and the risk is recorded.

Table 72. Open decisions

ID	Decision	Options	Recommended answer	Owner	Needed by
OD-01	Pega Support answers (GQ-01 to GQ-07)	Answers received; conservative plan	Raise now; plan conservatively until answered	Pega LSA	M2
OD-02	Confluent cluster types	Enterprise; Dedicated	Enterprise for NP1; PROD type for NP2 and PROD after PERF measurement	Kafka team	M2 (NP1), M3 (NP2, PROD)
OD-03	OpenSearch provider	Third-party managed on Azure; self-managed on AKS	Provider that passes Section 7.3	Enterprise architect	M2
OD-04	SRS per environment or shared	Per environment; per group	Per environment	Search team	M2
OD-05	Okta design	Server per environment; shared non-production server	PROD dedicated; shared non-production with client-based claim if proven	Identity team	M2

ID	Decision	Options	Recommended answer	Owner	Needed by
OD-06	Environment grouping	Options A to D (Section 5.1)	Option B	Enterprise architect	M1
OD-07	Masking approach	Before upgrade; after upgrade before run D	Before upgrade	DBA team	M2
OD-08	Treatment of each application Kafka data set	Keep; new cluster; Cluster Linking	Per data set (Table 42)	Application owner	M4
OD-09	Quota approach	Quotas per service account; no quotas	Quotas in NP1 and NP2 from measured peaks	Kafka team	M3
OD-10	Kafka authentication	SASL/PLAIN; SASL/OAUTHBEARER	SASL/PLAIN	Security architect	M2
OD-11	Topic creation	Pega creates (CREATE ACL); pre-created topics	Pega creates	Security architect	M2
OD-12	Outage and rollback windows	From rehearsal	Rehearsal time plus 25 %	Business owner	M5
OD-13	OpenSearch snapshots	None; provider snapshots	Provider snapshots for PROD and NP2; none for NP1	Search team	M3

18.3 Assumptions

Table 73. Assumptions

ID	Assumption	Check that would prove it wrong
AS-01	The customer's Okta tenant includes API Access Management	Okta admin console shows no custom authorization servers
AS-02	Confluent Cloud offers Enterprise and Dedicated with Private Link in the AKS region	Confluent console region list
AS-03	The application does not need Cassandra (no Customer Decision Hub or Process AI)	Application stack review
AS-04	The database engine and version of the clone are on the 26.1.1 Platform Support Guide [R1]	DBA check against [R1]
AS-05	8.8 queues can be drained within the planned window	Rehearsal drain time
AS-06	Non-production environments have their own AKS namespaces and network policies	Platform design review
AS-07	The firewall supports FQDN rules for <okta-domain>	Firewall policy test

18.4 Source reconciliation

These are the places where sources disagree or leave a gap, with the action taken.

Table 74. Source reconciliation register

ID	Topic	What the sources say	Action
RC-01	SRS version	The '26 page lists SRS 1.44.3 or later and OpenSearch best practice 2.15 [R14] ; the chart README lists a newer SRS and OpenSearch 2.19 [R24]	Choose OpenSearch 2.15 or 2.19; use the SRS tag the README lists for chart 4.13.0
RC-02	Kafka SASL mechanisms	Pega lists JAAS, OAUTHBEARER, PLAIN and SCRAM [R11] ; the chart's saslMechanism lists PLAIN and SCRAM [R23]	Use PLAIN (OD-10)
RC-03	Broker settings on Confluent Cloud	Pega lists replica.fetch.* and unclean.leader.election.enable [R11] ; not editable on Confluent Cloud [R34]	Written confirmation from Confluent; topic max.message.bytes set

ID	Topic	What the sources say	Action
RC-04	Kafka version	The chart's Kafka requirements page names an older Kafka [R26] ; the '26 page names client 4.0.0 [R3]	Follow the '26 page
RC-05	Default customerDeploymentId	Pega page: generated automatically by the chart [R14] ; README: defaults to the namespace name [R23]	Always set it explicitly
RC-06	Hazelcast removal for a cloned database	Removal required before updating to '25 [R5] ; steps assume a running system [R6]	GQ-03 to Pega Support
RC-07	DSS and Helm precedence	Not settled by the pages read	FS-24; GQ-05
RC-08	SRS checks on issuer and audience	Pega documents the signature and guid checks only [R23]	FS-12 in DEV
RC-09	Okta app attributes in client credentials tokens	Expression language lists app attributes [R46] ; client credentials page does not mention them [R43]	DEV test before choosing option 2

Appendix A. Configuration inventory

This appendix holds every environment-specific value. The names in the first table follow the convention in Section 2.6 and are fixed once the environment is built. The values in the later tables are recorded during the build and kept under change control. Secrets are never recorded here; only the Key Vault secret names are.

A.1 Names per environment

Table 75. Names and sizes per environment

Item	DEV	SIT	UAT	PERF	PREPROD	PROD
Code	dev	sit	uat	perf	ppd	prd
Service group	NP1	NP1	NP1	NP2	NP2	PRD
Pega namespace	pega-dev	pega-sit	pega-uat	pega-perf	pega-ppd	pega-prd
SRS namespace and release	srs-dev	srs-sit	srs-uat	srs-perf	srs-ppd	srs-prd
Key Vault	kv-pega-dev	kv-pega-sit	kv-pega-uat	kv-pega-perf	kv-pega-ppd	kv-pega-prd
Confluent cluster	cc-np1	cc-np1	cc-np1	cc-np2	cc-np2	cc-prd
Service account	sa-pega-dev	sa-pega-sit	sa-pega-uat	sa-pega-perf	sa-pega-ppd	sa-pega-prd
Topic and group prefix	pega-dev-	pega-sit-	pega-uat-	pega-perf-	pega-ppd-	pega-prd-
customerDeploymentId and index prefix	pega26-dev	pega26-sit	pega26-uat	pega26-perf	pega26-ppd	pega26-prd
OpenSearch service	os-np1	os-np1	os-np1	os-np2	os-np2	os-prd
OpenSearch user	srs-dev	srs-sit	srs-uat	srs-perf	srs-ppd	srs-prd
OpenSearch role	pega26-dev-srs	pega26-sit-srs	pega26-uat-srs	pega26-perf-srs	pega26-ppd-srs	pega26-prd-srs
Okta client	pega-srs-dev	pega-srs-sit	pega-srs-uat	pega-srs-perf	pega-srs-ppd	pega-srs-prd
Web replicas (min to max)	1 to 2	1 to 3	2 to 3	3 to 6	3 to 6	3 to 6
Batch replicas (min to max)	1 to 2	1 to 3	1 to 3	3 to 6	3 to 6	3 to 6
SRS replicas	1	1	1	3	3	3

The replica counts are starting values. PERF measurements (M3) set the PROD and PREPROD values.

A.2 Values recorded per service group

Table 76. Values recorded per service group

Value	Where used	NP1	NP2	PRD
Confluent environment and cluster ID (lkc-...)	Commands; Confluent console			
Cluster type and capacity	OD-02; Section 6.3			
Bootstrap host (<bootstrap-host>)	stream.bootstrapServer			
Confluent network ID and private endpoint	Section 6.4			
Private DNS zone for Confluent	Section 4.4			
Partition limit and measured count	Section 6.7			
OpenSearch provider and version	OD-03; Section 7.2			
OpenSearch endpoint (<search-host>) and port	srs.srsStorage.domain, port			

Value	Where used	NP1	NP2	PRD
OpenSearch data nodes, storage and shards	Section 7.4			
Disk watermarks (low, high, flood stage)	Section 7.4; Section 17.1			
Okta authorization server ID (<auth-server-id>)	Token URL			
Token endpoint and JWKS URL (<okta-jwks-url>)	pegasearch.srsAuth.url; 0AuthPublicKeyURL			
Access token lifetime	Okta access policy rule			

A.3 Values recorded per environment

Table 77. Values recorded per environment (one copy per environment)

Value	Where used	Recorded value and date
Confluent API key ID (not the secret)	confluent-jaas in Key Vault	
Client quota (produce and consume)	Section 5.4	
Okta client ID (<okta-client-id>)	pegasearch.srsAuth.clientId	
Okta client key ID (<kid>) and expiry	Section 8.3	
JDBC URL, driver class and database type	global.jdbc	
Rules and data schema names	global.jdbc.rulesSchema, dataSchema	
Pega host name (<pega-host>)	Web tier ingress	
Application Gateway request timeout	Ingress annotation	
Pega, installer and SRS image digests	Pipeline record	
Helm chart version	Pipeline	4.13.0
SRS image tag (<tag>)	srs.srsRuntime.srsImage	
Index shard and replica counts after the first build	Section 7.4	
Full index build time and document counts	Section 7.8	
Cloned DSS for stream and search (CD-01, CD-03)	Section 10.3	

Appendix B. Complete Helm values files

These files are generated from the same source as the tables in Section 9, for chart version 4.13.0 [R23, R24]. SIT stands for the NP1 environments and PROD for PROD; the other environments differ only in the code, the names in Table 75 and the replica counts. Values in angle brackets come from Appendix A. No secret is written in these files: each `external_secret_name` and `authSecret` names a Kubernetes secret created by the External Secrets Operator (Section 8.2). Render every file with `helm template` and review the output before use.

B.1 SIT: pega chart, upgrade run

`pega-sit-upgrade.yaml`. Upgrade run: installer job only, no Pega pods.

```
global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "upgrade"
  customerDeploymentId: "pega26-sit"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret"
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
      imagePullSecretNames: []
    pega:
      image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      replicas: 1
      service:
        port: 80
        targetPort: 8080
        tls:
          enabled: true
          external_secret_names: ["pega-tier-tls"]
          port: 443
          targetPort: 8443
      ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
          appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
        tls:
          enabled: true
      resources:
        requests:
          memory: "12Gi"
          cpu: 3
        limits:
          memory: "12Gi"
          cpu: 4
      hpa:
        enabled: true
        minReplicas: 1
        maxReplicas: 3
      pdb:
        enabled: true
        minAvailable: 1
    - name: "batch"
      nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
        Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
      replicas: 1
```

```

    hpa:
      enabled: true
      minReplicas: 1
      maxReplicas: 3
    resources:
      requests:
        memory: "12Gi"
        cpu: 3
      limits:
        memory: "12Gi"
        cpu: 4
    pdb:
      enabled: true
      minAvailable: 1
cassandra:
  enabled: false
hazelcast:
  enabled: false
  clusteringServiceEnabled: false
stream:
  enabled: true
  bootstrapServer: "<bootstrap-host>:9092"
  securityProtocol: SASL_SSL
  saslMechanism: PLAIN
  trustStore: ""
  keyStore: ""
  jaasConfig: ""
  streamNamePattern: "pega-sit-{stream.name}"
  replicationFactor: "3"
  external_secret_name: "pega-stream-secret"
pegasearch:
  externalSearchService: true
  externalURL: "https://srs-sit.srs-sit.svc.cluster.local"
  srsAuth:
    enabled: true
    url: "https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
    clientId: "<okta-client-id>"
    scopes: "pega.search:full"
    authType: "private_key_jwt"
    privateKeyAlgorithm: "RS256"
    external_secret_name: "pega-srs-oauth"
  srsMTLS:
    enabled: false
installer:
  image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
  upgrade:
    upgradeType: "in-place"
    targetRulesSchema: ""
    targetDataSchema: ""

```

B.2 SIT: pega chart, first-start run

pega-sit-first.yaml. First-start run: web tier only, batch tier held at zero.

```

global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "deploy"
  customerDeploymentId: "pega26-sit"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret"
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
    imagePullSecretNames: []
  pega:

```

```

    image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      replicas: 1
      service:
        port: 80
        targetPort: 8080
        tls:
          enabled: true
          external_secret_names: ["pega-tier-tls"]
          port: 443
          targetPort: 8443
      ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
          appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
        tls:
          enabled: true
      resources:
        requests:
          memory: "12Gi"
          cpu: 3
        limits:
          memory: "12Gi"
          cpu: 4
      hpa:
        enabled: true
        minReplicas: 1
        maxReplicas: 3
      pdb:
        enabled: true
        minAvailable: 1
    - name: "batch"
      nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
        Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
      replicas: 0
      hpa:
        enabled: false
      resources:
        requests:
          memory: "12Gi"
          cpu: 3
        limits:
          memory: "12Gi"
          cpu: 4
      pdb:
        enabled: true
        minAvailable: 1
  cassandra:
    enabled: false
  hazelcast:
    enabled: false
    clusteringServiceEnabled: false
  stream:
    enabled: true
    bootstrapServer: "<bootstrap-host>:9092"
    securityProtocol: SASL_SSL
    saslMechanism: PLAIN
    trustStore: ""
    keyStore: ""
    jaasConfig: ""
    streamNamePattern: "pega-sit-{stream.name}"
    replicationFactor: "3"
    external_secret_name: "pega-stream-secret"
  pegasearch:
    externalSearchService: true
    externalURL: "https://srs-sit.srs-sit.svc.cluster.local"
    srsAuth:
      enabled: true
      url: "https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
      clientId: "<okta-client-id>"
      scopes: "pega.search:full"
      authType: "private_key_jwt"
      privateKeyAlgorithm: "RS256"

```

```

    external_secret_name: "pega-srs-oauth"
  srsMTLS:
    enabled: false
  installer:
    image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
    upgrade:
      upgradeType: "in-place"
      targetRulesSchema: ""
      targetDataSchema: ""

```

B.3 SIT: pega chart, steady-state deploy run

pega-sit-deploy.yaml. Steady-state deploy run.

```

global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "deploy"
  customerDeploymentId: "pega26-sit"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret"
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
    imagePullSecretNames: []
    pega:
      image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      replicas: 1
      service:
        port: 80
        targetPort: 8080
        tls:
          enabled: true
          external_secret_names: ["pega-tier-tls"]
          port: 443
          targetPort: 8443
      ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
          appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
        tls:
          enabled: true
      resources:
        requests:
          memory: "12Gi"
          cpu: 3
        limits:
          memory: "12Gi"
          cpu: 4
      hpa:
        enabled: true
        minReplicas: 1
        maxReplicas: 3
      pdb:
        enabled: true
        minAvailable: 1
    - name: "batch"
      nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
        Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
      replicas: 1
      hpa:
        enabled: true

```

```

    minReplicas: 1
    maxReplicas: 3
  resources:
    requests:
      memory: "12Gi"
      cpu: 3
    limits:
      memory: "12Gi"
      cpu: 4
  pdb:
    enabled: true
    minAvailable: 1
cassandra:
  enabled: false
hazelcast:
  enabled: false
  clusteringServiceEnabled: false
stream:
  enabled: true
  bootstrapServer: "<bootstrap-host>:9092"
  securityProtocol: SASL_SSL
  saslMechanism: PLAIN
  trustStore: ""
  keyStore: ""
  jaasConfig: ""
  streamNamePattern: "pega-sit-{stream.name}"
  replicationFactor: "3"
  external_secret_name: "pega-stream-secret"
pegasearch:
  externalSearchService: true
  externalURL: "https://srs-sit.srs-sit.svc.cluster.local"
  srsAuth:
    enabled: true
    url: "https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
    clientId: "<okta-client-id>"
    scopes: "pega.search:full"
    authType: "private_key_jwt"
    privateKeyAlgorithm: "RS256"
    external_secret_name: "pega-srs-oauth"
  srsMTLS:
    enabled: false
installer:
  image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
  upgrade:
    upgradeType: "in-place"
    targetRulesSchema: ""
    targetDataSchema: ""

```

B.4 SIT: backingservices chart (SRS)

backingservices-sit.yaml

```
global:
  k8sProvider: "aks"
  imageCredentials:
    registry: "<acr-name>.azurecr.io"
srs:
  enabled: true
  deploymentName: "srs-sit"
  srsRuntime:
    replicaCount: 1
    srsImage: "<acr-name>.azurecr.io/platform-services/search-n-reporting-service-os:<tag>"
  env:
    AuthEnabled: true
    OAuthPublicKeyURL: "<okta-jwks-url>"
  ssl:
    enabled: true
    clientAuthentication: "want"
    keystore:
      file: "srs-keystore.p12"
      type: "PKCS12"
    truststore:
      file: "srs-truststore.jks"
      type: "JKS"
    certsSecret: "srs-runtime-certs"
  srsStorage:
    provisionInternalESCluster: false
    domain: "<search-host>"
    port: 443
    protocol: https
    tls:
      enabled: true
    basicAuthentication:
      enabled: true
    authSecret: "srs-search-credentials"
    requireInternetAccess: false
    networkPolicy:
      enabled: true
```

B.5 PROD: pega chart, upgrade run

pega-prd-upgrade.yaml. Upgrade run: installer job only, no Pega pods.

```
global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "upgrade"
  customerDeploymentId: "pega26-prd"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret"
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
    imagePullSecretNames: []
    pega:
      image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      replicas: 3
      service:
        port: 80
        targetPort: 8080
      tls:
```

```

        enabled: true
        external_secret_names: ["pega-tier-tls"]
        port: 443
        targetPort: 8443
    ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
            appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
        tls:
            enabled: true
    resources:
        requests:
            memory: "12Gi"
            cpu: 3
        limits:
            memory: "12Gi"
            cpu: 4
    hpa:
        enabled: true
        minReplicas: 3
        maxReplicas: 6
    pdb:
        enabled: true
        minAvailable: 1
- name: "batch"
  nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
    Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
  replicas: 3
  hpa:
    enabled: true
    minReplicas: 3
    maxReplicas: 6
  resources:
    requests:
        memory: "12Gi"
        cpu: 3
    limits:
        memory: "12Gi"
        cpu: 4
  pdb:
    enabled: true
    minAvailable: 1
cassandra:
  enabled: false
hazelcast:
  enabled: false
  clusteringServiceEnabled: false
stream:
  enabled: true
  bootstrapServer: "<bootstrap-host>:9092"
  securityProtocol: SASL_SSL
  saslMechanism: PLAIN
  trustStore: ""
  keyStore: ""
  jaasConfig: ""
  streamNamePattern: "pega-prd-{stream.name}"
  replicationFactor: "3"
  external_secret_name: "pega-stream-secret"
pegasearch:
  externalSearchService: true
  externalURL: "https://srs-prd.srs-prd.svc.cluster.local"
  srsAuth:
    enabled: true
    url: "https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
    clientId: "<okta-client-id>"
    scopes: "pega.search:full"
    authType: "private_key_jwt"
    privateKeyAlgorithm: "RS256"
    external_secret_name: "pega-srs-oauth"
  srsMTLS:
    enabled: false
installer:
  image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
  upgrade:
    upgradeType: "in-place"

```

```
targetRulesSchema: ""
targetDataSchema: ""
```

B.6 PROD: pega chart, first-start run

pega-prd-first.yaml. First-start run: web tier only, batch tier held at zero.

```
global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "deploy"
  customerDeploymentId: "pega26-prd"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret"
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
    imagePullSecretNames: []
    pega:
      image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      replicas: 3
      service:
        port: 80
        targetPort: 8080
      tls:
        enabled: true
        external_secret_names: ["pega-tier-tls"]
        port: 443
        targetPort: 8443
      ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
          appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
        tls:
          enabled: true
      resources:
        requests:
          memory: "12Gi"
          cpu: 3
        limits:
          memory: "12Gi"
          cpu: 4
      hpa:
        enabled: true
        minReplicas: 3
        maxReplicas: 6
      pdb:
        enabled: true
        minAvailable: 1
    - name: "batch"
      nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
        Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
      replicas: 0
      hpa:
        enabled: false
      resources:
        requests:
          memory: "12Gi"
          cpu: 3
        limits:
          memory: "12Gi"
          cpu: 4
```

```

    pdb:
      enabled: true
      minAvailable: 1
  cassandra:
    enabled: false
  hazelcast:
    enabled: false
    clusteringServiceEnabled: false
  stream:
    enabled: true
    bootstrapServer: "<bootstrap-host>:9092"
    securityProtocol: SASL_SSL
    saslMechanism: PLAIN
    trustStore: ""
    keyStore: ""
    jaasConfig: ""
    streamNamePattern: "pega-prd-{stream.name}"
    replicationFactor: "3"
    external_secret_name: "pega-stream-secret"
  pegasearch:
    externalSearchService: true
    externalURL: "https://srs-prd.srs-prd.svc.cluster.local"
    srsAuth:
      enabled: true
      url: "https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
      clientId: "<okta-client-id>"
      scopes: "pega.search:full"
      authType: "private_key_jwt"
      privateKeyAlgorithm: "RS256"
      external_secret_name: "pega-srs-oauth"
    srsMTLS:
      enabled: false
  installer:
    image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
    upgrade:
      upgradeType: "in-place"
      targetRulesSchema: ""
      targetDataSchema: ""

```

B.7 PROD: pega chart, steady-state deploy run

pega-prd-deploy.yaml. Steady-state deploy run.

```

global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "deploy"
  customerDeploymentId: "pega26-prd"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret"
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
    imagePullSecretNames: []
    pega:
      image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      replicas: 3
      service:
        port: 80
        targetPort: 8080
      tls:
        enabled: true
        external_secret_names: ["pega-tier-tls"]

```

```

        port: 443
        targetPort: 8443
    ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
            appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
        tls:
            enabled: true
    resources:
        requests:
            memory: "12Gi"
            cpu: 3
        limits:
            memory: "12Gi"
            cpu: 4
    hpa:
        enabled: true
        minReplicas: 3
        maxReplicas: 6
    pdb:
        enabled: true
        minAvailable: 1
-   name: "batch"
    nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
        Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
    replicas: 3
    hpa:
        enabled: true
        minReplicas: 3
        maxReplicas: 6
    resources:
        requests:
            memory: "12Gi"
            cpu: 3
        limits:
            memory: "12Gi"
            cpu: 4
    pdb:
        enabled: true
        minAvailable: 1
cassandra:
    enabled: false
hazelcast:
    enabled: false
    clusteringServiceEnabled: false
stream:
    enabled: true
    bootstrapServer: "<bootstrap-host>:9092"
    securityProtocol: SASL_SSL
    saslMechanism: PLAIN
    trustStore: ""
    keyStore: ""
    jaasConfig: ""
    streamNamePattern: "pega-prd-{stream.name}"
    replicationFactor: "3"
    external_secret_name: "pega-stream-secret"
pegasearch:
    externalSearchService: true
    externalURL: "https://srs-prd.srs-prd.svc.cluster.local"
    srsAuth:
        enabled: true
        url: "https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
        clientId: "<okta-client-id>"
        scopes: "pega.search:full"
        authType: "private_key_jwt"
        privateKeyAlgorithm: "RS256"
        external_secret_name: "pega-srs-oauth"
    srsMTLS:
        enabled: false
installer:
    image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
    upgrade:
        upgradeType: "in-place"
        targetRulesSchema: ""
        targetDataSchema: ""

```

B.8 PROD: backingservices chart (SRS)

backingservices-prd.yaml

```
global:
  k8sProvider: "aks"
  imageCredentials:
    registry: "<acr-name>.azurecr.io"
srs:
  enabled: true
  deploymentName: "srs-prd"
  srsRuntime:
    replicaCount: 3
    srsImage: "<acr-name>.azurecr.io/platform-services/search-n-reporting-service-os:<tag>"
  env:
    AuthEnabled: true
    OAuthPublicKeyURL: "<okta-jwks-url>"
  ssl:
    enabled: true
    clientAuthentication: "want"
    keystore:
      file: "srs-keystore.p12"
      type: "PKCS12"
    truststore:
      file: "srs-truststore.jks"
      type: "JKS"
    certsSecret: "srs-runtime-certs"
  srsStorage:
    provisionInternalESCluster: false
    domain: "<search-host>"
    port: 443
    protocol: https
    tls:
      enabled: true
    basicAuthentication:
      enabled: true
    authSecret: "srs-search-credentials"
    requireInternetAccess: false
    networkPolicy:
      enabled: true
```

Appendix C. Command reference

Run these commands from a tooling pod in `pega-<code>`, so they use the same network path, DNS and egress rules as Pega, unless the command says otherwise. Use a tooling image mirrored into the customer registry. Read secrets from Key Vault at run time and keep them out of shell history. Check option names against the installed tool versions before use.

C.1 Kafka and Confluent Cloud

Kafka commands K-1 to K-7

```
# K-1 DNS: bootstrap and broker names must resolve to private IPs
nslookup <bootstrap-host>
nslookup <broker-host>

# K-2 TLS: certificate chain and protocol
openssl s_client -connect <bootstrap-host>:9092 -servername <bootstrap-host> </dev/null

# K-3 Authentication and metadata: every broker listed and reachable
kcat -b <bootstrap-host>:9092 -X security.protocol=SASL_SSL -X sasl.mechanisms=PLAIN \
-X sasl.username="$API_KEY" -X sasl.password="$API_SECRET" -L

# K-4 Topics under the environment prefix
confluent kafka topic list --cluster <lkc-id> | grep "pega-<code>-"

# K-5 Partition count under the prefix
kcat -b <bootstrap-host>:9092 -X security.protocol=SASL_SSL -X sasl.mechanisms=PLAIN \
-X sasl.username="$API_KEY" -X sasl.password="$API_SECRET" -L -J \
| jq --arg p "pega-<code>-" \
' [.topics[] | select(.topic | startswith($p)) | .partitions | length] | add'

# K-6 Topic configuration and the message size limit
confluent kafka topic describe <topic> --cluster <lkc-id>
confluent kafka topic update <topic> --config max.message.bytes=5000000 \
--cluster <lkc-id>

# K-7 ACLs for the environment's service account
confluent kafka acl list --service-account <sa-id> --cluster <lkc-id>
```

K-8 runs against the customer's own Kafka cluster that 8.8 application data sets read, with that cluster's client properties. K-9 runs only during a refresh or retirement, after the environment's Pega tiers are at zero.

Kafka commands K-8 and K-9

```
# K-8 Committed offsets of an application consumer group (customer cluster)
kafka-consumer-groups.sh --bootstrap-server <customer-bootstrap> \
--command-config customer-client.properties --describe --group <group>

# K-9 Delete the environment's topics and consumer groups (refresh or retirement)
confluent kafka topic list --cluster <lkc-id> -o json \
| jq -r '.[].name | select(startswith("pega-<code>"))' > topics.txt
xargs -n1 confluent kafka topic delete --force --cluster <lkc-id> < topics.txt
kafka-consumer-groups.sh --bootstrap-server <bootstrap-host>:9092 \
--command-config sa-pega-<code>.properties --list | grep "^pega-<code>-" > groups.txt
xargs -n1 kafka-consumer-groups.sh --bootstrap-server <bootstrap-host>:9092 \
--command-config sa-pega-<code>.properties --delete --group < groups.txt
```

Caution. Review `topics.txt` and `groups.txt` before running the delete lines. Every name must start with this environment's prefix. The prefixed ACLs stop the service account from deleting another environment's topics, but the review is still required.

C.2 OpenSearch and SRS

Search commands S-1 to S-6

```
# S-1 DNS and version: private IP; version 2.15 or 2.19
nslookup <search-host>
curl -s -u "$OS_USER:$OS_PASSWORD" https://<search-host>/

# S-2 Cluster health (green expected)
curl -s -u "$OS_USER:$OS_PASSWORD" "https://<search-host>/_cluster/health?pretty"

# S-3 Settings required by SRS (administrator)
curl -s -u "$OS_ADMIN:$OS_ADMIN_PASSWORD" \
  "https://<search-host>/_cluster/settings?pretty&include_defaults=false"

# S-4 Indexes under the environment's customerDeploymentId
curl -s -u "$OS_USER:$OS_PASSWORD" "https://<search-host>/_cat/indices/pega26-<code>?*v"

# S-5 SRS from the Pega namespace: refused without a token, accepted with one
SRS=https://srs-<code>.srs-<code>.svc.cluster.local
curl -s -o /dev/null -w "%{http_code}\n" "$SRS/"
curl -s -o /dev/null -w "%{http_code}\n" -H "Authorization: Bearer $TOKEN" "$SRS/"

# S-6 Delete the environment's indexes (refresh or retirement, as the SRS user)
curl -s -u "$OS_USER:$OS_PASSWORD" \
  "https://<search-host>/_cat/indices/pega26-<code>?*h=index"
curl -s -u "$OS_USER:$OS_PASSWORD" -X DELETE "https://<search-host>/pega26-<code>?"
```

For S-5, the first call must return 401. The second call must not return 401 or 403; the exact code depends on the path, which is not part of the check. S-6 relies on `destructive_requires_name` being false (OS-2) and on the index-scoped role, which limits the delete to this environment's indexes.

C.3 Okta tokens

O-1 builds the signed client assertion that `private_key_jwt` needs and requests a token, as Pega does [R43, R45]. The assertion carries the client ID as issuer and subject, the token endpoint as audience, a unique ID and a short expiry. Run it from the Pega namespace, so the egress path to `<okta-domain>` is tested too. It needs Python with the PyJWT and cryptography packages in the tooling image.

Okta commands O-1 and O-2

```
# O-1 Token request with private_key_jwt
export TOKEN_URL="https://<okta-domain>/oauth2/<auth-server-id>/v1/token"
export CLIENT_ID="<okta-client-id>"
ASSERTION=$(python3 - <<'EOF'
import os, time, uuid, jwt
now = int(time.time())
claims = {"iss": os.environ["CLIENT_ID"], "sub": os.environ["CLIENT_ID"],
  "aud": os.environ["TOKEN_URL"], "iat": now, "exp": now + 300,
  "jti": str(uuid.uuid4())}
key = open("/tmp/okta-srs-private-key.pem").read()
print(jwt.encode(claims, key, algorithm="RS256", headers={"kid": "<kid>"}))
EOF
)
TOKEN=$(curl -s -X POST "$TOKEN_URL" \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d grant_type=client_credentials -d scope=pega.search:full \
  -d client_assertion_type=urn:ietf:params:oauth:client-assertion-type:jwt-bearer \
  -d client_assertion="$ASSERTION" | jq -r .access_token)
rm -f /tmp/okta-srs-private-key.pem

# O-2 Decode the token claims without verifying them.
# Check iss, scp (contains pega.search:full), guid (pega26-<code>) and exp.
# Do not paste production tokens into web tools.
python3 - "$TOKEN" <<'EOF'
import base64, json, sys
payload = sys.argv[1].split(".")[1]
payload += "=" * (-len(payload) % 4)
print(json.dumps(json.loads(base64.urlsafe_b64decode(payload)), indent=2))
EOF
```

C.4 Kubernetes and Helm

Kubernetes and Helm commands P-1 to P-6

```
# P-1 Pod status
kubectl get pods -n pega-<code> -o wide
kubectl get pods -n srs-<code> -o wide

# P-2 Pod logs: stream and search errors
kubectl logs -n pega-<code> <pod> --since=30m | grep -Ei "stream|kafka|srs|search"

# P-3 Rolling restart of a tier
kubectl rollout restart deployment/pega-<tier> -n pega-<code>
kubectl rollout status deployment/pega-<tier> -n pega-<code>

# P-4 External secret synchronization (SecretSynced expected)
kubectl get externalsecret -n pega-<code>
kubectl get externalsecret -n srs-<code>

# P-5 Installer job log
kubectl get jobs -n pega-<code>
kubectl logs -n pega-<code> job/<installer-job> -f

# P-6 Render and compare values before any Helm run
helm template pega pega/pega --version 4.13.0 -n pega-<code> \
  -f pega-<code>-<run>.yaml > rendered.yaml
helm diff upgrade pega pega/pega --version 4.13.0 -n pega-<code> \
  -f pega-<code>-<run>.yaml
```

P-6 uses the helm-diff plug-in. If it is not allowed, compare the rendered output with the output of the previous run instead.

Appendix D. Evidence and rehearsal templates

Each environment build, refresh, rehearsal and cutover produces an evidence pack, stored with the change record. The templates fix what each record holds, so rehearsal and production results can be compared line by line.

D.1 Evidence pack contents

Table 78. Evidence pack items

Item	Content	Produced by
E-01 Values diff	P-6 output against the previous run, secrets excluded	Platform team
E-02 Kafka connectivity	K-1 to K-3 output from a pod in <code>pega -<code></code>	Kafka engineer
E-03 Kafka objects	K-4 to K-7 output	Kafka engineer
E-04 Search connectivity	S-1 to S-3 and S-5 output	Search engineer
E-05 Token claims	O-2 output (claims only; never the token)	Identity team
E-06 Clean-up record	One line per CD item (template below)	Pega LSA
E-07 Landing pages	Stream and search landing page screenshots showing the Helm values	Pega LSA
E-08 Index record	S-4 output, build time and count sheet	Pega LSA
E-09 Test records	Isolation, functional, performance and failure scenario results	Test lead
E-10 Timings	Start and end time of each step	Cutover manager

D.2 Clean-up record

Table 79. Clean-up record (one per clone)

CD item	Found in the clone	Action taken	Done by and time	Check
CD-01				
CD-02				
CD-03				
CD-04				
CD-05				
CD-06				
CD-07				
CD-08				
CD-09				
CD-10				
CD-11				
CD-12				
CD-13				
CD-14				

D.3 Check record

Table 80. Check record

Field	Entry
Check or scenario ID	
Environment and occasion	
Run by and date	
Command or screen	
Expected result	
Actual result	
Result (Pass, Fail, Pass with note)	
Evidence item	
Deviation and action	

D.4 Timed cutover record

Table 81. Timed cutover record (rehearsals and production)

Step	Planned start	Actual start	Actual end	Duration	Notes
C-1					
C-2					
C-3					
C-4					
C-5					
C-6					
C-7					
C-8					
C-9					
C-10					

D.5 Go/No-Go record

Table 82. Go/No-Go record

Gate	Criteria met	Evidence items	Decision owner	Decision and time
Go/No-Go 1			Business owner with platform lead	
First-start gate			Cutover manager	
Go/No-Go 2			Cutover manager	
Go/No-Go 3			Cutover manager with business owner	

D.6 Retirement record

Table 83. Retirement record

Item	Deleted by and date	Evidence
Helm releases and namespaces		
Topics and consumer groups (K-9)		
ACLs, API keys and service account		
Indexes (S-6), OpenSearch user and role		
Okta client and <code>guid</code> expression entry		
Key Vault secrets and vault		
Firewall rules and DNS records		

Appendix E. Glossary

Table 84. Glossary of terms

Term	Meaning
ACL	Access control list. In Kafka, a rule that allows a principal an operation on a resource such as a topic or consumer group.
AKS	Azure Kubernetes Service.
Authorization server (Okta)	The Okta component that issues access tokens. A custom authorization server allows custom scopes and claims.
Clone-and-upgrade	The path in this document: copy the 8.8 database, upgrade the copy to 26.1.1 with the installer job, and start 26.1.1 on it.
CodeSet	A Pega rule that groups Java libraries imported into Pega Platform.
Confluent Cloud	Managed Apache Kafka service from Confluent.
customerDeploymentId	The ID that SRS uses to separate each Pega environment's data. It becomes the index name prefix and must match the guid claim.
Data set (Kafka)	A Pega rule that reads from or writes to a Kafka topic for an application integration. It is separate from the stream service.
DSS	Dynamic System Setting, a configuration record in the Pega database.
External Secrets Operator	A Kubernetes operator that copies secrets from Azure Key Vault into Kubernetes secrets.
Go/No-Go	A decision point with written criteria, at which the cutover continues or stops.
guid	The token claim that SRS compares with the customerDeploymentId of the environment.
Hazelcast	The in-memory clustering library used by Pega Platform up to '24, removed in '25.
Helm	The package manager for Kubernetes used to deploy the pega and backingservices charts.
Index build	Full reindex of Pega data into SRS from the upgraded database.
Installer job	The Kubernetes job the pega chart runs to install or upgrade the Pega database.
JAAS	Java Authentication and Authorization Service. The Kafka client reads its SASL credentials from a JAAS configuration string.
JWKS	JSON Web Key Set, the public keys that SRS uses to check token signatures.
Masking	Replacing personal or sensitive data in a non-production copy of the database.
OpenSearch	Open-source search engine. SRS stores Pega indexes in it.
Private Link	Azure service that exposes a service through a private IP in the customer VNet.
private_key_jwt	Client authentication in which the client signs an assertion with its private key instead of sending a secret.
Queue processor	A Pega rule that processes queued items asynchronously through the stream service.
SASL_SSL	Kafka security protocol that combines TLS encryption with SASL authentication.
Service group	The set of environments that share one Confluent cluster and one OpenSearch service (NP1, NP2) or the dedicated production set (PRD).
SRS	Search and Reporting Service, the Pega backing service between Pega Platform and OpenSearch.
Stream service	The Pega service that uses Kafka for queue processors, job schedulers, data flows and, from '25, cluster messaging.
streamNamePattern	Helm setting that defines how Pega names its Kafka topics. It carries the environment prefix.

Appendix F. References

Sources were read between 28 September and 1 October 2026. Vendor pages change; check each page again before each rehearsal (RK-12).

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Appendix G. Image sources and attribution

Figures marked "Prepared for this implementation" were drawn for this document from the facts cited in the surrounding text. They are not Pega-published diagrams. The figures below are reproduced from public vendor pages, unchanged except for scaling.

Table 85. Third-party images

Figure	Publisher	Source page
Pega Platform Kubernetes architecture for '26	Pega Documentation	[R21]
Earlier Pega cluster layout with embedded Kafka and in-cluster Elasticsearch	Pega Academy	[R30]
Externalized services in a Pega deployment	Pega Academy	[R29]
The two Kafka use cases in Pega Platform	Pega Documentation	[R11]
Azure Private Link between a customer VNet and Confluent Cloud	Confluent Documentation	[R32]
DNS records that Confluent lists for Azure Private Link	Confluent Documentation	[R32]
Client-managed cloud deployment with SRS as a backing service	Pega Academy	[R28]
OpenSearch cluster roles: cluster-manager and data nodes	OpenSearch Documentation	[R37]

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